



PES
UNIVERSITY

B. TECH IN COMPUTER SCIENCE AND ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES

- Prepare and train students in theoretical foundations to work with cutting edge computing technologies and design solutions to complex engineering problems, making them ready to work in industrial environment.
- Develop all round skills such as team building, inter-personal skills, and leadership qualities in order to effectively communicate with engineering community and with society at large.
- Promote research culture through internships, research assistantships, research-oriented projects, sponsored and collaborative research and enable them to pursue higher studies in computer science and related fields.
- To inculcate social concern meeting the requirements of prospective employers and to develop an ability to innovate efficient computing solutions for a better society.
- Create professionally superior and ethically strong globally competent employees and entrepreneurs.

PROGRAM OUTCOMES

- Apply mathematical and theoretical principles in the modeling and design of high-quality computer-based systems using state-of-the-art computer technology.
- Conduct in-depth study of research literature in the area of Computer Science, analyse problems in order to arrive at substantiated conclusions using first principles of mathematics, and allied sciences.
- Design, implement and evaluate Computer Systems, programs and processes that meet partial/ complete specifications with concern for society, environment and culture.
- Design and conduct experiments, collect data, analyze and interpret the results to investigate complex engineering problems in the field of Computer Science.
- Apply state-of-the-art techniques and modern computer-based tools in prediction, comparison and modelling of complex engineering activities.
- Have a sound understanding of professional, legal, security and social issues and responsibilities in engineering activities involving Computer Science.
- Understand societal and environmental concerns and demonstrate responsibility in sustainable development of computer-based solutions.
- Be aware of ethical and professional responsibilities in engineering situations; make informed judgments regarding intellectual property and rights in relation to computer-based solutions in global, economic, environmental and societal contexts.
- Able to function effectively in teams to establish goals, plan tasks, meet deadlines, manage risk and produce high-quality technical solutions.
- Contribute and communicate effectively with the society, be able to write effective reports and design documents by adhering to appropriate standards, make effective presentations, give and receive clear instructions.
- Apply skills in clear communication, responsible teamwork and time management by, for example, managing a team or project and communicating with external stakeholders.
- Recognize the need for and demonstrate an ability to engage in continuing professional development in its broadest sense.

B.TECH IN COMPUTERSCIENCEANDENGINEERING

I SEMESTER (2025-29 BATCH)

Sl. No	Course Code	Course Title	Hours per week				Credits	AI Tool/ Tools/Languages	Course Type
			L	T	P	S	C		
1	UE25CS151A	Python for Computational Problem Solving	4	0	2	5	5	Python interpreter 3.8 and above. IDLE and any IDE like Jupyter Copilot aided teaching - Not copilot driven course Chatgpt	FC- Lab Integrated

II SEMESTER (2025-29 BATCH)

Sl. No	Course Code	Course Title	Hours per week				Credits	AI Tools/ Tools/Languages	Course Type
			L	T	P	S	C		
1	UE25CS151B	Problem Solving with C	4	0	2	5	5	C Programming Language GCC Compiler, GDB Debugger AI Tools- Code 5, DeepCode, Codium, Alcode Helper	FC- Lab Integrated

III SEMESTER(2024-28 BATCH)

Sl. No	Course Code	Course Title	Hours per week				Credits C	AI Tools/ Tools/Languages	Course Type
			L	T	P	S			
1	UE24CS251A	Digital Design and Computer Organization	4	0	2	5	5	Logisim- for circuit simulation, Logically- for circuit simulation	CC- Lab Integrated
2	UE24CS252A	Data Structures and its Applications	4	0	2	5	5	C-Programming language Visu Algo (Interactive Visualizations), Algorithm Visualizer (AI Explanations)	CC-Lab Integrated
3	UE24MA242 A	Mathematics for Computer Science and Engineering	4	0	0	4	4	Jupyter Notebook, Python, Pandas, Matplotlib, Scipy, Seaborn, BeautifulSoup, Numpy, Scikit learn. R (R Studio) and Python, Orange	CC- Independent
4	UE24CS242A	Web Technologies	4	0	0	4	4	HTML, CSS, JavaScript, MERN Technologies. GitHub Copilot and Tabnine	CC- Independent
5	UE24CS243A	Automata Formal Languages and Logic	4	0	0	4	4	PLY, JFLAP AI Tool - https://www.opentrain.ai/glossary/automata-theory https://www.yeschat.ai/gpts-9t55kXLEufB-Automata-Expert	CC- Independent
6	UE24EC221A	CIE L1	2	0	0	2	2		ST
7	UE25MA201A*	Bridge Course Mathematics– I (Applicable for The Lateral Entry Students)	2	0	0	0	0		FC- Independent
Total			22/24	0	4	24	24		
*-Audit Course									

IV SEMESTER(2024-28 BATCH)

Sl. No.	Course Code	Course Title	Hours per week				Credits	AI Tools/ Tools/Languages	Course Type
			L	T	P	S			
1	UE24CS251B	Microprocessor and Computer Architecture	4	0	2	5	5	ARM Simulator, Arduino Microcontroller kit, Para Cache simulator. Compute intensive Application are run on GPU systems	CC-Lab Integrated
2	UE24CS252B	Computer Networks	4	0	2	5	5	Wireshark, Python. Advanced-Computer-Networking-Tutor Computer Network Professor (yeschat.ai)AI	CC-Lab Integrated
3	UE24CS241B	Design and Analysis of Algorithms	4	0	0	4	4	C-Programming language Visu Algo (Interactive Visualizations), Algorithm Visualizer (AI Explanations)	CC-Independent
4	UE24CS242B	Operating Systems	4	0	0	4	4	C, Linux/Unix OS for system call implementation	CC-Independent
5	UE24MA241B	Linear Algebra and its Applications	4	0	0	4	4	Python IDE, MathGPT, Deepseek	CC-Independent
6	UE24EC221B	CIE L2	2	0	0	2	2		ST
7	UE25MA201B *	Bridge Course Mathematics–II (Applicable to Lateral Entry Students)	2	0	0	0	0		FC-Independent
Total			22/24	0	4	24	24		

V SEMESTER(2023-27 BATCH)

Sl. No.	Course Code	Course Title	Hours per week				Credits	Tools/Languages	Course Type
			L	T	P	S			
1	UE23CS351A	Database Management System	4	0	2	5	5	MySQL Work bench, Python, ERwin, Any other tool for ER modeling By tebase SQL Editor, Chat2DB	CC-Lab Integrated
2	UE23CS352A	Machine Learning	4	0	2	5	5	NetworkX for statistical features of graphs, Tensor flow, Keras, and Scikit Learn for traditional graph ML, and Pytorch Geometric for Graph Neural Networks. Google cloud Vertex AI(end-to-end ML platform), Amazon Sagemaker, IBM Watson studio, Auto ML tools (H2O.ai (H2O AutoML), Google AutoML, Hugging Face, Haystack etc	CC-Lab Integrated
3	UE23CS341A	Software Engineering	4	0	0	4	4	Git, GitHub, docker, Jenkins, gtest, pytest, Jira/OpenProject, SonarQube.	CC-Independent
4	UE23CS342AAX	Elective I	4	0	0	4	4		EC
5	UE23CS343ABX	Elective II	4	0	0	4	4		EC
6	UE23CS320A	Capstone Project-Phase-I	0	0	8	2	2		PW
Total			20	0	12	22	24		
Elective-I									
7	UE23CS342AA1	Advanced Algorithms	4	0	0	4	4	VisuAlgo, AlgoExpert, Wolfram Alpha C or C++ Programming Language	EC-Independent
8	UE23CS342AA2	Data Analytics	4	0	0	4	4	R and Python, MS Excel tools	EC-Independent
9	UE23CS342AA3	Internet of Things	4	0	0	4	4	Python, Embedded- C ,Cloud Platforms, Single Board Computers	EC-Independent

10	UE23CS342AA4	Applied Cryptography	4	0	0	4	4	SEED labs, Python Programming Language	EC-Independent
11	UE23CS342AA5	Virtual and Augmented Reality	4	0	0	4	4	Virtual and Augmented Reality - C/ C++/ JAVA/ Python using OpenGL. Unit3D, ThreeJS, Blender, Cloud Compare, OpenGL, Unity ML Agents	EC-Independent
12	UE23CS342AA6	Human Computer Interaction	4	0	0	4	4	Human Computer Interaction-Figma, Adobe Creative Cloud.	EC-Independent
13	UE23CS342AA7	Topics in wireless and 5G Networks	4	0	0	4	4	Cisco Packet Tracer, OMNET++	EC-Independent
14	UE23CS342AA8	Bio Inspired Computing	4	0	0	4	4	Matlab, Inspyred pythonic tool	EC-Independent
15	UE23CS342AA9	Enterprise Business System	4	0	0	4	4	Oracle, Sales force, Open ERP	EC-Independent
16	UE23CS342AA10	Quantum Computing	4	0	0	4	4		EC-Independent
Elective-II									
17	UE23CS343AB1	Image Processing	4	0	0	4	4	MatLab, Python, ChatGPT	EC-Independent
18	UE23CS343AB2	Big data	4	0	0	4	4	Hadoop, HDFS, Spark, Streaming Spark, HIVE, HBase, Mllib.	EC-Independent
19	UE23CS343AB3	Graph Theory and its Applications	4	0	0	4	4	NetworkX, PyTorch, Deep Graph Library, Neo4j, Gephi, Graphistry	EC-Independent
20	UE23CS343AB4	Real Time operating system	4	0	0	4	4	FreeRTOS+Trace / Coverity / Valgrind / gprof / C Cursor AI, System View	EC-Independent
21	UE23CS343AB5	Advanced Computer network	4	0	0	4	4	GNS3, Cisco Packet Tracer, Mininet	EC-Independent
22	UE23CS343AB6	Computer Network security	4	0	0	4	4	SEED Ubuntu VM, Wireshark, Snort, Netwox, Scapy.	EC-Independent
23	UE23CS343AB7	ROS for Autonomous Systems.	4	0	0	4	4	C++, Python, Java, ROS, Rviz, Gazebo and Webots..	EC-Independent
24	UE23CS343AB8	Text & Speech Processing	4	0	0	4	4	Tensorflow, Scikit Learn, Python 3.x. CoreNLP, Natural Language Toolkit (NLTK), TextBlob, Gensim, SpaCy, PyTorch-NLP ,	EC-Independent

								OpenNLP, Hugging Face, OpenAI API.	
ELECTIVES TO BE OPTED FOR SPECIALIZATION									
Sl. No.	SPECIALIZATION	ELECTIVE-I				ELECTIVE-II			
A	System and Core Computing (SCC)	UE23CS342AA1 UE23CS342AA8 UE23CS342AA9 UE23CS342AA10				UE23CS343AB1 UE23CS343AB2 UE23CS343AB3 UE23CS343AB4			
B	Machine Intelligence and Data Science(MIDS)	UE23CS342AA2 UE23CS342AA5 UE23CS342AA6 UE23CS342AA8				UE23CS343AB1 UE23CS342AB2 UE23CS342AB7 UE23CS342AB8			
C	Cyber Security & Connected Systems(CSCS)	UE23CS342AA3 UE23CS342AA4 UE23CS342AA5 UE23CS342AA7				UE23CS343AB5 UE23CS342AB6 UE23CS342AB7			

VI - SEMESTER(2023-27BATCH)

Sl. No	Course Code	Course Title	Hours per week				Credits C	Tools/Languages	Course Type
			L	T	P	S			
1	UE23CS351B	Cloud Computing	4	0	2	5	5	Amazon AWS (or equivalent), AWS SkillBuilder, AWS Educate, Qwiklabs, Docker, Kubernetes, Jenkins, Zookeeper, Github, NoSQL database, Flask, Python, Go Lang. AWS, Sagemaker	CC
2	UE23CS352B	Object Oriented Analysis and Design	4	0	2	5	5	Star UML, Object Oriented Programming Language(Java/C++)	CC
3	UE23CS341B	Compiler Design	4	0	0	4	4	Lex/flex and YACC/Bison. Compiler Explorer + AI Explanation	CC
4	UE23CS342BA X	Elective III	4	0	0	4	4		EC
5	UE23CS343BBX	Elective IV	4	0	0	4	4		EC
6	UE23CS320B	Capstone Project Phase - II	0	0	8	2	2		PW
Total			20	0	12	24	24		
Elective-III									
7	UE23CS342BA1	Supply Chain Management for Engineers	4	0	0	4	4	SCM applications and tools, ML tools, Case studies, Webresources.	EC-Independent
8	UE23CS342BA2	Algorithms for Information Retrieval	4	0	0	4	4	Scikit, Tensor flow, Solr, Lucene Search Engines/ Python Programming Languages	EC-Independent
9	UE23CS342BA3	AI for sustainable development	4	0	0	4	4	OpenAI API/ChatGPT, Streamlit, Google Earth Engine, UN SDG Tracker, Google Colab.	EC-Independent
10	UE23CS342BA4	Generative AI and its Applications	4	0	0	4	4	Python, Hugging Face, LM Studio, Kaggle, Google Colab, DeepEval, TruLens	EC-Independent
11	UE23CS342BA5	Block Chain	4	0	0	4	4	Solidity, Ganache, Meta mask. RemixID, Ganache tool	EC-Independent
12	UE23CS342BA6	Digital Forensics and Incident responses	4	0	0	4	4	Open source tools on Forensics. Cellebrite, Deepfake Detection Tools	EC-Independent

13	UE23CS342BA7	Digital Twins	4	0	0	4	4	C/C++/JAVA/Python using OpenGL. Luma Labs Genie, 3D CSM, Skybox Blockade Labs, SIMENS Digital Twin Tool.	EC-Independent
14	UE23CS342BA8	Cloud security	4	0	0	4	4	C/ C++/ JAVA/ Python .AWS cloud services	
Elective–IV									
15	UE23CS343BB1	Heterogeneous Parallelism	4	0	0	4	4	Pthread, OpenMP, CUDA, Vivado HLS.	EC-Independent
16	UE23CS343BB2	Topics in Deep Learning	4	0	0	4	4	Pytorch, Tensorflow, Kaggle, Google Colab, Hugging Face models	EC-Independent
17	UE23CS343BB3	Database Technologies	4	0	0	4	4	MySQL, postgres, Oracle, Apache Spark, Apache Kafka, Amazon Kinesis.	EC-Independent
18	UE23CS343BB4	Machine Learning on Graphs	4	0	0	4	4	NetworkX for statistical features of graphs, Tensorflow, Keras, and Scikit Learn for traditional graph ML, and Pytorch Geometric for Graph Neural Networks.	EC-Independent
19	UE23CS343BB5	Embedded Systems with Micropython	4	0	0	4	4	Microcontroller: Pyboard, ESP32, BBC microbit Components: LEDs, resistors, buttons, temperature sensors (e.g., TMP36, DHT11), OLED display, potentiometer, buzzer, jumper wires, breadboard, USB cable	EC-Independent
20	UE23CS343BB6	Information Security	4	0	0	4	4	SEED Labs VM, Scapy, Burp Suite, Metasploit, Nmap, etc	EC-Independent
21	UE23CS343BB7	Mobile Autonomous Robotics	4	0	0	4	4	C,C++,Python, ROS, Webots, AirSim, Gazebo	EC-Independent
22	UE23CS343BB8	Security for IOT	4	0	0	4	4	Wireshark, Yersinia, VoIP Hopper, Bettercap,aircrack-ng	EC-Independent
23	UE23CS343BB9	Applied ML in IOT	4	0	0	4	4	Arduino Nano 33 BLE sense , Arduino IDE , Google Colab	EC-Independent
24	UE23CS343BB10	Video Processing & Understanding	4	0	0	4	4	Python with libraries like OpenCV, PyTorch, TensorFlow Ffmpeg, OpenCV, PyTorch, Hugging Face Transformers (DeepSort with YOLO integration)	EC-Independent

ELECTIVES TO BE OPTED FOR SPECIALIZATION									
Sl. No.	SPECIALIZATION		ELECTIVE–III		ELECTIVE–IV				
A	System and Core Computing (SCC)		UE23CS342BA1 UE23CS342BA2 UE23CS342BA3		UE23CS343BB1 UE23CS343BB3 UE23CS343BB5 UE23CS343BB10				
B	Machine Intelligence and Data Science (MIDS)		UE23CS342BA2 UE23CS342BA3 UE23CS342BA4 UE23CS342BA7		UE23CS343BB2 UE23CS343BB4 UE23CS343BB7 UE23CS343BB9 UE23CS343BB10				
C	Cyber Security & Connected Systems (CSCS)		UE23CS342BA5 UE23CS342BA6 UE23CS342BA7 UE23CS342BA8		UE23CS343BB6 UE23CS343BB7 UE23CS343BB8 UE23CS343BB9				

VII SEMESTER(2022-26BATCH)

Sl No.	Course Code	Course Title	Hours/week				Credits	Course Type
			L	T	P	S	C	
1	UE22CS441A	Capstone Project Phase-III	0	0	16	4	4	PW
2	U22CS426AX	Special Topic/ Directed Independent Study - Swayam/MOOC	2/4/4			2	6	ST
Total			2/4/4	2/4/4	20	6	10	

VIII SEMESTER (2022-26 BATCH)

Sl. No.	Course Code	Course Title	Hours/week				Credits	Course Type
			L	T	P	S	C	
1	UE22CS421B	Capstone Project Phase-IV	0	0	8	2	2	PW
2	UE22CS421XB	Special Topic/ Directed Independent Study– Swayam/MOOC.	2	0	0	2	2	ST
3	UE22CS461XB	Internship	0	0	12	6	6	PW
Total			2	0	20	10	10	

Course Code	UE25CS151A	Course Title	Python For Computational Problem Solving				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	1	Type of Course	Foundation				
AI Tools /Tools/Languages	Python interpreter 3.8 and above. IDLE and any IDE like Jupyter. Copilot aided teaching	Desirable Knowledge	-				
Prelude	Python is an easy to learn, general-purpose, powerful programming language. It has efficient high- level data structures and a simple but effective approach to object-oriented programming. Python’s elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms.						
Course Objectives:	• Understand the syntax and semantics of Python programming language. • Apply the process of structuring the data using lists, tuples, sets and dictionaries. • Utilize the use of built-in functions to navigate the file system. • Explore various paradigms of programming and implement the Object-Oriented Programming concepts in Python.						
Course Contents	Unit 1: Introduction Computer, Computational Problem Solving, Algorithm, Computer Hardware & Software(Application Software, System Software - Operating System),Syntax, semantics and program translation. Introduction to Python Programming Language, IDLE Python Development Environment, Output function, variables, types and id, input function, operators and expressions, Control structures. Standards and Guidelines for pythonic solution 14 Hours Unit 2: Collections &Functions Lists, Tuples, Dictionaries, Sets, Strings and text file manipulation: reading and writing files. Functions: Definition, call, Positional and keyword parameter, Default parameters, Variable number of arguments. 14 Hours Unit 3: Functions, GUI, Modules, Testing and Debugging Recursion, Call-backs, Closure, Decorators, generators. Graphical User Interface with WxPython, import mechanisms. Introduction to modules – Numpy, PyImage, nltk, sklearn, Regular Expressions – re, Testing- Pytest , Function testing with Doctest, pdb debugger commands, Debugging using IDLE 14 hours Unit 4: Functional & Object Oriented Programming Lambda function, Map, filter, and reduce, max, min, Zip, list comprehension. Classes and objects - inheritance, polymorphism, iterators, Error handling & Exceptions - try, except and raise, exception propagation. 14 Hours						

Laboratory	• Exploring IDLE • Programs on Input Output Functions, Operators and Expressions and Usage of Libraries • Programs on Control Structures. • Programs on Collections (Lists, Tuples, Sets ,Dictionaries , Strings). • Programs on Files and File manipulations. • Programs on Functions, Recursion and Callback • Programs on Functional Programming. • Programs on Object Oriented Programming. • Programs on Pytest and pdb and one application module • GUI programming (using wxPython)
Text Book(s):	1: “Think Python: How to Think Like a Computer Scientist”, Allen B. Downey, 2nd Edition, Green Tea Press, 2015. (Available under CC-BY-NC license at http://greenteapress.com/thinkpython2/thinkpython2.pdf (Chapters 13, 15, 16, 17, 18) (Download pdf/html files from the above link). 2: Automate the Boring Stuff with Python”, Al Sweigart, 1st Edition, No Starch Press, 2015. (Available under CC-BY-NC-SA license at https://automatetheboringstuff.com/). 3: “Python Programming using Problem Solving approach”, Reema Thareja, Second Edition, Oxford University Press, 2017. 4: “Fundamentals of Python Programming” by Richard L. Halterman, Course Technology ptr, 2018.
Reference Book(s):	1: “Introduction to Computer Science Using Python: A Computational Problem- Focus”, Charles Dierbach, Wiley India Edition, John Wiley, 2015. 2. “Learn python Programming”, Fabrizio Romano, 2nd Edition, Packet Publishing, 2018. 3. “Fundamentals of Python: First Programs”, Kenneth A. Lambert, Cengage, 2019. 4. “Introduction to Computation and Programming Using Python: With Application to Understanding Data”, John V. Guttag, MIT Press, MIT with Library of Congress Cataloguing- in-PublicationData, 2016. 5. “WxPython in action”, By NOEL RAPPIN ROBIN DUNN,
Course Outcome	• Develop efficient programs using the Python language. • Demonstrate the ability to create and manipulate lists, tuples and dictionaries. • Perform common file system operations using built-in Python functionalities. • Apply different paradigms of programming and interpret the concepts of Object-Oriented Programming as used in Python.

Course Code	UE25CS151B	Course Title	Problem Solving with C				
Program	B.Tech	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	2	Type of Course	Foundation				
AI Tools /Tools/Languages	C Programming Language GCC Compiler, GDB Debugger AI Tools- Code 5, Deep Code, Codium, Alcode Helper	Desirable Knowledge	-				
Prelude	The Problem Solving with C course introduces students to fundamental techniques for solving Common computational problems. Students will learn to analyse problem statements, design appropriate algorithms, and implement efficient solutions. Emphasis is placed on using C language constructs to develop well-structured and maintainable programs, thereby laying a strong foundation in both problem-solving and programming skills.						
Course Objectives:	- Develop the ability to solve relevant and logical problems using a computing machine through algorithmic thinking. - Translate algorithmic solutions into programs by effectively utilizing C programming language constructs. - Acquire a solid understanding of C language syntax, semantics, and its supporting ecosystem. - Recognize and appreciate the nuances of C standards and understand their implications on program behaviour..						

Course Contents	Unit1: Problem Solving Fundamentals Introduction to Programming, Salient Features of 'C', Program Structure, Variables, Data Types & arrange of values ,Qualifiers, Operators and Expressions, Control Structures, Input/Output Functions, Language Specifications -Behaviors, Single character input and output, Coding standards and guidelines 14 Hours Unit2: Counting, Sorting and Searching Arrays–1D and 2D, Pointers, Pointer to an array, Array of pointers, Functions, Call back, Storage classes, Recursion, Searching, Sorting Unit 3: Functions, GUI, Modules, Testing and Debugging 14 Hours Unit3: Text Processing and User-Defined Types Strings, String Manipulation Functions & Error handling, Command line arguments, Dynamic Memory Management functions & Error handling, Structures, #pragma, Array of Structures, Pointer to structures, Passing Structure and Array of structure to a function, Bit fields, Unions, Enums, Lists, Stack, Queue, Priority Queue. 14 Hours Unit4: File Handling and Portable Programming File IO using redirection, File Handling functions of C, Searching, Sorting, Header files, Comparison of relevant User defined and Built-in functions, Variable Length Arguments, Environment variables, Preprocessor Directives, Conditional Compilation. 14 Hours
Laboratory	• Programs on IO, Operators and Control structures. • Programs on Arrays and Pointers, Functions using arrays and pointers • Programs on Strings, Structures and Dynamic Memory Management functions • Programs on Structures and Array of structures • Programs on Inclusion of files using redirection operator. • Programs using File handling functions in C - Sorting and Searching using Array of pointers to structures. • Implementation of Ordered List • Implementation of Priority based scheduling. • Programs on unions, enums and Preprocessor directives.
Text Book(s):	1: "The C Programming Language", Brian Kernighan and Dennis Ritchie, Prentice Hall PTR, 2nd Edition, 1988.
Reference Book(s):	1: "How To Solve It By Computer", R G Dromey, Pearson, 2011. 2: "C Programming: A Modern Approach", 2nd Edition by K.N. King. W. W. Norton & Company. 3: "Learn C the Hard Way": Zed Shaw's Hard Way Series, 1st Edition. 4: "C Puzzle Book" by Alan R. Fever, Pearson Education. 5: "Expert C Programming: Deep C Secrets" by Peter Van Der Linden, Pearson Education.

Course Outcome	• Understand and apply algorithmic solutions to counting and other basic computational Problems using appropriate C construct. • Understand, analyse, and implement fundamental sorting and searching techniques. • Apply string manipulation and text processing methods using arrays, pointers, and functions. • Design and implement user-defined data types using structures, unions, and other mechanisms in C, and demonstrate the ability to read from and write to secondary storage in a portable manner.
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Course Code	UE24CS251A	Course Title	Digital Design & Computer Organization				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	3	Type of Course	Core				
AI Tools /Tools/Languages	Icarus Verilog Simulator, GTK Wave waveform viewer. (Open–Source Tool) Logisim- for circuit simulation, Logically- for circuit simulation	Desirable Knowledge	Basic Electronics				
Prelude	This course focuses on the structure, design and operation of a computer system at different levels of abstraction. The digital design part of the course describes low level digital logic building blocks while the computer organization part explains the structure and operation of microprocessors						
Course Objectives:	- Fundamental (combinational and sequential) building blocks of digital logic circuits. - Design of more complex logic circuits such as adders, multipliers and register files. - Design of Finite State Machines based on problem specification. - Construction, using above logic circuits, of a microprocessor, and its functioning at the clock cycle level.						
Course Contents	Unit 1: Gate-Level Minimization and Combinational logic-1 Introduction, The map method Four variable K-map, Product of Sums simplification, Don't Care conditions, NAND and NOR implementation, Combinational circuits, Analysis procedure Design Procedure, Combinational logic-1: Binary Combinational logic: Adder- Subtractor, Decimal Adder, Binary multiplier, Magnitude comparator Decoders Encoders, Multiplexers. 14 Hours Unit 2: Synchronous Sequential Logic-I Synchronous Sequential Logic: Introduction, Sequential circuits, Storage elements: Latches, Flip flops, Analysis of clocked sequential circuits, State reduction and assignment, Design procedure Registers and counters: Registers, Shift register, Ripple counters, Synchronous counters Other counters. 14 Hours Unit 3: Basic structure of computers, Standard I/O interface, Interrupts. Memory System Computer Types, Functional Units: Input Unit, Memory Unit, ALU, Output Unit, Control Unit, Basic operational concepts, Number representation and arithmetic Operations, Character representation, Memory locations and addresses, Memory Operations, Instruction and instruction sequencing , Addressing modes, Assembly Languages, Accessing I/O Devices, Interrupts Standard I/O Interfaces. Semiconductor RAM memories 14 Hours Unit 4: Arithmetic Processing Unit and Control Unit Design Arithmetic: Multiplication of Positive numbers, Signed operand Multiplication, Fast multiplication, Integer division, floating point numbers operation and Architecture, Some fundamental concepts, Execution of a complete instruction, Multiple Bus Organization, Hardwired control , Single-cycle, Case Study: Multi-cycle processor data path and control. 14 Hours						

Laboratory	• Implementation of Basic gates. • Implementation of Half adder and full adder. • Implementation of Decoder , encoder and Priority encoder. • Implementation of 4-to-1 multiplexer and 1-to-4 demultiplexer. • Implementation of 4-bit magnitude comparator • Implementation of BCD adder. • Implementation of Flip-Flops(JK,T and D). • Implementation of counters(Ripple, updown). • Implementation of counters(Ring, Johnson). • Implementation of Universal Shift register. • Implementation of 4-bit serial adder. • Implementation of Sequential Binary multiplier.
Text Book(s):	1. "Digital Design", M Morris Mano, Michael D Ciletti, Pearson, 5th Edition, 2012. 2. "Computer Organization", Carl Hamacher, Zvonko Vranesic, Safwat Zaky, McGraw Hill, 5th Edition, 2002.
Reference Book(s):	1: "Digital Design & Computer Architecture", David Money Harris, Sarah L. Harris, 2nd Edition, Elsevier, 2013. 2: "Computer Organization and Design", David A. Patterson, John L. Hennessey 5th Edition, Elsevier.
Course Outcome	• Perform analysis of given synchronous digital logic circuit. • Design and implement small to medium scale data path logic circuits from given specification. • Design and implement control logic using Finite State Machines. • Understand hardware level microprocessor operation, providing a foundation for the higher layers.

Course Code	UE24CS252A	Course Title	Data structures & its Applications				
program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	3	Type of Course	Core				
AI Tools /Tools/Languages	C-Programming language Visu Algo (Interactive Visualizations), Algorithm Visualizer (AI Explanations)	Desirable Knowledge	Problem Solving with C				
Prelude	This course provides a comprehensive introduction to fundamental data structure concepts, emphasizing both their theoretical foundations and practical applications. It begins with abstract representations of data and explores how these abstractions facilitate effective problem solving. The course then bridges the gap between theory and practice by demonstrating how these concepts can be concretely implemented using a programming language. Equal importance is given to understanding the underlying principles, their implementation techniques, and their real-world applications, enabling students to not only grasp the concepts but also apply them effectively in software development.						
Course Objectives:	- To introduce the fundamental concepts and methodologies for analyzing and designing data structures, fostering problem-solving skills for efficient data storage, retrieval, and manipulation. - To explore and utilize various data structures such as arrays, linked lists, stacks, queues, trees, heaps, and graphs, emphasizing their suitability for different computational tasks and operations. - To implement core operations—insertion, deletion, searching, and modification—across a variety of data structures in both linear and non-linear forms. - To design and develop applications using appropriate data structures, demonstrating the ability to select and apply the most suitable structure to solve real-world problems effectively.						
Course Contents	Unit 1: Linked List and Stacks Review of C , Static and Dynamic Memory Allocation. Linked List: Doubly Linked List, Circular Linked List – Single and Double, Multilist: Introduction to sparse matrix (structure). Skip list Case study: Dictionary implementation using skip list Stacks: Basic structure of a Stack, Implementation of a Stack using Arrays & Linked list. Applications of Stack: Function execution, Nested functions, Recursion: Tower of Hanoi. Conversion & Evaluation of an expression: Infix to postfix, Infix to prefix, Evaluation of an Expression, Matching of Parenthesis. 14 Hours Unit 2: Queues and Trees Queues & Dequeue: Basic Structure of a Simple Queue, Circular Queue, Priority Queue, Dequeue and its implementation using Arrays and Linked List. Applications of Queue: Case Study – Josephus problem, CPU scheduling- Implementation using queue (simple /circular). General: N-ary trees, Binary Trees, Binary Search Trees (BST) and Forest: definition, properties, conversion of an N-ary tree and a Forest to a binary tree. Traversal of trees: Preorder, Inorder and Postorder.						

	14 Hours Unit 3: Application of Trees and Introduction to Graphs Implementation of BST using arrays and dynamic allocation: Insertion and deletion operations, Implementation of binary expression tree., Threaded binary search tree and its implementation. Heap: Implementation using arrays. Implementation of Priority Queue using heap - min and max heap. Applications of Trees and Heaps: Implementation of a dictionary / decision tree (Words with their meanings). Balanced Trees: definition, AVL Trees, Rotation, Splay Tree, Graphs: Introduction, Properties, Representation of graphs: Adjacency matrix, Adjacency list. Implementation of graphs using adjacency matrix and lists. Graph traversal methods: Depth first search, Breadth first search techniques. Application: Graph representation: Representation of computer network topology. 14 Hours Unit 4: Applications of Graphs , B-Trees, Suffix Tree and Hashing Application of BFS and DFS: Connectivity of graph, finding path in a network. Suffix Trees: Definition, Introduction of Trie Trees, Suffix trees. Implementations of TRIE trees, insert, delete and search operations. Hashing: Simple mapping / Hashing: hash function, hash table, Collision Handling: Separate Chaining & Open Addressing, Double Hashing, and Rehashing. Applications: URLs decoding, Word prediction using TRIE trees / Suffix Trees. 14 Hours
Laboratory	1: Linked List and advanced operations. 2: Stack and applications based on it. 3: Queue and applications based on it. 4: Binary Tree, Binary Search Tree and applications based on it. 5: Graph Data structure and applications based on it. 6: Hashing Techniques.
Text Book(s):	1: "Data Structures using C / C++", Langsum Yedidyah, Moshe J Augenstein, Aaron M Tenenbaum Pearson Education Inc, 2nd edition, 2015.
Reference Book(s):	1: "Data Structures and Program Design in C", Robert Kruse, Bruce Leung, C.L Tondo, Shashi Mogalla, Pearson, 2nd Edition, 2019
Course Outcome	• Select and apply appropriate data structures for solving problems in a variety of application domains. • Implement fundamental data structures and their operations using suitable programming constructs and techniques. • Utilize data structures effectively in competitive programming to design optimal and efficient solutions. • Design and develop efficient software systems by leveraging an in-depth understanding of data structure principles and their applications.

Course Code	UE24MA242A	Course Title	Mathematics for Computer Science Engineers				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	3	Type of Course	Core				
AI Tools /Tools/Languages	Jupyter Notebook, Python, Pandas, Matplotlib, Scipy, Seaborn, BeautifulSoup, Numpy, Scikit learn. R (R Studio) and Python, Orange	Desirable Knowledge	Engineering Math -1 , Engineering Math - 2				
Prelude	This course covers both Descriptive statistics to understand the data and the inferential statistics which seeks to infer something about a population on the basis of a statistical sample and build a simple linear regression model. This course will also cover engineering optimization techniques like unconstrained, constrained, and discrete optimization techniques.						
Course Objectives:	• Provide insights about the basic roles of a Data Scientist. Develop a greater understanding of the importance of summary statistics and Random Variables and their Distributions. • Provide students with knowledge of Confidence Intervals and their importance. Make inferences about the population parameters using sample data and test it to draw meaningful conclusions. • Provide an understanding on the importance and techniques of predicting a relationship between the two sets of data and determine the goodness of fit model. • Provide an understanding on the importance and techniques of engineering optimization.						
Course Contents	Unit 1: Descriptive Statistics and Sampling Distributions Sampling and Descriptive Statistics: Introduction, Motivating Examples and Scope. Statistics: Introduction, Types of Statistics, Types of Data, Types of Experiments – Controlled and Observational study, Sampling: Sampling Methods, Sampling Errors, Case Study. Probability and commonly used Distributions: Chebyshev's inequality, Linear function of random variables, Normal Probability Plots, Introduction to Generation of Random Variates and mention the types, Acceptance-Rejection method, Sampling Distribution, The Central Limit Theorem and Applications. Self-Learning: Generation of Random Variates -Inverse Transform Method. Applications: 1. Poisson distribution, calculation of number of calls received in a specified time duration in call centres. 2. Variance, standard deviation, identifying the customer satisfaction in online shopping 3. Central limit theorem, Load Balancing in distributed systems and internet traffic prediction 4. Sampling mean, Estimating database query response times 14 Hours Unit 2: Point Estimation, Confidence Intervals and Hypothesis Testing Commonly used Distributions: Principles of Point Estimation - Mean Squared Error for Bernoulli, Binomial, Poisson, Normal, Maximum Likelihood Estimate for Bernoulli, Binomial, Poisson, Normal and Case Study. Introduction to multivariate normal distribution, MAP distribution. Confidence Intervals: Interval Estimates for Mean of Large and Small Samples, Student's t Distribution, Interval Estimates for Proportion of Large and Small Samples, Confidence Intervals for the Difference between Two Means, Interval Estimates for Paired Data. Factors affecting Margin of Error,						

	Hypothesis Testing: Introduction, Large-Sample tests for a Population Mean, Drawing conclusions from the results of Hypothesis tests, Tests for population proportion, Small-Sample tests for a Population Mean, Case Study. Self-Learning: Confidence interval for difference between two proportions. Applications: 1. t-distribution, confidence interval, students' performance analysis based on hours of study 2. z-test, application form processing in banking system. 3. Hypothesis testing, randomly trained students' placement into tier-I and tier-II companies. 14 Hours Unit 3: Hypothesis Testing and Multiple Linear Regression Hypothesis Testing: Distribution Free Tests, Chi-squared Test, Fixed Level Testing, Type I and Type II Errors, Power of a Test, Factors Affecting Power of a Test. Correlation and Simple Linear Regression: Introduction, Correlation, the Least Square Lines, Predictions using regression models - Uncertainties in Regression Coefficients, Checking Assumptions and transforming data Multiple regression: Introduction to the Multiple Regression Model, Case Study. Self-Learning: F test for equality of Variance. Applications: 1. Linear regression, stock market prediction 2. using Chi-Square Test, Analyzing the association between vaccination and recovery of the patients considering COVID data. 3. Chi-Square Test and Test of Independence, Analyzing the relationship between gender and preference for a product purchase. 4. Identifying Type 1 and Type 2 Errors in Spam mail classification. 14 Hours Unit 4: Engineering optimization Introduction to optimization, Types of optimization, Mathematical concepts of objective function, constraints and decisions, Constrained Linear Optimization- LPP – Formulation, Graphical Method, The Simplex Method, Discrete variable optimization - Integer Programming Problem, Branch and Bound Method, Constrained Non-linear Optimization - Lagrangian Multipliers method, Kuhn Tucker Conditions, Unconstrained non-linear optimization - Gradient Descent Method, Evolutionary computation - Genetic Algorithms. Self-Learning: Newton Rapson Method. Applications: 1. Minimize a Loss functions in Neural Networks using Batch gradient descent (Unconstrained Optimization) 2. Lagrange Multipliers to find local maxima and minima of a function subject to equations constrains (Constrained Optimization) 3. Case study on Bayesian Optimization with Discrete Variables (Discrete Variable optimization) 4. Use Genetic Algorithms to optimize Production Scheduling in a manufacturing environment, focusing on minimizing total production costs while meeting job deadlines and machine constraints. Evaluate the GA's effectiveness against traditional scheduling methods. 14 Hours
Text Book(s):	1: "Statistics for Engineers and Scientists", William Navidi, McGraw Hill Education, India, 4th Edition, 2015. 2: "Engineering Optimization Theory and Practice", Singiresu S. Rao, John Wiley & Sons, Inc, 4th Edition, 2009.
Reference Book(s):	1: "Fundamentals of Data Science", Sanjeev J. Wagh, Manisha S. Bhende, and Anuradha D. Thakare, First edition, CRC Press, 2022.

	2: "Sampling- Design and Analysis", Sharon L. Lohr, 2nd edition (stats), Cengage, 2010. 3: "Data Science from Scratch", Joel Grus, O'Reilly, 1st Edition, 2015. 4: "A Hands-On Introduction to Data Science", Chirag Shah, Cambridge University Press, First edition, 2020.
Course Outcome	• Use Python and other tools to extract, clean and analyze data from several data sources (files, web) analyze an extremely large dataset and perform exploratory data analysis to extract meaningful insights. • Analyze a real-world problem and solve the same with the knowledge gained from various distribution studies. • Develop and test a hypothesis about the population parameters to draw meaningful conclusions and fit a regression model to data and use it for prediction. • Analyze a real-world problem and solve the same with the knowledge gained from various engineering optimization techniques.

Course Code	UE24CS241A	Course Title	Web Technologies				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	3	Type of Course	Core				
AI Tools /Tools/Languages	HTML, CSS, JavaScript, MERN Technologies GitHub Copilot and Tabnine	Desirable Knowledge	-				
Prelude	Web Technologies course demonstrates an in-depth understanding of the technologies necessary for designing and developing a rich web application in an efficient way.						
Course Objectives:	- Basic web technologies and building blocks of a website using HTML, CSS, JavaScript and Advanced JavaScript - The B.Tech CSE concepts of HTML5, JQuery and AJAX, MERN (MongoDB, ExpressJS, ReactJS and NodeJS) stack and build an UI of the application using React JS - Building a multi-tier application by interfacing UI to NodeJS - Integrate database MongoDB through Express JS Framework and Web services						
Course Contents	Unit 1: HTML, CSS and Client-Side Scripting Introduction to Web Architecture and Web protocols (HTTP Request Response Formats, URLs), Basic Mark- ups & syntax, HTML elements & attributes, Web Form, HTML5 (New Tags, Inputs, Elements and Controls), CSS3.0 - Styles and Style sheets, Selectors, Style properties, Box Model, JavaScript Basics (variables, scope, Built-in Objects), JavaScript objects, DOM Manipulations, Events and Event Handling in JavaScript 14 Hours Unit 2: HTML5 and ReactJS HTML5 (APIs), Audio, Video and Progress, Geolocation, Callbacks & Promises, Single Page Application, XML Vs JSON, , Async/Await, ReactJS - MERN Introduction, JSX, rendering of Elements, Set-up React Installation, React Component and JSX, React Component Styling, Properties, States and Context – (Arrays), How state works in functional component 14 Hours Unit 3: ReactJS and NodeJS State Management – Complex components, Keys,Event Handling, React Forms, React Hook – useState, useRefs and UseEffect, React Hook – useContext and useReducer, React Router, Introduction to NextJS, Understanding Node JS Architecture, callbacks, Node Modules, Buffers and Streams, File system, Axios API,Nodejs 14 Hours Unit 4: MongoDB MongoDB-Documents, Collections, Reading and Writing to MongoDB, MongoDB NodeJS Driver, running a react application on NodeJS, React Router. ExpressJS – Introduction to Web services and REST API’s, Express Framework Overview, Routing and URL building, Error Handling, Express Middleware, Form Data and File Upload. 14 Hours						

Text Book(s):	1: "Learning PHP, MySQL & JavaScript", Robin Nixon., 5th edition, O'Reilly Media, Inc. ISBN: 9781491978917, 2018. 2: "Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node", Vasan Subramanian, Apress, 2017.
Reference Book(s):	1: "Beginning Node.js, Express & MongoDB Development", Greg Lim, July 2019. 2: "Learning React, Functional Web Development with React and Redux", Alex Banks and Eve Porcello, O'Reilly Media, May 2017.
Course Outcome	• Understand basic web technologies like HTML, CSS and JavaScript • Achieve rich user experience by implementing HTML5 features and Asynchronous communication using MERN stack layers (MongoDB, ExpressJS, ReactJS and NodeJS) and Create rich User Interface using React JS • Understand and Integrate the UI with NodeJS • Create RESTful Web services using ExpressJS and MongoDB database

Course Code	UE24CS243A	Course Title	Automata Formal Languages &Logic				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	3	Type of Course	Core				
AI Tools /Tools/Languages	PLY,JFLAP AI Tool - https://www.opentrain.ai/glossary/automata-theory https://www.yeschat.ai/gpts-9t55kXLEUfB-Automata-Expert	Desirable Knowledge	-				
Prelude	The course introduces fundamental concepts in Automata and Formal Languages and their application to Logic. The course covers the notions of Finite State Automaton, Regular expression, Push down Automaton, Context Free Languages and Turing Machines. These abstract and formal models and their usage in Propositional and First Order Predicate Logic, allow for solving problems in Formal language Generation and Recognition.						
Course Objectives:	- Teach students to construct basic machines like DFA, NFA which represent Regular Languages, Regular Expressions, and Regular Grammars and to identify Non – Regular Languages. - To familiarize students to construct Context Free Languages, to construct Push down Automata which represent Context Free Languages, to convert the given grammar to various normal forms and to make use of Membership Algorithm. - Teach students to understand closure properties of Context Free Languages, to identify Non – Context Free Languages and to construct Turing Machines and - To familiarize students with concepts like Recursively Enumerable languages, Recursive Languages, Undecidable Problems. And to familiarize notions of mathematical logic: logical notations (syntax) and how to assign meaning to them (semantics).						
Course Contents	Unit 1: Introduction Mathematical Preliminaries and Notation, Three Basic Concepts. Finite Automata: Deterministic Finite Accepters, Non-Deterministic Finite Accepters, Equivalence of Deterministic and Non-Deterministic Finite Accepters, Reduction of the number of states in Finite Automata. Regular Expressions, Connection between Regular Expressions and Regular Languages, Usage of Regex in practical programming/Regex in practice, Regular Grammars. 14 Hours Unit 2: Regular Languages and Context Free Languages Properties of Regular Languages: Closure Properties of Regular Languages, Elementary Questions about Regular Languages, Identifying Non-Regular Languages. Definitions of PDA and CFL, Deterministic Pushdown Automata, Non-Deterministic Pushdown Automata, Pushdown Automata and Context Free Languages,ContextFreeGrammars. 14 Hours Unit 3: Properties of Context Free Languages and Turing Machine Parsing and Ambiguity: Parse tree, ambiguous grammars, removing ambiguity, inherently ambiguous grammars. Simplification of Context–Free Grammars and Normal Forms: Methods for Transforming Grammars, Two Important Normal Forms, A Membership algorithm for Context Free Grammar. Properties of Context-Free Languages: Closure Properties and Questions about Context–Free Languages,						

	Pumping Lemma for Context-Free Languages. Turing Machines: The Standard Turing Machine, Constructing Turing Machines, Combining Turing Machines for Complicated Tasks, Turing's Thesis. 14 Hours Unit 4: Undecidability and design of a declarative language. Hierarchy of Formal Languages and Automata: Recursive and Recursively Enumerable Languages, the Chomsky Hierarchy. Limits of Algorithmic Computation: Some Problems that cannot be solved by Turing Machines, Undecidable Problem for Recursively Enumerable Languages, idea of reduction, Rice's theorem, its proof and its applications. A very simple Logic, Syntax, Semantics, A simple knowledge Base, A simple inference procedure. Propositional Theorem Proving: Inference and Proofs, Proof by Resolution, Conjunctive Normal Form, A resolution algorithm. Syntax and Semantics of First Order Logic: Models for First Order Logic Symbols and interpretations, Terms, Atomic Sentences, Complex Sentences Quantifiers, Equality, Numbers, sets and Lists. Example - The electronic circuits' domain 14 Hours
Text Book(s):	1: "An Introduction to Formal Languages and Automata", Peter Linz, Jones and Bartlett, New Delhi, India, 6th Edition, 2016. 2: "Artificial Intelligence – A Modern Approach", Stuart Russell and Peter Norvig, Pearson, 3rd Edition (Paperback), 2016.
Reference Book(s):	1: "Theory of Computation", Michael Sipser, Cengage Learning, New Delhi, India, 2008. 2: "Introduction to Automata Theory, Languages, and Computation", John E Hopcroft, Rajeev Motwani, Jeffrey DULLman, Pearson Education, New Delhi, India, 3rd Edition, 2009. 3: "Theory of Computation: A Problem-Solving Approach", Kavi Mahesh, Wiley India, New Delhi, 2012.
Course Outcome	- Design simple machines like DFA, NFA, convert NFA to DFA and minimize a given DFA, construct regular expressions for different languages, verify that some languages are regular and some are not. - Analyze the difference between Regular Languages and Context Free Languages, design Push Down automata, construct Context Free Grammars, and convert one form of the grammar to another form. - Enumerate the properties of Context Free Grammars, verify that some languages are context free and some are not, design Turing Machines, and analyze the difference between acceptability and decidability and - Analyze the difference between Recursive and Recursively Enumerable Languages, Decidable Languages, Turing – Recognizable and Co-Turing – Recognizable, some problems that cannot be solved by Turing Machines, reduce one Undecidable Problem to another, Undecidable Problems for Recursively Enumerable Languages. And make use of Propositional Logic and Predicate Logic in knowledge representation and truth verification

Course Code	UE24CS251B	Course Title	Microprocessor & Computer Architecture				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	4	Type of Course	Core				
AI Tools /Tools/Languages	ARM Simulator, Arduino Microcontroller kit, Para Cache simulator. Compute intensive Application are run on GPU systems	Desirable Knowledge	Digital Design and Computer Organization				
Prelude	This course will give you an in-depth understanding of the inner-workings of modern digital computer systems and trade-offs present at the hardware-software interface. The course focuses on key topics in microprocessor such as the system architecture, low level programming aspects and interface with other key components. Also, the course will help in understanding the B.Tech CSE computer architecture concepts such as multilevel in memory hierarchies, pipelining, and super scalar techniques. A desirable knowledge of Digital Design and Computer organization is required.						
Course Objectives:	- Introduce concepts of basic processor architecture and its design. - Understanding the concept of concepts of pipeline architecture and hazards. - Study of memory hierarchy, cache memory and its optimizations. - Introduce advanced concepts in processor architecture like multi-B.Tech CSE/ many B.Tech CSE Processor architectures.						

Course Contents	Unit 1 : Architecture and Assembly Language Programming Introduction, ISA Classification – RISC and CISC, Memory Addressing, Operands - Types and Size, Instruction Set - Operations, Control Flow, Instruction Encoding, Case Study - ARM/ MIPS/ x86 Processor. 16 Hours Unit 2 : Pipelining 3- Stage Pipelining, 5 - Stage Pipelining, Pipeline Datapath and Control, Data Hazards – Forwarding vs. Stalling, Control Hazards, Branch Prediction Mechanisms and Exceptions, Performance Metrics. 12 Hours Unit 3 : Basics of Cache and Cache Optimization Basics of Caches - Fully Associative, Direct Mapped and Set Associativity, Cache Performance, Basic Cache Optimization- Reduce in Miss Rate. Basic Cache Optimization- Reduce Miss Penalty, Reduce Hit Time. 14 Hours Unit 4 : Advances in Architecture Introduction to Parallel Computing, Parallel Computer Memory Architecture and Flynn's Taxonomy, Parallel Computing Concepts and Terminology, Design Issues & Constraints, performance in parallel computing : Amdahl's Law ; Gustafson's Law , Multicore Processors and OpenMP programming , Realization of ML operations on Modern Architecture, Introduction to Heterogenous Parallel Computing ,Introduction to GPU Computing 14 Hours
Laboratory	• Introduction to ARM Simulator and sample programs • Implementation of Data Processing Instructions, Programs on usage of addressing modes • Programs on functions and software interrupts, Matrix Operations – Addition and Multiplication • Introduction to PARACACHE simulator – Direct Mapping, Associative Mapping • Introduction to Plugins interface. • Project work
Text Book(s):	1: "Computer Organization and Design", Patterson, Hennessey, 5th Edition, Morgan Kaufmann, 2014. 2. "Computer Organization and Design – ARM Edition", Patterson, Hennessey, 4th Edition, Morgan Kaufmann, 2010. 3. "ARM System-on-Chip Architecture", Steve Furber, 2nd Edition, Pearson India, 2015. 4. "Programming Massively Parallel Processors A Hands-on Approach", Third Edition David B. Kirk Wenmei W. Hwu, Morgan Kaufmann. 5. "Multicore and GPU Programming An Integrated Approach", Gerassimos Barlas, second edition, Morgan Kaufmann
Reference Book(s):	1: "Computer Architecture: A Quantitative Approach", Hennessey, Patterson, 5th Edition, Morgan Kaufmann, 2011. 2: "The Definitive Guide to the ARM Cortex-M0 and Cortex MO+ processors", Joseph Yiu, 2nd Edition, Newnes, 2015
Course Outcome	of the course, the student will be able to: • Demonstrate ability to understand the design of different instruction sets like RISC/ CISC And their addressing modes. • Demonstrate the ability to understand the design of a pipelined processor and its Challenges. • Demonstrate the use of tools to analyses the performance of programs on different Architectures. Design alternative memory hierarchy layouts and optimizations. • Demonstrate and appreciate modern trends in architecture such as multicore Architectures.

Course Code	UE24CS252B	Course Title	Computer Networks				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	4	Type of Course	Core				
AI Tools /Tools/Languages	Wireshark, Python. Advanced-Computer-Networking-Tutor Computer Network Professor (yeschat.ai)AI	Desirable Knowledge					
Prelude	This is a foundation course on Computer Networking which focuses on building blocks of the Internet. We trace the journey of messages sent over the Internet from an application residing on one host machine (source) to another (Destination) using a top-down layered approach. The course contents are organized based on TCP/IP Protocol stack.						
Course Objectives:	- To provide a comprehensive understanding of the fundamental principles and architectures of computer networks and the Internet. - To enable students to design, implement, and analyze network applications using socket programming and understand the role of application layer protocols. - To equip students with the knowledge to analyze and evaluate transport layer protocols, including their mechanisms for reliable data transfer and congestion control, and to introduce them to the concepts of Software Defined Networking (SDN). - To familiarize students with link layer protocols, error detection/correction techniques, and routing algorithms, enabling them to understand the operational aspects of local area networks and network routing.						
Course Contents	Unit 1- Computer Networks and the Internet, Application Layer Introduction to Computer Networks, Internet: A Nuts-and-Bolts Description, A Services Description, Protocol, The Network Edge: Access Networks, Physical Media, Introduction to physical layer devices, The Network Core, Packet Switching, Circuit Switching, A Network of Networks, Delay, Loss, and Throughput in Packet-Switched Networks, Overview of Delay in Packet Switched Networks – Queuing Delay and Packet Loss, End-to-End Delay, Throughput in Computer Networks, The OSI Model and the TCP/IP Protocol Suite, Protocol Layers, The OSI Model, TCP/IP Protocol Suite. Network Application Principles: Network Application Architectures, Processes Communication, Transport Services available to Applications, Transport Services provided by Internet. 14 Hours Unit 2 - Application Layer, Transport Layer – UDP The Web, HTTP and HTTPS, Non persistent and persistent connection, HTTP Message Format, User Server Interaction: Cookies, Web Caching. DNS, The Internet’s Directory Service: Services provided by DNS; How DNS works: DNS Records and messages; Peer to peer Applications; Socket Programming with TCP and UDP; Other Application Layer Protocols: FTP, SMTP, SNMP, Telnet, SSH. Introduction to Transport Layer Services: Relationship Between Transport and Network Layer, Overview of the Transport layer in the Internet, Multiplexing and Demultiplexing; Connectionless Transport UDP: UDP Segment Structure, UDP Checksum. Principles of Reliable Data Transfer: Building a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocol. 14 Hours Unit 3- Transport Layer – TCP, Network Layer and Internet Protocol Go-Back-N Protocol, Selective-Repeat; Connection Oriented Transport TCP: The TCP Connection, TCP Segment Structure, Flow Control, TCP Connection Management, TCP Congestion Control. Numerical on						

	TCP congestion control mechanisms–TCP Tahoe, Reno. Overview of Network Layer: Forwarding and routing, what’s Inside a Router? Introduction to SDN, Components and Architecture of SDN, The Internet Protocol (IP) IPV4: Datagram Format, Fragmentation, Addressing, NAT. 14 Hours Unit 4 - Network Layer and Internet Protocol, Link Layer and LAN Introduction to Network layer Protocols: DHCP, ICMP; IPv6 Protocol: Packet Format, Transition from IPv4 to IPv6; Introduction to Routing Algorithms: Link State: Dijkstra’s algorithm and Distance Vector: Bellman–Ford Algorithm. Link layer – Error–Detection and Correction techniques, Parity checks, Internet Checksum, Cyclic Redundancy Check, and Multiple Access Protocols: CSMA/CD, CSMA/CA; Switched LAN: Link layer addressing and ARP, Ethernet: Link–layer switches. Retrospective: A Day in the Life of a Web Page Request. Physical Layer – Purpose, Signals to Packets, Transmission media. Wireless LANs: IEEE 802.11 LAN architecture. 14 Hours
Laboratory	- Designing and Simulation of Network Topology using Cisco Packet Tracer. This experiment introduces students to Cisco Packet Tracer, enabling them to create basic network topologies, configure devices, and visualize data flow within a network. - Understanding Persistent and Non-persistent HTTP Connections. Students use Wireshark to analyze HTTP traffic and understand the differences between persistent and non-persistent connections, focusing on real-world scenarios like web page loading. - Exploring UDP with DNS and Sockets using Wireshark. This week is divided into two parts: analyzing DNS queries and responses to understand UDP traffic and implementing UDP socket communication for practical understanding. - Congestion Window Plotting using NS2. Students simulate TCP congestion control using NS2 and plot the congestion window, learning how TCP adapts to network conditions. - Understanding Hardware Components in Networking. Students explore physical networking components, such as routers, switches, and cables, gaining hands-on experience with hardware setup and troubleshooting.
Text Book(s):	1: “Computer Networking: A Top – Down Approach”, James F. Kurose, Keith W. Ross, 7 th Edition, Pearson Publication, 2017.
Reference Book(s):	1. “TCP IP Protocol Suite”, Behrouz Forouzan, 4 th Edition, McGraw–Hill, 2010.
Course Outcome	- Analyze and apply fundamental networking concepts, including protocol layers, network architectures, and addressing schemes (IPv4, IPv6). - Evaluate and implement key application layer protocols (HTTP, DNS, SMTP) and understand their role in network communication. Apply socket programming concepts with TCP and UDP to develop network applications. - Design and analyze transport layer protocols (UDP, TCP), including reliable data transfer, flow control, and congestion control mechanisms, and evaluate their impact on network performance. Additionally, gain an introductory understanding of Software Defined Networking(SDN) concepts. - Explain and apply link layer protocols (Ethernet, Wi-Fi), error detection/correction techniques, and routing algorithms (Dijkstra, Bellman-Ford) for efficient network operation

Course Code	UE24CS241B	Course Title	Design and Analysis of Algorithm				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	4	Type of Course	Core				
AI Tools /Tools/Languages	C-Programming language Visu Algo (Interactive Visualizations), Algorithm Visualizer (AI Explanations).	Desirable Knowledge	Problem solving with C, Data Structures and its Applications				
Prelude	Algorithms play a key role in science and practice of computing. Learning algorithm design technique is a valuable endeavor from practical standpoint and algorithm design techniques have considerable utility as general problem solving strategies, applicable to problems beyond computing. This course includes classic problems of computer science, application of design techniques and analysis in terms of time and space.						
Course Objectives:	- Learn various algorithm design techniques and apply appropriate algorithmic design techniques for specific problems. - Learn to design and analyze algorithms with an emphasis on resource utilization in terms of time and space. - Learn to trade space for time in algorithmic design using input enhancement and per-structuring. - Learn the limitations of algorithmic power and techniques to handle these limitations						
Course Contents	Unit1: Introduction and Brute Force Algorithms: Fundamentals of Algorithmic Problem Solving, Important Problem Types. Analysis of Algorithm Efficiency: Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, Mathematical Analysis of Non-Recursive and Recursive Algorithms. Brute Force: Selection Sort, Bubble Sort, Sequential Search, Brute Force String Matching, Exhaustive Search. 14 Hours Unit2: Decrease – and – Conquer & Divide-and-Conquer Decrease-and-Conquer: Decrease by constant number algorithms - Insertion Sort, Topological Sorting, Algorithms for Generating Combinatorial Objects, Decrease-by-a-Constant-Factor Algorithms – Fake Coin Problem, Russian Peasant Multiplication, Josephus problem, Decrease-by-Variable-Size Algorithms – Computing a median and the selection problem. Divide-and-Conquer: Master Theorem, Merge Sort, Quick Sort, Binary Search, Binary Tree Traversals, Complexity analysis for finding the height of BST, Multiplication of Large Integers, Strassen’s Matrix Multiplication. 14 Hours Unit3: Transform-and-Conquer Space and Time Tradeoffs& Greedy Technique Transform and Conquer: Pre-sorting, Heap Sort, Red-Black Tree Construction and Time complexity Analysis for insert and search operation, 2-3 Trees and B Tree: insertion, deletion, searching, and time complexity analysis. Space and Time Tradeoffs: Sorting by Counting, Input Enhancement in String Matching - Horspool’s and Boyer-Moore Algorithms. Greedy Technique: Prim’s Algorithm, Kruskal’s Algorithm and union-find algorithm, Dijkstra’s Algorithm, Huffman Trees 14 Hours Unit4: Limitations, Coping with the Limitations of Algorithm Power & Dynamic Programming Dynamic Programming: Computing a Binomial Coefficient, The Knapsack Problem and Memory Functions, Warshall’s and Floyd’s Algorithms. Limitations of Algorithm Power: Lower-Bound Arguments, Decision Trees, P, NP, and NPComplete, NP-Hard Problems. Coping with the Limitations of Algorithm Power: Backtracking						

	14 Hours
Text Book(s):	1: "Introduction to the Design and Analysis of Algorithms", Anany Levitin, Pearson Education, Delhi (Indian Version), 3rd Edition, 2012
Reference Book(s):	1: "Introduction to Algorithms", Thomas H Cormen, Charles E Leiserson, Ronald L Rivest and Clifford Stein, PrenticeHall India, 3rd Edition, 2009. 2: "Fundamentals of Computer Algorithms", Horowitz, Sahni, Rajasekaran, Universities Press, 2nd Edition, 2007. 3: "Algorithm Design", Jon Kleinberg, Eva Tardos, Pearson Education, 1st Edition, 2006.
Course Outcome	• Identify the design technique used in an algorithm • Design and implement efficient algorithms for practical and unseen problems and analyze these algorithms using quantitative evaluation. • Understand Time and Space Tradeoff's • Understand, and cope up with the limitations of algorithm power.

Course Code	UE24CS242B	Course Title	Operating System				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	4	Type of Course	Core				
AI Tools /Tools/Languages	C, Linux/Unix OS for system call implementation.	Desirable Knowledge	Data Structures and its Applications				
Prelude	This course focuses on fundamental operating systems concepts including various algorithms and trade-offs for efficient management of resources such as CPU, Memory, Storage and I/ O. This course requires the student to have a desirable knowledge of Data Structures and its Applications.						
Course Objectives:	• Focus on fundamental Operating System concepts. • Provide an understanding of various components of the Operating System (OS). • Delve deeper into various algorithms and associated trade-offs for efficient resource management such as process, disk, and memory management. • Introduce design principles and trade-offs in the design of Operating Systems.						
Course Contents	Unit 1 : Introduction and Process Management What Operating Systems Do, Operating-System Structure & Operations, Kernel Data Structures, Operating-System Services, Operating System Design and Implementation. Shell programming: Overview of bash shell programming – variables, control flow Processes: process concept, Process Scheduling, Operations on Processes, System calls for process management-fork (), vfork (), wait () and exec (), Life cycle of I/O request. CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms. Case Study: Linux Scheduling Policies.Shell programming –cron 14 Hours Unit 2 : IPC, Threads and Concurrency – 14 hours IPC – Introduction, Shared Memory systems, Message Passing, Communication in Client–Server Systems- Pipes, ordinary pipes and named pipes, system calls for shared memory, pipes and fifo’s. Threads: Overview, Multicore Programming, Multithreading Models, Thread Libraries, Thread Scheduling. Process Synchronization: Background, The Critical-Section Problem, Peterson’s Solution, Synchronization Hardware, Mutex Locks, Semaphores, Classic Problems of Synchronization- The Bounded-Buffer Problem, The Readers–Writers Problem, The Dining-Philosophers Problem, Synchronization Examples- Synchronization in Linux. System calls for threads creation and synchronization-POSIX Threads. Deadlocks: System Model, Deadlock Characterization, Deadlock avoidance, Banker’s Algorithm, Deadlock Detection. 14 Hours Unit 3: Memory Management – 14 hours Main Memory: Background- Basic Hardware, Address Binding, Logical Versus Physical Address Space,						

	Dynamic Loading, Dynamic Linking and Shared Libraries, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table. Virtual Memory: Background, Demand Paging, Copy-on-Write, Page Replacement Algorithms-FIFO, LRU, Optimal, Allocation of Frames, Thrashing. 14 Hours Unit 4 : File and Storage Management – 14 hours File-System Interface: File Concept, system calls for file operations-open(), read(),write(), lseek(), close() and system call to retrieve file attributes and file types-stat(), lstat(), Access Methods, Directory and Disk Structure, system calls for reading directories, system calls to create hard links (link()) and symbolic links-symlink().File-System Implementation: File-System Structure, File-System Implementation, Directory Implementation, Allocation Methods, File Sharing, Protection. Storage management: Overview of Mass-Storage Structure, Disk Scheduling, Swap-Space Management, RAID Structure.System Protection: Goals, Principles and Domain of Protection, Access Matrix, Implementation of the Access Matrix, Access Control Shell programming - awk, sed 14 Hours
Text Book(s):	1. “Operating System Concepts”, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne 10th Edition, John Wiley & Sons, Global Edition, 2023. 2. “Advanced Programming in the Unix Environment”, Richard Stevens and Stephen A Rago, Pearson, 3rd edition, 2017.
Reference Book(s):	1:“Operating Systems, Internals and Design Principles”, William Stallings, 9th Edition, Pearson, 2018. 2: “Modern Operating Systems”, Andrew S Tanenbaum, 3rd edition, Pearson, 2007. 3: “Learning the bash shell”, Cameron Newham, 3rd edition, O’Reilly, 2005
Course Outcome	• Understand the principles and modules of Operating Systems. • Understand the design of various algorithms for scheduling and their relative performance. • Understand the concept of Deadlocks that typically occur in OS. Deadlocks - Avoidance and Detection. • Implement Operating Systems Concepts related to process management, Concurrent processes, Threads and Memory Management.

Course Code	UE24MA241B	Course Title	Linear Algebra and its Applications				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	4	Type of Course	Core				
AI Tools /Tools/Languages	-	Desirable Knowledge	-				
Prelude	This is a basic subject on matrix theory and linear algebra. Emphasis is given to topics that will be useful in Engineering discipline, including system of equations, vector spaces, eigenvalues, similarity, and positive definite matrices. The course provides hands-on experience in basic programming concepts using PYTHON/MATLAB for solving problems relevant to these areas.						
Course Objectives:	- To teach students how to use linear algebra as a powerful tool for computation. - To show how these computations can be conceptualized in a geometric framework. - To give a gentle introduction to the theory of abstract vector spaces. - To visualize solution to linear system of equations with different approaches using Python/Matlab.						
Course Contents	Unit 1: Matrices, Gaussian Elimination Introduction, The Geometry of Linear Equations, Gaussian Elimination, Singular Cases, Elimination Matrices, Triangular Factors -LU decomposition and Row Exchanges, Inverses and Transposes, Inverse by Gauss -Jordan method, Symmetric Matrices. Self-Learning Component: Algebra of Matrices. Unit 1: Basic operations with matrices in Matlab/Python. Matrix operations and image manipulation Matrix multiplication, inversion, and photo filters Solving linear systems 14 Hours Unit 2: Vector Spaces & Linear Transformations Vector Spaces and Subspaces (definitions only), Linear Independence, Basis and Dimensions, Row reduced Echelon form, The Four Fundamental Subspaces, Rank-Nullity theorem. Linear Transformations, Algebra of Linear transformations, Self-Learning Component: Examples of Vector Spaces and Subspaces. Application Unit 2: Systems of linear equations and college football team ranking (with an example of the Big 12) Convolution, inner product, and image processing revisited Norms, angles, and your movie choices Interpolation, extrapolation, and climate change 14 Hours Unit 3: Orthogonalization, Eigenvalues and Eigen vectors Orthogonal Vectors and Subspaces, Orthogonal Bases, Cosines and Projections onto Lines, Projections and Least Squares. Orthogonalization, The Gram-Schmidt Orthogonalization process, Introduction to Eigenvalues and Eigenvectors, Properties of Eigenvalues and Eigenvectors, Cayley-Hamilton theorem (statement only), Diagonalization of a Matrix. Application Unit 3: Orthogonal matrices and 3D graphics Discrete dynamical systems, linear transformations of the plane, and the Chaos Game Projections, eigenvectors Matrix eigenvalues and the Google's PageRank algorithm 14 Hours Unit 4: Singular Value Decomposition Quadratic Forms, Definitions of Positive definite, negative definite, positive semi-definite, negative						

	semi-definite, indefinite forms and matrices, Tests for Positive Definiteness, Cholesky's method, Singular Values and Singular Vectors, Image Processing by Linear Algebra, Principal Component Analysis (PCA by the SVD and Covariance matrix), Minimum Principles. Application Unit 4: Principal Component Analysis, and face recognition algorithms Social networks, clustering, and eigenvalue problems Singular Value Decomposition and image compression 14 Hours Hands on learning using PYTHON and Applications to Computer Science Engineers.
Text Book(s):	1: Linear Algebra and its Applications, Gilbert Strang, Thomson Brooks/ Cole, 5th Edition, Second Indian Reprint 2007. 2: Application : Applied projects for an introductory linear algebra class by Anna Zemlyanova.
Reference Book(s):	1: Linear Algebra and its Applications, David. C lay, Publication by Pearson Education, 5th Edition, 2015 2: Linear Algebra, Schaum's outlines, Seymour Lipschutz and Marc Lipson, Tata McGraw- Hill publications, 4th Edition, 2009. 3: Higher Engineering Mathematics, B S Grewal, Khanna Publishers, 44th Edition, 2020,. 4: Practical Linear Algebra, Gerald Farin and Dianne Hansford, CRC Press, Taylor Francis Group, 3rd Edition.
Course Outcome	After completing this course, students will be able to: • Solve systems of Linear Equations using Matrix Transformations, Interpret the nature of Solutions, Visualize Consistency of Linear system of Equations and also compute inverse of a Matrix. • Demonstrate the ability to work within Vector Spaces, distil Vector Space properties and understand the concepts of the four fundamental Subspaces, Linear Span, Linear Independence, Dimension and Basis • Learn the concepts of Orthogonal Vectors and Orthogonal Subspaces and apply the Gram-Schmidt process to find an Orthonormal Basis in a Subspace, Eigenvalues, Eigenvectors and Diagonalization of a Matrix. • Apply the concept of Positive Definite Matrices, Singular Value Decomposition into application problems.

Course Code	UE23CS351A	Course Title	Database Management System				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	5	Type of Course	Core				
AI Tools /Tools/Languages	MySQL Workbench, Python, ERwin, Any other tool for ER modeling, RelaX - relational algebra calculator. Bytebase SQL Editor, Chat2DB	Desirable Knowledge	-				
Prelude	This course offers a solid theoretical foundation in Database Management System (DBMS) and explores the practical applications of DBMS in a real-world scenario. The course emphasizes on the creation and the design of relational database systems. It also introduces the databases that provides flexible schemas and scale easily with large amounts of data and high user loads.						
Course Objectives:	- Understand fundamental concepts, terminology, and application of relational databases and Construct ER diagrams for a desired application and transform the same into a relationalschema. - Understand database design concepts and design relational databases and construct basic and advanced SQL queries. - Understand, and apply Normal Forms in database design, Transactions, Concurrency control, Locking, and Recovery, implement them in a real-time setup, and transform the given Normal Form into the desired form. - Understand the concepts of Database Security, and NoSQL databases such as MongoDB, Neo4j, and DynamoDB						
Course Contents	Unit 1: Introduction to Database Management Introduction to databases, Database application architecture, Users and Administrators, E-R Model, reducing ER to a relational schema. Structure of relational databases, Database schema, constraints, Keys, Enhanced ER Model Subclasses, Superclasses, Inheritance, Specialization and Generalization and its Constraints and Characteristics. Relational Model: Relational operations (Algebra), Unary Operations - Unity, Binary, Aggregate Functions, Grouping. 14 Hours Unit 2: Database Design: Basic and Advanced SQL SQL overview, Data definition, Structure of SQL queries, Additional Basic Operations, Set Operations, Null Values, Aggregate Functions, Nested Subqueries, Database Modification, Join expressions, Views, Triggers, Functions, and Procedures, Event scheduler - Automated tasks, Replication – Master-slave, Multi-source, CTEs, Window functions, Full-Text Search Functions, Introduction to strategies of Query processing and Query optimization. 14 Hours Unit 3: Advanced Design Concepts and Implementation Functional Dependencies, Inference Rules, Closure, Equivalence, Minimal Cover Normal Forms Based on Primary Keys (1NF, 2NF, and 3NF), General Definitions of Second and Third Normal Forms Boyce-Codd Normal Form, Proper ties of Relational Decompositions, Overview of Higher Normal Forms. 14 Hours Unit 4: Advanced Databases Database transactions, Concurrency control, Locking, Recovery, Database Security, Security Threats (SQL Injection), Data protection at rest and in transit, Introduction to NoSQL databases, CAP theorem, Document database (MongoDB), Key-Value database (DynamoDB), Graph databases (Neo4j), Introduction						

	to Multi-model Databases.	14 Hours
Laboratory	1. Draw an ER diagram for a given problem statement, Conversion of an ER diagram into a Relational schema. 2. Execution of Relational algebraic expressions using Relax tool. 3. DDL – create constraints, alter, rename, drop, truncate table, Views. 4. DML – Insert, Update, Delete, Transactions - commit, rollback, savepoint. 5. SQL - Set operators: union, intersect, minus. SQL – Aggregate functions. 6. SQL – Joins: inner, outer; Sub queries: correlated and uncorrelated. 7. SQL – Full-Text Search Functions. 8. SQL - Event Scheduler-Automated tasks, Replication – master-slave, multi-source, CTEs, Window functions. 9. SQL – Creating Functions and Procedures. 10. SQL – Creating Triggers and Cursors. 11. XML- Database access. 12. NoSQL database queries. 13. High-level programming language accessing a database using an API.	
Text Book(s):	1: “Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017. 2: “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019.	
Reference Book(s):	1: Database Management Systems, R Ramakrishnan, J Gehrke, 3rd Edition, McGraw Hill, 2002. 2: Data on the Web: From Relations to Semistructured Data and XML, S Abiteboul, P Buneman, D Suciu, Morgan Kauffman, 1999.	
Course Outcome	• Demonstrate an ability to explain the basic concepts of database management, construct and transform ER diagrams into Relational Schema. • Design databases and construct simple and advanced SQL queries for given contexts. • Explain Normal Forms, employ them in Database Design, and apply database security concepts in application contexts. • Demonstrate the ability to use semi-structured and NoSQL databases.	

Course Code	UE23CS352A	Course Title	Machine Learning				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	5	Type of Course	Core				
AI Tools /Tools/Languages	Pytorch, SkLearn, Keras, Tensor flowGoogle cloud Vertex AI(end-to-end ML platform), Amazon Sagemaker, IBM Watson studio, Auto ML tools (H2O.ai (H2O AutoML), Google AutoML, Hugging Face, Haystack etc	Desirable Knowledge	Mathematics for Computer Science Engineers Linear Algebra & its Applications, Design and Analysis of Algorithms.				
Prelude	Machine Learning surrounds us today: in phones that respond to voice commands, programs that beat humans at Chess and Go, robots that assist surgeries, vehicles that drive in urban traffic, and systems that recommend products to customers on e-commerce platforms. This course aims to familiarize students with the breadth of modern ML, to impart an understanding of the dramatic surge of ML in the last decade, and to foster an appreciation for the distinctive role that ML can play in shaping the future of our society. This course requires the student to have a desirable knowledge of Statistics for Data Science, Linear Algebra and its Applications and Design and Analysis of Algorithms.						
Course Objectives:	- Formulate a well-defined Machine Learning problem with clear metrics. - Understand the notions of Hypotheses Space, Hypotheses Structure and Search. - Become conversant with types of Machine Learning Algorithms, their applicability and Inductive Bias. - Familiarize with techniques for Ensemble Learning and deep learning.						
Course Contents	Unit1: Foundations of Machine Learning and Supervised learning Models: Introduction to Machine Learning- Machine Learning Models-Supervised, Unsupervised, Semi-supervised, Reinforcement Learning. Concept Learning, Hypotheses and Hypothesis Space, Version space, Inductive bias. Decision Tree Learning- Decision Trees, ID3 and dealing with continuous values for classification in DTs, Decision boundary for decision trees(x-y axis), Bias Variance decomposition. Issues in Decision Tree Learning – Overfitting, Solutions to overfitting. Other Core Supervised Learning Approaches- Instance-based learning: k-nearest neighbor learning (Classification & Regression), Decision boundary for KNN.- 14 Hours Unit2: Neural Networks, Ensemble Learning, and Kernel-Based Methods for Classification: Artificial Neural Networks- Introduction, Perceptrons, Multi-layer Networks and Backpropagation, Activation Functions (Step, Sigmoid, Tanh, ReLU), Various Optimizers (Gradient Descent, Stochastic Gradient Descent, Momentum-based, Adagrad, Adam), Ensemble Models- Combining Weak Learners, Improving Performance with Gradient Boost. Support Vector Machines – Margin and Maximization, SVM – The Primal Problem, The Lagrangian Dual, SVM – Solution to the Lagrangian Dual (Hard Margin), 14 Hours Unit3: Stochastic Models and Unsupervised learning Bayesian Learning- Bayes’ Theorem, Maximum Likelihood Estimation (MLE), Bayes Optimal Classifier, Unsupervised Learning-Hierarchical vs. Non-Hierarchical Clustering, Agglomerative and Divisive Clustering, Bisecting K-Means, and Expectation Maximization (EM), K-Means as a Special Case of EM, Gaussian Mixture Models (GMMs), Hidden Markov Models (only Forward Algorithm derivation. Backward, Viterbi and Baum Welsch Algorithms applications only). 14 Hours						

	Unit4 : Introduction to Deep Learning Architectures from CNNs to Transformers Convolution Neural Networks-Introduction to Convolution Operation, Convolutional Neural Network (CNN) Architecture, Parameter Calculation in CNNs, Max Pooling, Average Pooling, Recurrent Neural Networks- Introduction to Recurrent Neural Networks (RNN), Vanishing and Exploding Gradient Problems, Variants of RNN – LSTM and GRU (mention of gates), Advanced Deep Learning Architectures- Basic GAN and Its Architecture, Basic Architecture of Transformers (Encoder-Decoder Perspective with Basics of Attention Mechanism). 14 Hours
Lab	1. Introduction to SciKit, Performance Metrics 2. Decision Trees. 3. ANN 4. Random Forest. - Concept & Implementation 5. SVM (Kernel Functions/soft margin SVM-Concept & implementation) 6. Naïve Bayes (concept and Implementation) 7. HMM 8. K means Clustering (Concept and Implementation) 9. CNN model.
Text Book(s):	1: "Introduction to Machine Learning", Ethem Alpaydin, Third edition, MIT Press Ltd, 2015 2: "Machine Learning: The Art and Science of Algorithms that Make Sense of Data", Peter Flach, Cambridge University Press (2012). 3: Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville (MIT Press, 1st Ed., 2016)
Reference Book(s):	1: "Machine Learning", Tom Mitchell, McGraw Hill Education (India), 2013. 2: Pattern Recognition and Machine Learning by Christopher M. Bishop (Springer, 1st Ed., 2006) 4: Machine Learning: A Probabilistic Perspective by Kevin P. Murphy (MIT Press, 1st Ed., 2012)
Course Outcome	• Distinguish categories of Data Attributes, Dimensions, and Sample Sizes. • Acquire a thorough understanding of Supervised, Unsupervised Learning, • Apply Ensemble Methods and deep learning methods for solving the problems. • Design and implement a solution to real world machine learning problem.

Course Code	UE23CS341A	Course Title	Software Engineering				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Core				
AI Tools /Tools/Languages	Git, GitHub, docker, Jenkins, gtest, pytest, Jira/OpenProject, SonarQube.	Desirable Knowledge	-				
Prelude	In this course, students learn and gain practical experience with software engineering principles, techniques, and tools. Topics in this course include requirements analysis, specification, architecture, design, programming style, testing, quality management, security validation, devOps, teamwork, and software project management. The practical experience centers on a semester-long team project, in which a software development project is carried through all the stages of the software life cycle. Particular emphasis is placed on performing all SDLC activities similar to industry setup. Assignments further provide a hands-on experience in using state-of-the-art techniques and tools.						
Course Objectives:	• Learn Software Engineering concepts. Ensure the relevance and need of an engineering approach to software development. Expose students to the software project management and Life Cycles. • Expose students to agile development methodology using standard tools and associated specifications • Enable the students to understand the principles of Software Configuration Management and importance of software quality for Software Product Development. • Expose students to the continuous integration and continuous development, build, test and release of software products through a semester long project.						
Course Contents	Unit 1: Introduction to Software Engineering and Requirements Engineering Introduction, Case studies and motivation for SE, DevOps and CI/CD tools introduction and setup, Software development lifecycles overview (waterfall etc) and motivation for agile methodology. Agile approach of software development, Contrasting Agile and Plan driven approaches, Agile SCRUM model and exposure to other Agile approaches like Lean Agile, XP etc. Hands on Jira training. Security development lifecycle. Requirements Engineering (user Stories for AGILE) : Requirement’s introduction and properties, Feasibility, Requirements Elicitation, Analysis and modelling, Specification and Verification, Requirement Management and Requirements Traceability; Introduction to UML and development of use cases. Validation planning and security validation plan. 16 Hours Unit 2: Software Project Management and Software Architecture, Design and Quality Fundamentals, Software Project Management Lifecycle, Planning activities, Choice of Lifecycles, Project Organization, Software Estimation for AGILE, Scheduling, Risk Management, Quality Management, Gantt charts basics, Monitoring of Execution and Control, Project Closure. RACI matrix with example Software Architecture: Case study for motivation of SW Arch, Architectural View, Styles, Architectural and Design Patterns. Modularity, interface definition, microservices intro, Security architecture, Agile context. Software Design: Modularity, importance of interfaces (API and ABI), Error definition, handling and management, Forward and backward compatibility, usage of UML.						

	Quality and Test case development: Case studies and motivation, types of testing, technical debt, Test cases documentation, Black and white box testing. 12 Hours Unit 3: AI, Implementation, SCM, DevOps AI for SW Engineering and SE for AI/ML systems. CI/CD pipeline and tools for development, Introduction to software construction, Coding principles, standards and guidelines, Factors for effective coding -Defensive, Secure and Testable programming, Code Review / Peer Review, Security code reviews; logging, tracing, debugging techniques. Test case development and dynamic analysis, functional and non functional testing, security testing, Automation; mutation testing to validate test suite Software Configuration Management (SCM): Elements of a Configuration Management System, Configuration Items, Baselines, Repositories, Branch Management, Build Management, Install, Change and Release Management, Patching and Patch Management, 16 Hours Unit 4: Usability, Quality and System Testing I Case study discussion, Industry talk, Usability, UX design, Code coverage, branch coverage; Ethics and Software Engineering; Opensource Introduction; SW maintenance and support; Security and Privacy aspects; Measurements and Matrix for Quality; Risk management for Quality Security Validation – Penetration Testing, Fuzzing; System testing – Regression testing, Integration testing Demo and student presentations for the mini-project 12 Hours
Text Book(s):	1. "Software Engineering", International Computer Science Series, Ian Somerville, Pearson Education, 10th Edition, 2021. 2. "Software Engineering: A Practitioner's Approach", Roger S Pressman, McGraw Hill, 9th Edition, 2019.
Reference Book(s):	1. "Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling" by By Jennifer Davis, Ryn Daniels, O' Reilly Publications, 2018 2. "Software Engineering: Principles and Practice", Hans van Vliet, WileyIndia, 3rd Edition, 2010. 3. "Foundations of Software Testing ", Aditya Mathur, Pearson, 2008. 4. "Software Testing, A Craftsman's Approach ", Paul C. Jorgensen, Auerbach, 2008. 6:IEEE SWEBOK, PMBOK, BABOK and Other Sources from Internet 5. "Software Testing – Principles and Practices", Srinivasan Desikan and Gopalaswamy Ramesh, Pearson, 2006.
Course Outcome	• Relate to the challenges of Software Development and Software Engineering as a methodical approach for development. Use state-of-the-art tools and techniques for large-scale software systems development. • Use Software project management ideas and apply suitable architecture and design approaches for software development • Implement the major software development methods in practical projects including CI/CD, various devOps tools etc • Follow industry accepted documentation and review processes to develop a high quality SW project.

Course Code	UE23CS342AA1	Course Title	Advanced Algorithms				
Program	Elective	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1 : SCC				
AI Tools /Tools/Languages	VisuAlgo, AlgoExpert, Wolfram Alpha C or C++ Programming Language	Desirable Knowledge	Design and Analysis of Algorithms.				
Prelude	Algorithm Design and Analysis is a fundamental and important part of computer science. The course on Advanced Algorithms introduces the learner to advanced techniques for design and analysis of algorithms and explores a variety of applications. This course requires the student to have a desirable knowledge of Design and Analysis of Algorithm.						
Course Objectives:	• Enable the learner with basics of Amortized Complexity Analysis of Data Structures. • Empower the learner with String Matching/ Prediction Algorithms and hone the problem-solving skills of the learner by introducing them to Flow Networks, Bipartite Matching and DFT. • Introduce the learner to Number Theoretic Algorithms. • Enable the learner with design strategy of Dynamic Programming, Randomized Algorithms and Approximation Algorithms.						
Course Contents	Unit 1: Basics of Complexity Asymptotic Notations, Standard notations and common functions, Recurrences and Solution of Recurrence equations- The Substitution method, the Recurrence tree method, the Master method, Amortized Complexity Analysis: Aggregate, Accounting and Potential Methods.NP-Completeness, NP Reduction. 14 Hours Unit 2: String Matching Algorithms, Suffix Trees and Maximum Flow, Naive String Match, Rabin_Karp, String matching with Finite State Automata, and Knuth_Morris_Pratt Algorithms, Suffix Arrays, Suffix Trees - Applications of Suffix Trees, Regular Expression Searches Using Suffix Trees, Flow Networks, The Ford-Fulkerson method, The Edmonds-Karp algorithm, Maximum Bi-Partite Matching. 14 Hours Unit 3: Polynomials and FFT &Number-Theoretic Algorithms Polynomials and FFT: Representation of Polynomials, Efficient Polynomial Multiplication, DFT and FFT, Efficient Implementation of FFT. Number-Theoretic Algorithms: Elementary notions; GCD, Modular Arithmetic, Solving modular linear equations, Modular Inverse, The Chinese remainder theorem, Powers of an element; RSA cryptosystem; Primality testing; Integer factorization. 14 Hours Unit 4: Dynamic Programming Randomized Algorithms and Approximation Algorithms Elements of Dynamic Programming, Problems - Coin- Row, Rod-Cutting, Matrix-Chain Multiplication, Longest Common Subsequence. Randomized Algorithms: Hiring Problem, Indicator random variables, Approximation Algorithm: Vertex Cover Problem, TSP, The Subset Sum Problem. 14 Hours						

Text Book(s):	1. "Introduction to Algorithms", T H Cormen, C E Leiserson, R L Rivest and C Stein, MIT Press, 4th Edition, 2022.
Reference Book(s):	1. The Algorithm Manual", Steven Skiena, Springer. ISBN: 9788184898651, 2nd Edition, Springer, 2008 2. "Randomized Algorithms", R Motwani and P Raghavan, Cambridge University Press, 2011
Course Outcome	• Perform Amortized Analysis of complex Data Structures. • Apply String Matching Algorithms to solve string related problems and implement Max Flow and FFT Algorithms. • Apply Number Theoretic concepts in designing Cryptographic Algorithms. • Solve complex problems using Dynamic Programming, Randomized Algorithms and Approximate Algorithms.

Course Code	UE23CS342AA2	Course Title	Data Analytics				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	R and Python, MS Excel tools	Desirable Knowledge	Mathematics for Computer Science Engineers				
Prelude	The course explores the data analytics lifecycle: question formulation, data collection and cleaning, exploratory, analysis, visualization, statistical inference, prediction, and decision-making. Focuses on building analytical models using key principles and techniques. This course requires the student to have a desirable knowledge of Statistics for Data Science.						
Course Objectives:	- Assess a dataset for the type of data, inter-relationships between features and interpret the data meaningfully - Understand the assumptions that underlie regression models and determine the appropriate model and parameters for a problem - Recognize patterns such as trend and seasonality in a time series data and design appropriate forecasting models. - Understand the nuances of recommender systems, build and validate appropriate models for recommendations.						
Course Contents	Unit 1: Exploratory data analysis, ANOVA and correlation analysis Introduction to business analytics, review of descriptive analysis and dimensionality reduction, ANOVA, correlation analysis, hidden variables. - 14 hours Unit 2: Regression Analysis Simple linear regression (alternative computational approaches and an emphasis on model diagnostics and validation), multiple linear regression (alternative computational approaches and an emphasis on model interpretation, diagnostics and validation) and variations (such as ridge regression and lasso regression), an introduction to multivariate and nonlinear regression. 14 Hours. Unit 3: Time Series Analysis and Markov Chains Time series data components, moving average and exponential smoothing methods, regression models for forecasting, AR model identification using ACF/ PACF, concept of stationarity – its importance, testing for stationarity and converting a non-stationary signal to stationary, ARMA and ARIMA modelling, Discrete Markov chains, Markov chains with Absorbing States, expected duration to reach a state from other states, computing the retention probability and customer lifetime value, confounding variables. 14 Hours Unit 4: Recommender Systems Collaborative filtering, knowledge-based filtering: constraint-based and case-based (with a mention of how traditional supervised learning approaches and unsupervised learning such as, DBSCAN, can help arrive at recommendations), a brief introduction to text classification/ clustering for content-based filtering, interpreting ML models (LIME and SHAP), mining of association rules and evaluation of recommender systems, A/B testing. 14 Hours						

Text Book(s):	1; Business Analytics, The Science of Data-Driven Decision Making”, U. Dinesh Kumar, Wiley, second edition, 2022 2: “Recommender Systems: The Textbook”, Charu C. Agarwal, Springer 2016.
Reference Book(s):	1: Data Mining: Concepts and Techniques by Jiawei Han, Micheline Kamber and Jianei, The Morgan Kaufmann Series in Data Management Systems, Elsevier publications, 3rd Edition, 2012. 2: The Elements of Statistical Learning, Trevor Friedman, Robert Tibshirani and Jerome Hastie, Data Mining, Inference and Prediction, Springer 2001. 3: Practical Data Science with R, Nina Zumel and John Mount, Manning Publications, 2014.
Course Outcome	• Perform exploratory data analysis on a given set of data and identify/ interpret correlation between features. • Narrow down a subset of regression techniques to model data and evaluate the efficacy of these models. • Analyse a time series and build appropriate time series models for prediction and evaluate them. • Analyse data to infer underlying patterns, select the appropriate techniques and formulate recommendations.

Course Code	UE23CS342AA3	Course Title	Internet of Things				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	Python, Embedded- C ,Cloud Platforms, Single Board Computers	Desirable Knowledge	-				
Prelude	The Internet of Things is already changing the way people live and interact with humans and machines. Businesses in every vertical have already started to leverage the power of IoT platforms to increase their efficiency and performance. This course introduces the students to the three layer architecture of the IoT exploring all the major connectivity options and application layer protocols. It also encourages the students to develop interesting applications using different development boards, sensors and actuators. The course introduces the students to cloud platforms and analytics including security aspects of IoT Application development.						
Course Objectives:	• Learn the fundamentals of the Internet of Things. • Learn the concepts of smart objects & IoT architecture. • Compare different application and network layer protocols for Internet of Things. • Appreciate the role and importance of Data Analytics and Security in real word problems and know the role of IoT in various verticals.						
Course Contents	UNIT I:Introduction to IoT & Architecture: Introduction IoT Traffic Model, IoT Connectivity, IoT Verticals, Use Cases & Applications, IoT Value Chain. Examples of IoT Use Cases &Applications, IoT Project Implementation, IoT Standards. IoT Architecture: Introduction, Factors Affecting an IoT Architectural Model, IoT Architectural Model, IoT WF Model, Data Centre & Cloud, Computing (Cloud, Fog & Edge). 14 hours UNIT 2 IoT Sensors: Introduction, Sensors & Its Performance Metrics, Sensor Selection, Smart Sensors, MEMS, Sensor Fusion, Self-Calibration, Sensors of the Future. Introduction to Embedded Programming, Designing IoT applications using the STM32 development board. 14 hours UNIT 3:IoT-Protocols: IoT Wired Connectivity: Introduction, Ethernet, Ethernet TSN. Unlicensed-band Wireless Connectivity: Introduction, Zigbee, BLE, WiFi, LoRaWAN. Cellular IoT Technologies: Introduction, GSM-IoT, LTE, Practical Use Cases. Application Protocols for IoT: The Transport Layer-IoT Application Transport Methods, SCADA, Generic Web-Based Protocols, IoT Application Layer Protocols-CoAP and MQTT. 14 hours UNIT 4: IoT – Analytics, Security & Privacy: Introduction, Data Pipeline, AI, Machine Learning, Supervised Learning Technique: Classification: Decision Tree Example. IoT Cloud-Based Services & Platforms: Introduction to thinger.io and ThingSpeak for IoT. IoT Security - Introduction, IoT Threats, IoT Vulnerabilities, IoT Threat Modelling & Risk, IoT Security Regulations, IoT Privacy Concerns & Regulations Security & Privacy Examples, IoT & Blockchain 14 hours						

Text Book(s):	1; Dian, F. J, “Fundamentals of Internet of Things”, Wiley Professional Development (P&T), 2022. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, "IoT 2: Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things”, 1st Edition, Pearson Education (ISBN: 978-9386873743)
Reference Book(s):	1: “Building Enterprise IoT Applications”, Chandrasekar Vuppalapati, CRC Press, Taylor & Francis Group, 2020 2: Arshdeep Bahga, Vijay Madiseti, “Internet of Things – A hands-on approach”, UniversitiesPress, 2015 3: Embedded Systems And Internet Of Things by Dr. Sonu Kumar, Dr. M. Venkatanarayana, ISBN:978-93-5747-591-4 4: "Embedded Software Development for the Internet Of Things: The Basics, the Technologies and Best Practices" by Klaus Elk
Course Outcome	• Recognize the principles of smart objects and the potential for IoT and comprehend IoT architecture from sensors to the cloud, including edge gateways. • Determine the static and dynamic performance metrics for core sensors and to understand the architecture of smart sensors for IoT system design. • Learn how to use various wired and wireless technologies to provide connectivity for IoT applications and select a suitable application-layer protocol among the ones used for IoT applications. • Recognize how analytics and artificial intelligence algorithms are used in the IoT ecosystem along with IoT security and privacy concerns.

Course Code	UE23CS342AA4	Course Title	Applied Cryptography				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	SEED labs, Python Programming Language	Desirable Knowledge	-				
Prelude	Cryptography is the science of securing data by using mathematical concepts. This course will present the fundamentals of cryptography, as well as its applications and issues of how cryptography is used in practice. Students will have opportunities to dwell well into problem solving and hands-on sessions.						
Course Objectives:	- To enable understanding foundations of Cryptography - To discuss about various symmetric encryption techniques. - To understand the concept of public key cryptography. - To introduce message authentication and hash function Explore emerging cryptography and future Directions						
Course Contents	Unit 1: Classical Ciphers. Introduction to Cryptography, Cryptanalysis and brute-force attack, Basic cryptographic primitives. Mathematical background for cryptography: Modulo arithmetic, GCD, Euclidean algorithm, Introduction to probability, Conditional probability, Law of total probability. Classical ciphers: Substitution - Caesar, Playfair and Hill cipher, Transposition cipher - Rail fence, Columnar and Double columnar, Cryptanalysis of classical ciphers. Shannon’s theorem and Perfect secrecy, One-time pad encryption and its limitations. 14 hours Unit 2: Symmetric Key Cryptography Introduction to symmetric key cryptography and Pseudo Random Number Generator (PRNG), The Feistel cipher, Data Encryption Standard (DES), Analysis, Multiple DES, Security and the avalanche effect, Mathematics- I: Galois fields, Polynomials- Advanced Encryption Standard (AES), Transformation functions, Key expansion, Analysis of AES, Block and Stream ciphers, Block cipher modes of operation, Drawback of symmetric key cryptosystem, Need for asymmetric key cryptography. 14 hours Unit 3: Public Key Cryptography and Hashing techniques. Introduction, Mathematics- II: Prime number, Primitive root, Prime factorization, The RSA algorithm, Diffie-Hellman key exchange, Elgamal cryptographic system, Elliptic curve cryptosystem, Digital Certificates, Cryptographic hash functions: Introduction, Applications, Collision resistance, Secure Hash Algorithm (SHA), Message Digest algorithm (MD5). 14 hours Unit 4: Hashing Techniques and key. Key-Distribution Centre (KDC), Kerberos, Public Key Infrastructure (PKI) and standard (PKCS), Birthday attack, Entity authentication methods: Password, Challenge-Response, Zero knowledge protocols. Introduction to Post-Quantum cryptography, Types of Quantum-Resistant Algorithms, Key Algorithms in PQC, PQC in Emerging Technologies. 14 hours						

Text Book(s):	1: Understanding Cryptography: A Textbook for Students and Practitioners, Christoff Paar and Jan Pelzl, Springer 2010. 2: Cryptography and Network Security: Principles and Practice, William Stallings, 7th Edition, Pearson, 2017. 3: Cryptography and Network Security, Behrouz A. Forouzan, 3rd Edition, Tata McGraw Hill, 2017.
Reference Book(s):	1: Introduction to Modern Cryptography”, Jonathan Katz, Yehuda Lindell, 2nd Edition, CRC Press, 2015.
Course Outcome	• Explain classical encryption techniques, cryptographic primitives, and cryptanalysis methods, along with the mathematical foundations that support secure communication. [Bloom’s Level: Understanding] • Implement and analyse symmetric encryption algorithms such as DES and AES, including their structure, operation modes, and cryptographic strength. [Bloom’s Level: Applying, Analysing] • Apply number-theoretic concepts to design and evaluate public-key algorithms including RSA, Diffie-Hellman, ElGamal, and elliptic curve cryptography. [Bloom’s Level: Applying] • Analyze and apply cryptographic solutions in real-world contexts such as secure messaging, e-payments, blockchain, HTTPS, and metaverse environments. [Bloom’s Level: Analyzing, Applying] • Discuss and explore post-quantum cryptographic principles and their role in securing future-proof cryptosystems. [Bloom’s Level: Understanding, Evaluating]

Course Code	UE23CS342AA5	Course Title	Virtual and Augmented Reality				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	Virtual and Augmented Reality - C/ C++/ JAVA/ Python using OpenGL. Unit3D, ThreeJS, Blender, Cloud Compare, OpenGL, Unity ML Agents	Desirable Knowledge	Data Structures & its Applications				
Prelude	This course presents an introduction to virtual and augmented reality technologies, with an emphasis on designing and developing interactive virtual and augmented reality experiences using blender and Unity3D.						
Course Objectives:	• Introduce the use of geometric transformations on graphics objects, their application in composite form and its implementation. • Impart the basics of computer graphics and Introduce graphics programming using OpenGL and Graphics Pipeline. • Introduce Virtual and Augmented Reality essentials and understanding human physiological aspects with respect to virtual reality applications and user interfaces. • Understanding of use of Artificial Intelligence in the field of Virtual and Augmented Reality and applying the concepts to create creative virtual experiences.						
Course Contents	Unit 1: Geometric Objects and Transformations Scalars, Points and Vectors, Three-Dimensional Primitives, Coordinate Systems and Frames, Modelling a Coloured Cube, Overview of 2D Transformations: Rotation, Translation and Scaling, Affine transformations, Transformation in Homogeneous Coordinates, Concatenation of Transformations, OpenGL Transformation Matrices, Interfaces to Three Dimensional Applications, Quaternion’s 14 Hours Unit 2: Graphical System and Programming and 3D Modelling. The Programmer’s Interface, Graphics Architectures, Programmable Pipelines. Graphics Programming: Programming Two Dimensional Applications. The OpenGL: The OpenGL API, Primitives and Attributes, Colour, Viewing, Control Functions, Polygons, Viewing, Control Functions, the gasket Program, Polygon and Recursion, The Three-dimensional gasket, Creating Scene on ThreeJS, Working with Three.Js Material and Geometries, Animations, Adding Physics, Working with Blender, Three.JS with React and WebXR 14 Hours Unit 3: Augmented and Virtual Reality. Introduction to Augmented Reality: Definition and Scope, A Brief History, Examples, Requirements and Characteristics: Methods of Augmentation, Spatial Display Models, Visual Display, Stationary Tracking Systems, Mobile Sensors. Introduction: What is Virtual Reality, Modern VR Experience. Bird’s Eye View: hardware, Software. Eye movement and its implications or VR. Tracking: 2D and 3D orientation, Tracking Position and Orientation, Tracking Attached bodies. 14 Hours Unit 4: IO modalities, AI, and Behaviour in VR Computer Vision and Augmented Reality; marker tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Incremental Tracking, Simultaneous Localization and Mapping, Outdoor Tracking, Interaction: Output and input modalities, Haptic interaction and Multimodal interaction. 3D						

	Scanning of environments. Reactive AI: Adaptability, Complexity and Universality, Feasibility, More Intelligence in the System: Deliberative AI, Reinforcement learning through interaction, Imitation Learning through human demonstration. 14 Hours
Text Book(s):	1: "Interactive Computer Graphics - A top-down approach with shader-based OpenGL", Edward Angel and Dave Shreiner, Pearson Education, Sixth edition, 2012. 2: Steven M. LaValle. Virtual Reality. Cambridge University Press, 2017, http://vr.cs.uiuc.edu/ (Links to an external site.) (Available online for free) 3: Creating Augmented and Virtual Realities, by Erin Pangilinan, Steve Lukas, Vasanth Mohan, PUBLISHED BY: O'Reilly Media, Inc. PUBLICATION DATE: March 2019
Reference Book(s):	1: "Interactive Computer Graphics: A Top -Down Approach with WebGL", Edward Angel, Pearson Education, 7th Edition, 2015. 2: Unity Game Development in 24 Hours , Geig, Mike. Sams Teach Yourself . Pearson Education, 2014. 3: "OpenGL Programming Guide": Mason Woo, Jackie Neider, Tom Davis, Dave Shrenier: 3rd Edition, OpenGL version 1.2, Addison Wesley, 1999. 4: Blender 3D Basic, Gordon Fisher, PACKT Publishing, 2nd Edition (Note: For working with recent version the course material for UNIT 4 can be substituted with appropriate web content. 5: D. Schmalstieg and T. Höllerer. Augmented Reality: Principles and Practice. Addison-Wesley, Boston, 2016, ISBN-13 978-0-32-188357-5
Course Outcome	• Apply techniques and methods of augmenting virtual objects in real space. • Demonstrate the fundamentals of computer graphics and display pipeline systems and use OpenGL for complex 3D graphical visualization and demonstrate its applications. • Apply techniques and tool to design a immersive virtual reality experience and graphics in greater depth to more complex aspects of Image Processing, Tracking and Human Computer Interfaces, etc. • Apply AI models and methods for creating dynamic interactive and adaptable virtual spaces.

Course Code	UE23CS342AA6	Course Title	Human Computer Interaction				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	Human Computer Interaction- Figma, Adobe Creative Cloud.	Desirable Knowledge	-				
Prelude	This course provides an overview and introduction to the field of human-computer interaction, with an emphasis on what HCI methods and HCI-trained specialists can bring to design and development teams for all kinds of products. The course will introduce students to tools and techniques for creating or improving user interfaces.						
Course Objectives:	- Understand the fundamental principles of Human-Computer Interaction (HCI) and its importance in designing user-friendly systems. - Analyse user needs and apply interaction design methodologies to develop effective interfaces. - Evaluate usability and accessibility aspects in interactive systems using various HCI evaluation techniques. - Explore emerging trends in HCI, including AI-driven interfaces, VR/AR, and ethical considerations in interaction design.						
Course Contents	Unit 1: Introduction to HCI and User-Centred Design Definition, Importance, and Goals of HCI, Human Factors in HCI - Perception, Cognition, Memory, Interaction Models -Gulf of Execution & Evaluation, Norman’s Model, User-Centred Design (UCD) Process, Understanding User Needs - Personas, Scenarios, Task Analysis, Design Principles -Affordances, Feedback, Consistency, Universal Design Principles. 14 Hours Unit 2: Interaction Design & Prototyping Interaction Styles -Command Line, GUI, Voice, Gesture, Design Thinking & Ideation Techniques, Wireframing and Prototyping -Low-Fidelity , High-Fidelity, Figma as a Prototyping Tool, Interface Design Guidelines – Material Design, Apple HIG, Case Studies in Interaction Design 14 Hours Unit 3: Usability Evaluation & Accessibility Usability Principles - Learnability, Efficiency, Memorability, Evaluation Methods - Heuristic Evaluation, Cognitive Walkthrough, User Testing - Think-Aloud Protocol, A/B Testing, Web & Mobile Accessibility - WCAG Guidelines, Inclusive Design & Assistive Technologies, Analyzing Usability Data - Metrics, Surveys, Interviews 14 Hours Unit 4: Emerging Trends & Future of HCI AI in HCI - Chatbots, Recommender Systems, Virtual & Augmented Reality (VR/AR) Interfaces, Brain-Computer Interfaces (BCI) & Wearable Computing, Ethical & Social Implications of HCI, Cross-Cultural HCI & Globalization Challenges 14 Hours						
Text Book(s):	1: Interaction Design: Beyond Human-Computer Interaction, Helen Sharp, Jennifer Preece, Yvonne Rogers,5th Edition (2019), Wiley						

Reference Book(s):	1: Designing the User Interface: Strategies for Effective Human-Computer Interaction, Ben Shneiderman, Catherine Plaisant, 6th Edition, Pearson Education, 2016. 2: “Human Computer Interaction”, Dix A., Finlay J, Abowd G. D. and Beale R., 3rd Edition, Pearson Education, 2005. 3: “Don’t Make Me Think, Revisited: A Common Sense Approach to Web and Mobile Usability”, Steve Krug(Author), 3rd Edition, 2014 (e-book)
Course Outcome	• Explain core HCI concepts, including usability, user experience (UX), and interaction design principles. • Apply user-centered design (UCD) methodologies to prototype and evaluate interactive systems. • Conduct usability testing and heuristic evaluations to assess interface effectiveness. • Discuss advanced HCI topics such as multimodal interfaces, AI in HCI, and ethical implications.

Course Code	UE23CS342AA7	Course Title	Topic in Wireless Network and 5G Networks				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective – 1				
AI Tools /Tools/Languages	Cisco Packet Tracer, OMNET++	Desirable Knowledge	Computer Networks.				
Prelude	This course offers an in-depth understanding of wireless communication systems, covering WLAN, WPAN, and cellular networks from 1G to 5G. Students will explore key topics like wireless access methods, modulation schemes, MIMO, LTE, and 5G network architecture. The course also delves into emerging areas such as Network slicing, OpenRAN, SDN/NFV, 5G security, and applications in IoT, AI/ML, and digital twins. With a forward-looking perspective on 6G, the course balances theoretical knowledge with practical skills, preparing students for careers in wireless communication and next-generation mobile networks.						
Course Objectives:	- Understand fundamental wireless technologies, IEEE 802.11 standards, and access methods. - Learn modulation techniques, cellular architecture, and evolution of mobile communication 1G to 5G. - Explore 4G and 5G architecture, protocol stack, authentication, and signaling mechanisms. - Introduce 5G applications, security, SDN/NFV, and emerging trends toward 6G.						
Course Contents	Unit 1 : Overview of Wireless communication Introduction, Different Wireless Networks – WWAN, WBAN, WPAN, WLAN , Different Wireless Standards of 802.11, Wireless Local Area Networks: IEEE 802.11, 802.11 Frame Format, Basic Access Methods – CSMA/CA, ALOHA, Distribution Coordinating Function (DCF), DIFS, SIFS, Point Coordination Function (PCF), Multiple Access for Wireless Systems – FDMA, TDMA, CDMA, OFDMA, 802.11 Physical Media, WPAN Technologies: Bluetooth, Bluetooth Protocol Stack, NFC, 6LOWPAN, LORAWAN, WiMAX, Adhoc Routing Protocols – DSDV, AODV, Dynamic Source Routing (DSR), MANET & VANET, Wireless networking and security issues in 802.11 – Static filtering based on MAC address, Wired Equivalent Privacy (WEP), WPA & WPA2, Wireless Link Characteristics, RFID: Concept, frequency band, classification of RFID tags, applications. 14 hours Unit 2 : Advanced Wireless Technologies and Cellular Standards Wireless Modulation Schemes – Amplitude Modulation (AM), Phase Modulation (PM), Frequency Modulation (FM), Binary Phase-Shift Keying (BPSK), Quadrature Phase-Shift Keying (QPSK), Frequency Spectrum, Cellular Concept –basic mobile communication, Mobile systems concept: mobile terminal, base station (BS), mobile switching centre (MSC), Frequency Reuse, Techniques: Multiple-Input and Multiple-Output (MIMO), mmWave, massive MIMO, Small Cells, Handoff, Mobile Radio standards – AMPS, GPRS, GSM, UMTS, CDMA 2000, Evolution of Wireless Network generations (1G, 2G, 3G, 4G, 5G), How does a mobile connectivity work? Global System for Mobile Communication (GSM), 3G UMTS – architecture, Need for 3G, 4G and 5G technologies – architectural difference 14 hours						

	Unit 3 : Fundamentals of Cellular System 4G – LTE – EPC architecture, LTE protocol stack, UMTS vs LTE, Configuring LTE Control and Data Plane, Authentication Key Agreement in 4G LTE, Introduction to 5G, 5G Architecture, features, specifications and other procedures, 5G-SA/NSA, 5G vs 4G architectural differences, Simple call registration flow in 5G – AKA Authentication, Session establishment in 5G call flow, Signaling System No. 7 (SS7). 14 hours Unit 4 : 5G Applications and Evolution to 6G OpenRAN, Network Slicing, Long Range Communication- Satellite Communication, VoLTE, SDN, NFV, Private 5G networks, Application use cases: Digital Twin, IoT, AI/ML, Satellite Communications, 5G Security and Beyond, Mobile Forensics, 5G Security, Introduction to 6G. 14 hours
Text Book(s):	1: "Computer Networking: A Top Down Approach", James Kurose and Keith Ross, Pearson, 8 th Edition
Reference Book(s):	1: "Wireless and Mobile Network Architectures", Lin Yi-Bang and Clamta Imrich, John Wiley & Sons 2001 2: "Wireless Communications and Networking", Vijay Garg, Morgan Kaufmann Publishers 3: "Wireless Network Security: Second Edition, by Wolfgang Osterhage, CRC Press 4: "Wireless and Mobile Networks: Concepts and Protocols", Dr. Sunil Kumar S. Manvi, Mahabaleswar S. Kakkasageri, Wiley India 5: "Wireless Communication and Networks: 3G and Beyond", ITI Saha Mishra, McGraw-Hill Education 6: "Wireless Communications: Principles and Practice", Theodore S. Rappaport 7: "Introduction to Mobile Network Engineering: SGM, 3G-WCDMA, LTE and the Road to 5G", Alexander Kukushkin, 2018 John Wiley & Sons Ltd 8: "5G Mobile and Wireless Communications Technology", AFIF OSSEIRAN, JOSE F. MONSRAT, PATRICK MARSCH, Cambridge University Press 9: "5G Mobile Networks: A Systems Approach", Larry Peterson and OgMuz Sunay, 2020, Morgan & Claypool Publishers.
Course Outcome	• Explain wireless standards, access protocols, and security in wireless networks. • Compare wireless modulation schemes and mobile communication generations. • Analyze 4G and 5G network architectures and key protocols. • Evaluate 5G applications, network slicing, and the transition to 6G technologies.

Course Code	UE23CS343AA8	Course Title	Bio-Inspired Computing				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	Matlab, Inspyred pythonic tool	Desirable Knowledge	Design and Analysis of Algorithms				
Prelude	The field of natural computing has been the focus of a substantial research effort in recent decades. These bio inspired computing algorithms have proven to be successful problem solvers across domains as varied as management science, telecommunications, business analytics, bioinformatics, finance, marketing, engineering, architecture and design, to name but a few. This course provides a comprehensive introduction to bio inspired computing algorithms. This course requires the student to have a desirable knowledge of Design and Analysis of Algorithms.						
Course Objectives:	- To introduce the fundamental concepts of bio-inspired computing. - To explore various evolutionary and swarm-based algorithms. - To demonstrate the application of nature-inspired models to real-world optimization problems. - To enable students to analyze and implement algorithms using computational tools.						
Course Contents	Unit 1: Introduction to Natural Computing Algorithms Overview, Evolutionary Computing, Canonical Genetic Algorithm, Design Choices, Representation, Fitness, Diversity, Parameter Tuning, Memetic Algorithms, Linkage Learning, Estimation of Distribution Algorithms. 14 Hours Unit 2: Evolution Strategies and Genetic Programming Evolution Strategies, Evolutionary Programming, Differential Evolution (Canonical & Extended), Discrete DE, Genetic Programming, Bloat in GP, GP Variants, GP Semantics. 14 Hours Unit 3: Social Computing and Swarm Intelligence Particle Swarm Optimization (PSO), Hybrid PSO, Ant Algorithms (Foraging, Optimization, Clustering, Classification), Honeybee Algorithms, Firefly Optimization, Non-uniform Oscillators. 14 Hours Unit 4: Immune Systems and Other Social Algorithms Artificial Immune Systems (Negative Selection, Clonal Selection, Immune Programming), Glowworm, Bat, Fish School, Locust Algorithms, Future Trends and Open Research Issues.. 14 Hours						
Text Book(s):	1: Natural Computing Algorithms, Anthony Brabazon, Michael O’Neill, Seán McGarraghy, Springer, 2015.						
Reference Book(s):	1:Fundamentals of Natural Computing, Leandro Nunes de Castro, CRC Press, 2007 2: Bio-Inspired Artificial Intelligence: Theories, Methods, and Technologies, Dario Floreano & Claudio Mattiussi, MIT Press, 2008 3: Metaheuristics: From Design to Implementation, El-Ghazali Talbi, Wiley, 2009 4: Swarm Intelligence, James Kennedy, Russell Eberhart, and Yuhui Shi, Morgan Kaufmann, 2001						

Course Outcome	• Understand the essential features and behaviour of biologically inspired systems. • Model and simulate biological processes using computational techniques. • Apply evolutionary and swarm-based algorithms to solve complex optimization problems. • Evaluate and compare the effectiveness of nature-inspired algorithms in different domains.
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Course Code	UE23CS342AA9	Course Title	Enterprise Business Systems for Engineers				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	Oracle, Sales force, Open ERP	Desirable Knowledge	-				
Prelude	The course in Enterprise Business Systems (EBS) offers students a comprehensive understanding of integrated business management systems and their role in streamlining organizational processes. Through theoretical insights, application examples, and case studies, students will understand the functionalities, implementation methodologies, challenges and strategic implications of ERP systems in modern enterprises including associated systems like Supply Chain Management [SCM] and Customer Relationship Management [CRM].						
Course Objectives:	- To learn the strategic importance of Enterprise Resource Planning, Supply Chain and CRM systems in the industry. - To learn the basics of ERP, the modules of ERP, their purpose and functions. - To learn the key selection criteria and issues & risks of ERP implementation. - To be aware of current trends like Role of Analytics, AI, Cloud and other technologies employed in EBS.						
Course Contents	Unit 1: Introduction to Enterprise Business Systems Introduction to EBS, Value Chain Framework, Problems with Disintegrated Data in an Organization, Evolution of ERP Systems, Role of ERP Systems in an Organization, Scope of ERP Systems, General Model of Business and Role of ERP, Major ERP Players, Implementations Types – On-prem, Cloud, Hybrid, Two-tier. Life Cycle of an ERP Implementation Project: Life Cycle of an ERP Project, ERP Project Teams, Implementation Methodologies, Deployment Methods. Benefits and Cost of an ERP System: Benefits, Cost of an ERP Implementation, Cost-Benefit Analysis. 14 hours Unit 2: Change Management, Re-Engineering and BPM: Change Management: People Issues: Are You Ready for ERP, Factors that Influence Pre-implementation Attitude, How to Enhance Attitude?, Change Management Strategies to Handle Organizational Issues, Creating a Change Management Strategy to Handle Organizational Issues, Tools for Assessing the Organizational Changes. Re-Engineering: Processes and their Characteristics, Life Cycle of a BPR Project, Life Cycle of an IT-driven BPR Project, Re-engineering Examples, Case Studies. Business Process Modelling and Business Modelling: BPM Introduction, Business Process Hierarchy, Standards for Business Processes and Modelling, Process Modelling Maturity and Multi-Dimensional Modelling, Process Modelling Software, Business Modelling, Integrated Data Modelling. 14 hours Unit 3: ERP Functional Modules: Human Capital Management: HCM Systems, Leading HR Solutions from ERP Vendors, Strategic Vs. Operational HR Processes and HR Outsourcing, Employee Health and Safety. Financial Management: ERP Financial Application, Financial Modules in Detail. Procurement and Inventory Management: Procurement, Inventory Management. Production Planning and Execution: Understanding MRP II Concepts, How ERP PP module supports MRP II Processes, Critical Master Data Elements, Managing different Production Scenarios.						

	14 hours Unit 4: ERP Selection and Emerging Trends: ERP Solution and Vendor Selection, Selection Criteria, Request for Proposal (RFP), Gap Analysis, ERP Selection Team. Managing an ERP Project: Critical Success Factors, Project Risks, Performance Metrics. ERP and Related Technologies: Supply Chain Management (SCM), Customer Relationship Management (CRM), Product Life Cycle Management (PLM), Data Warehousing and Business Intelligence (DW-BI). Introduction to Commercial ERP Related Software: Indian ERP Market, SAP, Oracle, MS Dynamics, Salesforce CRM. Implementations in India; Articles and Case Studies on ERP Trends like Analytics, Cloud, Digital Transformation, Embedding AI & ML capabilities, IoT, Automation and other technology integrations. Explore Emerging Trends like Gen-AI and web resources like Gartner Magic Quadrant and HypeCycle. 14 hours
Text Book(s):	1. "Enterprise Resource Planning- Text & Cases", Rajesh Ray, Tata McGraw Hill, New Delhi, 2011.
Reference Book(s):	1: "ERP Demystified", Alexis Leon, McGraw Hill Education, 4th Ed, 2019. 2: "Modern ERP: Select, Implement, and Use Today's Advanced Business Systems" by Marianne Bradford; Pearson, 4th Edition, 2020. 3: "Enhancing Enterprise Intelligence: Leveraging ERP, CRM, SCM, PLM, BPM, and BI", Vivek Kale, Auerbach Publications, 2016. 4: "Enterprise Resource Planning: A Managerial Perspective", Veena Bansal, Pearson Education India, 2013. 5: "Supply Chain Management: Strategy, Planning and Operation", Sunil Chopra, Pearson Education, 7th Edition, 2019. 6: Common Sense ERP: The Insider's Guide to Selecting ERP and Business Software, Daniel A. Carr, June 2016. 7: Ultimate Guide to ERP by TechTarget. 8: Web Resources for Current Trends.
Course Outcome	- Identify typical functionality of ERP, SCM & CRM sub-systems. - Apply criteria to select ERP Package & Consulting Partner. - Systematically develop plans for an ERP Implementation project. - Identify critical success factors and associated risks of an ERP Implementation.

Course Code	UE23CS343AA10	Course Title	Quantum Computing				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective – 1				
AI Tools /Tools/Languages	IBM Qiskit	Desirable Knowledge	-				
Prelude	This course is meant to give an overview of the field of quantum technologies and make the students familiar with the state-of-the-art in all four verticals. The emphasis is not on depth in this course, but on covering the exciting aspects of the field.						
Course Objectives:	• The general physical principles of realising qubits for computation • The various hardware implementations of qubits for computation • The basic ideas of quantum sensing • The applications of quantum sensing						
Course Contents	Unit 1Introduction : Introduction to the course, Expectation from the course, What this course is NOT about, Quantum and classical physics, Example Double slit Experiment with bullets and waves Double slit Experiment with electrons and concept of probability density, Dirac Notation, State of a quantum system Hilbert Space representation Quantum Mathematics Discrete and continuous base representations Basic postulates of Quantum Mechanics Matrix representation of state vectors Python implementation of some of the basic quantum examples More examples / tutorial 14 hours Unit 2: Quantum Entanglement background Bloch sphere Representations Pure and mixed states Ensemble and density matrix space of density matrices Cbits and Qbits Quantum logic gates Introduction to IBM/Microsoft Quantum computer environment Continuation of previous class topic Simulation of Quantum circuits on IBM Quantum environment Examples 14 hours Unit 3: Entanglement EPR paradox Density matrices and entropies Superposition of States Distinguishable and Indistinguishable states and consequences Quantum Entanglement Quantum Entanglement examples Sources of Entangled photons Time, momentum and polarization entanglement Consequences of entanglement on Quantum computation and Quantum information. Quantum teleportation and protocol Implementation of Quantum entanglement on Qiskit Examples 14 hours Unit 4Quantum information Introduction to classical information theory Entropy and information Quantum measurement Quantum circuits Quantum error correction introduction to quantum information Quantum dense coding implementation of quantum computing and studying Bloch spheres on Qiskit and Quantum computers Case studies Student paper presentation 14 hours						

Text Book(s):	1: Quantum Information Science – Manenti R., Motta M., 1st Edition, Oxford University Press (2023)
Reference Book(s):	1: Quantum computation and quantum information – Nielsen M. A., and Chuang I. L., 10th Anniversary edition, Cambridge University Press (2010) 2: Elements of Quantum Computation and Quantum Communication, A. Pathak, Boca Raton, CRC Press (2015) 3: An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, and Michele Mosca, Oxford University Press (2006) 4. Quantum computing explained, David McMahon, Wiley (2008)
Course Outcome	• The various hardware implementations of qubits for computation • The basic ideas of quantum sensing • The applications of quantum sensing • The implementations of quantum communications protocols in fibre-based and free-space

Course Code	UE23CS343AB1	Course Title	Image Processing				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective - 2				
AI Tools /Tools/Languages	MatLab, Python, ChatGPT	Desirable Knowledge	Design and Analysis of Algorithms.				
Prelude	Digital Image Processing deals with processing images that are digital in nature. Improving the quality of images for human perception and understanding, extracting useful information for decision making and efficient storage are some of the driving factors behind image processing techniques/algorithms. The course on Digital Image Processing introduces the learner to various image processing techniques, algorithms and their applications.						
Course Objectives:	- Understand the principles underlying imaging and assess the quality of an image based on its visual content, histogram and statistical measures - Gain an insight to image transforms and enhancement techniques in the frequency domains - Be able to apply morphological operations and segmentation techniques on monochrome and color images - Learn different feature extraction techniques for classification						
Course Contents	Unit 1: Introduction and Enhancement in the Spatial Domain What is Digital Image Processing, examples of fields that use DIP, Fundamental Steps in Digital Image Processing, elements of visual Perception, Basic Concepts in Sampling and Quantization, Representing Digital Images, Spatial and Gray-level Resolution, Zooming and Shrinking Digital Images, Some Basic Relationships Between Pixels, Linear and Nonlinear Operations. Image Enhancement in the Spatial Domain: Some Basic Gray-level Transformations, Histogram Processing, Enhancement Using Arithmetic/Logic Operations, Basics of Spatial Filtering, Smoothing Spatial Filters, Sharpening Spatial Filters, and Combining Spatial Enhancement Methods. 14 Hours Unit 2: Image Enhancement in the Frequency Domain, Transforms and Compression Image Enhancement in the Frequency Domain: Introduction to the Fourier Transform and the Frequency Domain, Smoothing Frequency-Domain Filters, Sharpening Frequency Domain Filters, Homomorphic filtering. Image transforms (such as, DCT, Harr, DWT) for applications. Image Compression: Fundamentals - Image Compression Models, Some encoding techniques (including, block processing and transform-based techniques). 14 Hours Unit 3: Morphological Image Processing, Preliminaries of segmentation and Color Image Processing Dilation and Erosion, Opening and Closing, the Hit-or-Miss Transformation, Some Basic Morphological Algorithms. Image Segmentation: Detection of Discontinuities, Edge Linking and Boundary Detection, Thresholding, Region-Based Segmentation. Color Models, Pseudocolor Image Processing, Basics of Full-Color Image Processing, Color Transformations, Smoothing and Sharpening, Color Segmentation, Noise in Color Images.						

	14 Hours Unit 4: Feature extraction, pattern classification Boundary Preprocessing, Boundary Feature Descriptors, Region Feature Descriptors, Principal Component Analysis (use of Eigen images), Whole image features, Scale Image Feature Transform, Image pattern classification using minimum distance classifier and prototype matching, deep convolutional neural networks and some of its recent variations for computer vision applications (UNet for segmentation, ResNet for feature extraction, YoLo for object detection) and recent advances in computer vision. 14 Hours
Text Book(s):	1: "Digital Image Processing", Rafael C Gonzalez and Richard E. Woods, Prentice Hall, 4th Edition, 2018.
Reference Book(s):	1. Stan Birchfield, "Image Processing and Analysis", 2017, First Edition, CL Engineering, ISBN number: 978-1285179520 2: "Digital Image Processing and Analysis", Scott E. Umbaugh, CRC Press, 2014. 3: "Digital Image Processing", S. Jayaraman, S. Esakkirajan, T. Veerakumar, McGraw Hill Ed. (India) Pvt. Ltd., 2013. 4: "Digital Signal and Image Processing", John Wiley, 2003.
Course Outcome	- Assess the quality of an image and identify the appropriate processing technique that needs to be applied in the spatial domain - Apply image enhancement techniques frequency (Fourier) domain and interpret images in the transform domain - Design and evaluate methodologies for segmentation of region of interest from binary, gray scale or color images - Use appropriate feature extraction techniques for tasks such as classification and be able to work with data-driven approaches (deep learning architectures) for computer vision applications

Course Code	UE23CS343AB2	Course Title	Big Data				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective - 2				
AI Tools /Tools/Languages	Hadoop, HDFS, Spark, Streaming Spark, HIVE, HBase, Mllib.	Desirable Knowledge	Data Structures and its Applications.				
Prelude	The course introduces various Big Data technologies that are used to analyze large amounts of data either in batch mode or streaming mode. It focuses on both processing and storage technologies and looks at how algorithms need to be modified to work with large amounts of data. This course requires the student to have a desirable knowledge of Data Structures and its Applications and Design and Analysis of Algorithm.						
Course Objectives:	• Provide an introduction to Big Data technologies. • Introduce computational and storage technologies for Big Data Ecosystem. • Introduce algorithms for processing Big Data and programming tools focussing on practical issues in working with Big Data. • Learn application of Big Data techniques to various real-life problems.						
Course Contents	Unit 1 : Introduction Big Data definition, Challenges and opportunities with Big Data, Data intensive scientific discovery and the role of Big Data, History, Map Reduce – Storage (HDFS), Computation model, Map Reduce architecture, Overview of Hadoop Ecosystem, Apache Flume as a data ingestion and log aggregator tool, YARN introduction, Case Study: HIVE, Google File System, Google Big Query. 14 Hours Unit 2 : Big data infrastructures (Compute/Storage) Apache airflow basics and working, Introduction to sample Big Data Algorithms – matrix multiplication and pagerank, Issues with Hadoop, Complexity of Big Data algorithms – Communication Cost complexity model. case study: HBase , Cassandra. 14 Hours Unit 3 : In memory computation Kafka – use cases, architecture, Spark and Scala/PySpark programming model, Transformations and Actions, Spark SQL, Relational operators with Spark, Spark architecture – RDD, DataFrames, Wide and Narrow dependencies, Streaming Algorithms –sampling, set membership – Bloom Filters counting, counting unique elements – Flajolet Martin Algorithm. 14 Hours Unit 4 : Streaming analysis and advanced analytics on Big Data Streaming analytics use cases, Streaming Spark, , Apache Flink, Clustering algorithms - k means and Collaborative filtering, Scaling Neural Networks for Big Data, Case Study: MLLib, practical examples of Big Data success stories 14 Hours						
Text Book(s):	1: “Big Data Analytics”, Rajkamal, Preeti Saxena, 1st Edition, McGraw Hill Education, 2019. 2: “Big Data Simplified”, Sourabh Mukherjee, Amit Kumar Das, Sayan Goswami, 1stEdition, Pearson, 2019.						

Reference Book(s):	1: "Mining of Massive Datasets", Anand Rajaraman, Jure Leskovec, Jeffrey D. Ullman, Cambridge Press, 2014. 2: "Big Data Analytics Beyond Hadoop: Real-Time Applications with Storm, Spark, and More Hadoop Alternatives", Vijay Srinivasa Agneeswaran, Pearson Education, 2014.
Course Outcome	• Explore various characteristics of Big Data Problems. • Evaluate principles and design alternative computational/storage technologies for Big Data. • Design Big Data applications using available infrastructure for Big Data through practical assignments. • Applying Big Data techniques in real life problems through a group-based project.

Course Code	UE23CS343AB3	Course Title	Graph Theory and its Applications				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective - 2				
AI Tools /Tools/Languages	NetworkX, PyTorch, Deep Graph Library, Neo4j, Gephi, Graphistry	Desirable Knowledge	UE23CS151B – Problem Solving with C, UE23CS252A – Data Structures and its Applications.				
Prelude	This course focuses on mathematical structure to model the relations between the objects. It also discusses about the basics of graph theory together with a wide range of applications to different branches of Science and Technology, and to real-world problems. The course includes principles of Graph database and social networks.						
Course Objectives:	- Familiarize students with concepts and abstraction of Graph Theory. - Up skill students with computer representation of graphs and algorithms. - Introduce students with advanced concepts in Graph Theory including Vizing’s and Brooks’ Theorems. - Apply principles of graph theory for social network analysis and dynamic processes.						
Course Contents	Unit 1: Introduction, Paths, Cuts, Planar Graphs, and Vizing’s Theorem Introduction- Representation and Traversals, Walks, Paths, Circuits - Euler graphs, Hamiltonian paths and circuits - Directed graphs, Digraphs and binary relations - Trees: Properties, Rooted trees, Binary trees, Spanning trees - Cut sets, Fundamental circuits and cut sets - Connectivity and Separability - Network flows, Isomorphism - Vizing’s Theorem 14 Hours Unit2:Graph Coloring, Graph Applications, and Brooks' Theorem Planar Graphs – Different representations - Chromatic number, Chromatic partitioning, Chromatic polynomial - Matching, Covering - Four Colour Problem - Register allocation using graph coloring - Minimum Spanning Trees - Shortest Path Problem - Articulation Points - Network Flow Algorithms - Chinese Postman Problem - Time Table Problem - Strongly Connected Components - Brooks' Theorem on Colorings. 14 Hours Unit3: Graph Databases and Graphs in Social Networks Graph Database: RDBMS vs Graph Database - Neo4j Basics: Graph Data Model (Nodes, Properties, Relationships, Labels) - Basic CQL Commands - Graphs in Social Networks: Centrality Measures: Degree, Betweenness, Closeness, Eigenvector, PageRank - Measures of Cohesiveness: Degree Distribution, Diameter, Transitivity - Connected Components (in/out/weak/strong combined) - Groups and Substructures: Clique, N-clique, N-core. 14 Hours Unit4: Graphs in Social Networks and Social Graph Dynamics Community Detection: Cluster vs Community, Overlapped Communities (CPM), Girvan-Newman Algorithm, Louvain Algorithm - Generative Models: Small World: Milgram's Small World Experiment, Watts-Strogatz Model, Barabási-Albert Model - Dynamics on Graphs: Herd Behavior, Diffusion of Innovation (Bass Model), Epidemic Modeling, Cascades 14 Hours						

Text Book(s):	1. "Graph Theory: With Application to Engineering and Computer Science", Narsingh Deo, Prentice Hall of India, 2017. 2. "Graph Databases: New Opportunities for Connected Data", Ian Robinson, Jim Webber, Emil Eifrem, 2nd Edition, O'Reilly Publications, 2015. 3. "Networks, Crowds, and Markets: Reasoning about a Highly Connected World", David Easley, Jon Kleinberg, Cambridge University Press, 2010. 4. "Social Media Mining", Reza Zafarani, Mohammad Ali Abbasi, Huan Liu, Cambridge University Press, 2014.
Reference Book(s):	1. "Graph Theory", F. Harary, Addison-Wesley, 1969. 2. "Discrete and Combinatorial Mathematics", Ralph P. Grimaldi & B.V. Ramana, 5th Edition, PHI/Pearson Education. 3. "Graph Theory with Applications", J. A. Bondy and U. S. R. Murthy, Elsevier Science Publishing Co. 4. "Graph Representation Learning", William L. Hamilton, Morgan and Claypool Publishers, 2020. 5. "Introduction to Graph Theory", Douglas B. West, 2nd Edition, Prentice Hall.
Course Outcome	● Understand and model real-world problems using graph theory concepts. ● Implement efficient graph representations and algorithms. ● Apply graph coloring theorems like Vizing's and Brooks' to theoretical and practical problems. ● Analyze and model social networks and dynamic processes using graph principles.

Course Code	UE23CS343AB4	Course Title	Real Time Operating System				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective - 2				
AI Tools /Tools/Languages	FreeRTOS+Trace / Coverity / Valgrind / gprof /C Cursor AI, SystemView	Desirable Knowledge	Operating system				
Prelude	The goal of this course is to meet the basics of real time systems and to provide thorough discussion of the fundamentals of Operating system design and to relate these to contemporary design issues and to current directions in the development of operating systems						
Course Objectives:	- To understand basics of Operating System - To understand the process states, creation and to perform inter process communication - To understand necessary and sufficient conditions of real time tasks and scheduling algorithms - To understand memory management systems - To understand key characteristics of RTOS and hands on using Free RTX						
Course Contents	Unit1: Introduction to Real-Time Systems Introduction, types of RTOS and their usage, Definition and characteristics of real-time systems, Hard vs. soft real-time systems, Applications in various domains (automotive, aerospace, IoT), Overview of real-time operating systems, Kernel and Microkernel-based OS, System performance and user services, Real-time vs general-purpose systems, Introduction to safety-critical systems and certification (e.g., ISO 26262), Overview of POSIX-compliant real-time systems, Case Study Zephyr. 14 Hours Unit 2: RTOS Fundamentals and Scheduling in Real-Time Systems RTOS architecture and components, Task management and scheduling, Inter-process communication mechanisms, Interrupt handling and latency considerations, Scheduling algorithms: Rate Monotonic, Earliest Deadline First,Priority inversion and inheritance, Multiprocessor scheduling challenges, Case studies and performance analysis Case study- Zephyr 14 Hours Unit 3: Memory Management in RTOS andDevice Driver Development Memory allocation strategies, Memory protection and isolation, Dynamic vs. static memory management, Garbage collection and memory leaks, Flat memory vs Virtual memory Device Drivers:Fundamentals of Device Drivers, Character and Block Devices, Polling and Interrupt Mechanisms Hardware Enumeration and Configuration, Data Transfer and Management Mechanisms Interrupt handling and latency considerations. 14 Hours Unit 4: File System Support and Advanced Topics in RTOS File System Support :Fundamentals of Filesystems in Embedded Systems,Filesystem Requirements and Constraints in RTOS. Debugging and Optimization:Using Tracealyzer for system visualization, Debugging techniques for FreeRTOS applications, Performance profiling and optimization.Advanced topics:Real-time system design patterns, Security considerations in RTOS, Energy-efficient real-time systems, Software						

	Timers and Delays,Future trends and research directions. 14 Hours
Text Book(s):	1. Qing Li and Carolyn, “Real-Time Concepts for Embedded Systems” , CRC Press, 1st Edition, 2003.
Reference Book(s):	1. William Stallings, “Operating Systems” , Pearson Publications,2018 2. Silberschatz , Galvin, Gagne,John ,“Operating System Concepts” , Wiley & Sons,9th Edition,2013 3. J.P. Hayes, “Computer Architecture and Organisation”, Tata McGraw-Hill, Second edition, 1988. 4. Sam Siewert, “Real-Time Embedded Systems and Components” , Cengage Learning India, 2007.
Course Outcome	● Difference between General Operating system and Real time operating system ● The multitasking techniques in real time systems. ● Real time scheduling policies in applications. ● Memory management in RTOS ● Design embedded applications using RTOS.

Course Code	UE23CS343AB5	Course Title	Advanced Computer Networks				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective - 2				
AI Tools /Tools/Languages	GNS3, Cisco Packet Tracer, Mininet	Desirable Knowledge	Computer Networks.				
Prelude	The Advanced Computer Networks course is designed for students who have basic knowledge of networking (OSI model, TCP/IP, ARP etc.). The course builds its contents by considering a mixed approach involving theory and practical. From this perspective, the course introduces advanced topics that are basic blocks of networking and covers fundamentals that are used to develop few of the advanced technologies, including routing protocols and switching. The course aims to provide information so that of the course, the students find themselves comfortable in taking either the direction- industrial job or further research in networking.						
Course Objectives:	• Provide in-depth study of dynamic and distance vector routing protocols • To learn about Layer 2 Switching in detail • To get an insight of Link State routing protocols • To explore the new approach of Networking, SDN and NFV						
Course Contents	Unit 1: Dynamic and Distance Vector Routing, Access ListsIP Routing Process, Static routing – IPv4/IPv6, Floating Static Routes, Load Sharing, Fast and Process SwitchingDistance Vector Routing – RIPv2 packet format, classless Routing, compatibility with RIPv1, Metric Calculation, Route Summarization, Standard IP and Extended IP access lists, Calling access lists, Named access lists. 14 Hours Unit 2: Exterior Gateway Routing Protocol (EIGRP), Layer 2 Switching, Spanning Tree Protocol (STP), Virtual LAN (VLAN)EIGRP – Neighbor discovery, Reliable Transport Protocol (RTP), Diffusing Update Algorithm (DUAL), VLSM and Summary Routes, Metrics – Maximum Paths & Hop Count, Load Balancing with EIGRP, Discontiguous Networks, Auto summarization, passive interfaceLayer 2 switching, Bridging vs LAN Switching, Forwarding Table, Spanning Tree Protocol and its operations, VLAN Basics, Static and Dynamic VLANs, VLAN Trunking Protocol (VTP), Routing between VLANs, configure VLANs and Inter-VLAN routing, MAC VLANs, VXLAN 14 Hours Unit 3: Link State Routing Protocols – OSPF & BGOpen Shortest Path First (OSPF)v2 – OSPF packet format, Network Types, Designated Routers and Backup Designated Routers (DR & BDRs), OSPF Neighbors, Areas – Intra, Inter, External, Router Types, Virtual Links, Link State Database, LSA Types, Metric Calculation, Summary Routes, Route redistributionBorder Gateway Protocol (BGP) – BGP Packet Format, External and Internal BGP. 14 Hours Unit 4: Introduction to Software Defined Networks (SDN) and Network Function Virtualization (NFV)Segment Routing, SDN Control and Data Plane, SDN Controller and Network-control Applications, OpenFlow Protocol, Data and Control Plane Interaction Network Virtualization architecture, Management, Control and Data Planes, Virtual Switches, Micro-segmentation. 14 Hours						

Text Book(s):	1. "Routing TCP/IP volume 1", 2nd Edition, Jeff Doyle, Cisco Press 2005. 2. "Computer Networking: A Top-Down Approach", James F. Kurose, Keith W. Ross, 7th Edition, Pearson Publication, 2017 [For SDN]
Reference Book(s):	1: "Software-Defined Networks: A Systems Approach", Peterson, Cascone, O'Connor, Vachuska, and Davie, https://sdn.systemsapproach.org/index.html [For NFV] 2: "TCP IP Protocol Suite", Behrouz Forouzan, 4th Edition, McGraw-Hill, 2010 [For BGP]
Course Outcome	• Designing a network topology by demonstrating the ability to configure routers with different routing techniques • Understand the concepts of Layer 2 switching, VLANs and its usage • Designing networks with one or multiple routing protocols and investigating its behaviour/features • Comprehend features of Software Defined Networking (SDN) for next generation systems

Course Code	UE23CS343AB6	Course Title	Computer Network Security				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective - 2				
AI Tools /Tools/Languages	SEED Ubuntu VM, Wireshark, Snort, Netwox, Scapy.	Desirable Knowledge	Computer Networks.				
Prelude	This course will give an overview and conceptual understanding of network related security aspects. Students will have opportunities to dwell well into technical “how to” with hands-on sessions and case study discussions.						
Course Objectives:	- To provide an overall view of Network Security and introduce the concept of packet analysis. - To understand the security problems in the design and implementation of the TCP, IP/ICMP, ARP protocols. - To learn the vulnerabilities in DNS protocol and to implement and experiment with Firewall rules. - To provide an overview of network management techniques and implementation of VPN. To understand concepts of risk management and security aspects of wireless networks.						
Course Contents	Unit 1: Introduction, Packet Sniffing & Spoofing, MAC Layer Attacks, Network Layer Attacks CIA principles, Attack surface and types, Assets, Vulnerabilities and Threats, Countermeasures, Privacy, General Data Protection Regulation, Security vs Privacy, Data Breaches. Real Life Examples of Cyber Crime, Security framework, Job outlook. Packet Sniffing and Spoofing: Introduction, Sending packets: Network Interface Card (NIC), BSD packet filter (BPF). Packet sniffing: Receiving packets using sockets, Packet sniffing using Raw sockets, Packet sniffing using PCAP API, Processing captured packets. Packet spoofing: Sending normal packets using sockets, Constructing spoofed raw ICMP packets and UDP packets. Sniffing and then spoofing, Python vs Scapy, Endianness. MAC layer and attacks: The MAC layer, ARP protocol, ARP cache poisoning attacks, MITM using ARP cache poisoning, Demo, Countermeasure. 14 Hours Unit 2: TCP Attacks and DNS Attacks Network layer: IP, ICMP and attacks: IP protocol, IP fragmentation, Attacks using IP fragmentation: Problem and solution, Routing, and spoofing prevention, ICMP protocol, ICMP redirect attack, Smurf and other ICMP attacks. Attacks on the TCP protocols: TCP overview, Send and receive buffers, SYN flood attack: TCP 3-way handshake, the SYN flooding attack, Launching the attack using Netwox and C, Countermeasure. TCP reset attack: TCP reset attack on Telnet, SSH and video streaming connections. TCP session hijacking attack: TCP session and session hijacking, Launching the attack, Hijacked TCP connection. Reverse shell: working, redirecting IO to TCP connection, Creating reverse shell. Countermeasure. DNS Attacks: DNS hierarchy, zones and servers, DNS query process, Constructing DNS request and response using Scapy, DNS attacks: Overview, Local DNS cache poisoning attack, 12 Hours Unit 3: Firewall, IDS, IPS, Honeypots Remote DNS cache poisoning attack (Kaminsky attack), Reply forgery attacks from malicious DNS servers, Countermeasure against DNS spoofing attacks, DoS attacks on DNS servers. Case Study – 1.Firewall: Introduction, Requirements of a firewall, Firewall characteristics and Access policy, Types of firewalls, NG						

	firewall, Shortcomings, Firewall location and configuration: DMZ networks, Firewall topologies. Introduction, Build a simple firewall, Netfilter, iptables firewall in Linux, Stateful firewall and connection tracking, Application/Proxy firewall and Web proxy, Evading firewalls. Intrusion Detection and Prevention: Intruders, Intrusion detection, Analysis approaches, Host-based intrusion detection, Network-based intrusion detection, Distributed or hybrid intrusion detection, Honeypots, Example system: Snort. 14 Hours Unit 4: Virtual Private Networks, Wireless security Virtual Private Network: Introduction, Why VPN, analogy, and tunnelling. Overview of TLS/SSL VPN: Establishing a tunnel, Forwarding, and releasing IP packets, TLS/SSL VPN details. Building, Setup and Testing VPN. Bypassing Firewall using VPN. Case Study – 2. The Heartbleed Bug and Attack: Introduction and the Heartbeat protocol, Launching the attack, Fixing the Heartbleed bug. Wireless Security: Communications and 802.11 WLAN standards: Wired Equivalent Privacy (WEP), Wireless Protected Access (WPA), IEEE 802.1x, 802.11i/ WPA2, Wireless Network Threats. SOC and SIEM. 14 Hours
Text Book(s):	1. Internet Security: A Hands-on Approach, Wenliang Du, 2nd Edition, 2019 2. Internet Security: A Hands-on Approach, Wenliang Du, 3rd Edition, 2022 3. “Computer Security: Principles and Practice”, William Stallings and Lawrie Brown, 3rd edition, 2015
Course Outcome	• Sniff packets from clients and analyse them to extract important info such as headers, passwords etc. • Launch DoS and MITM attacks using various protocol vulnerabilities and mitigate them. • Configure firewalls on Linux machines and exploit vulnerabilities on DNS protocol. • Design and implement VPN for a secure connection over the internet. Master in wireless network security systems in depth and perform effective network management.

Course Code	UE23CS343AB7	Course Title	ROS for Autonomous Systems				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	5	Type of Course	Elective - 2				
AI Tools /Tools/Languages	C++,Python, Java, ROS, Rviz, Gazebo and Webots..	Desirable Knowledge					
Prelude	To introduce the foundational principles of autonomous systems, focusing on mobile robotics and programming using ROS (Robot Operating System). The course integrates theoretical understanding with simulation-based hands-on experience.						
Course Objectives:	- To introduce students to the fundamentals of autonomous robotics, including key components such as sensors, actuators, and controllers. - To familiarize students with the Robot Operating System (ROS) environment, including node creation, topic communication, and workspace setup. - To enable practical skills in integrating and programming common sensors and actuators in simulated robotic platforms. - To develop an understanding of robot perception, environment modeling, and mapping techniques such as SLAM. - To provide exposure to robot navigation using ROS frameworks and modern learning-based methods like reinforcement and imitation learning.						
Course Contents	Unit 1: Introduction to Autonomous Robotics & Simulation Tools-What is a robot? Definitions, types, and criteria, Sensors, actuators, controllers, Autonomous vs. telerobot systems; Environment classification. Overview of ROS, its architecture, and installation, RSVP: Robot Scenario Visual Planning; Hands-on Activities: Installing ROS and setting up workspace, Creating and launching ROS nodes (publisher/subscriber) 14 Hours Unit 2: Sensor Integration, Motor Control and Simulation -Programming sensors: Color, ultrasonic, IMU, camera, Motor/servo characteristics and programming, Sensor packages and interfacing in ROS; Hands-on Activities: Simulating robot with LiDAR/camera in Gazebo, Sensor visualization with RViz and ROS topics, Motor movement control in a ROS simulation 14 Hours Unit 3: Perception, Mapping, and Environment Modelling-Environment perception: Coordinate frames, TFs, Introduction to mapping: occupancy grids, SLAM, RSVP and SPACES methodology for decision making; Hands-on Activities: Building a 2D map using SLAM, Simulating perception using TurtleBot3 in Gazebo 14 Hours Unit 4: ROS Navigation-Localizing the robot in a map, ROS Navigation Stack-hardware requirement-navigation packages, path planning, motion planning of robot – software requirement and configuration.						

	Navigation using Reinforcement learning and Imitation learning. 14 Hours
Text Book(s):	1: Lentin Joseph, Robot Operating System (ROS) for Absolute Beginners: Robotics Programming MadeEasy, 1st Edition, APress, 2018. 2: Jonathan Cacace; Lentin Joseph, Mastering ROS for Robotics Programming: Design, build, and simulate complex robots using the Robot Operating System, 2nd Edition, Packt Publishing, 2018.
Reference Book(s):	1: Hughes, C. and Hughes, T., Robot programming: a guide to controlling autonomous robots. Que Publishing, 2016 2: Quigley, M., Gerkey, B. and Smart, W.D., Programming Robots with ROS: a practical introduction to the Robot Operating System. " O'Reilly Media, Inc.", 2015 Recommended Materials: Robot Operating System (ROS): http://wiki.ros.org/ROS/Tutorials ROS2 tutorials: https://docs.ros.org/en/foxy/Tutorials.html
Course Outcome	• Identify and describe the essential components and classifications of autonomous robotic systems. • Configure and implement ROS-based simulations involving sensor-actuator integration using standard tools like Gazebo and RViz. • Apply SLAM and environment modeling techniques for autonomous robot perception and mapping in simulated environments. • Develop and test navigation algorithms using the ROS Navigation Stack with path and motion planning capabilities. • Demonstrate foundational knowledge of learning-based navigation strategies such as reinforcement and imitation learning in robotics.

Course Code	UE23CS343AB8	Course Title	Speech and Text Processing				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective - 2				
AI Tools /Tools/Languages	Tensorflow, Scikit Learn, Python 3.x. CoreNLP, Natural Language Toolkit (NLTK), TextBlob, Gensim, SpaCy, PyTorch-NLP , OpenNLP, Hugging Face, OpenAI API.	Desirable Knowledge	Machine Learning				
Prelude	The goal of this course is to focus on the processing of text and speech data as found in natural language usage. The key problem discussed in this course is understanding the meaning of text and speech through various types of learning models, including recent approaches using deep learning, and the significance of the NLP and speech processing pipeline in that meaning disambiguation process. The course also covers disambiguation of syntax as a step in the meaning disambiguation process. Additionally, the course introduces speech processing concepts such as feature extraction and classification, which play a critical role in analyzing and interpreting speech data. This course requires the student to have a desirable knowledge of Machine Intelligence.						
Course Objectives:	● Introduce text and speech processing concepts, ambiguity, and language models. ● Explore lexical semantics, vector models, and speech feature extraction. ● Analyze sequence modeling, parsing, dialog systems, and speech recognition. ● Understand transformers, LLMs, fine-tuning, and prompt engineering.						
Course Contents	Unit 1: Introduction to Text and Speech Processing Introduction, Types of ambiguity in natural language processing, Text normalization, Morphological parsing of words – Porter stemmer, Lemmatization and Stemming, Sentence segmentation. Noisy Channel model: Real world spelling error, Minimum edit distance algorithm. Language Model: N- grams, ngram language model, smoothing, discounting and back-off, Kneser-Ney smoothing, interpolation, perplexity as an evaluation measure. Introduction to Speech Processing, and Phonetics. 14 Hours Unit 2 : Lexical and Vector Semantics and speech feature extraction Word senses and relations between word senses, WordNet: A Database of Lexical Relations; Word sense disambiguation, Semantic relatedness. Vector Semantics and Embeddings: Words and vectors, TF IDF, Pointwise Mutual Information, measuring similarity, using syntax to define a word’s context, Evaluating vector models, Dense vectors, Feature extraction for speech, Neural Embedding: skip gram and CBOW Pre-trained word representations: Word2Vec, Improving Word2vec, Limitation of distributional methods. 14 Hours Unit 3: Handling sequences of text and speech POS Tagging example, Sequence labelling applications – Named Entities and Named Entity Tagging, Constituency parsing: Ambiguity presented by parse trees, CCG Parsing, Partial parsing – chunking. Statistical Parsing Probabilistic Context Free Grammar, Probabilistic CKY parsing of PCFG, Problems with PCFG, Probabilistic Lexicalized CFG , Evaluating parsers. Dialog systems and chatbots (15-1,3,4) speaker Diarization, Speech recognition architecture (15.3) 14 Hours						

	Unit 4 : Coreference resolution, Transformers and Pretrained Language Models Coreference resolution: Forms of referring expression, algorithms for coreference resolution – mention pair and mention ranking model, mention detection, classifiers using hand-built features. Self-Attention Networks: Transformers, Transformers as Language Models, Sampling, Pretraining Large Language models. (Top notch/trending LLM model to be discussed during the course delivery), LLM's Vs SLM's. Fine-Tuning and Masked Language Models: Bidirectional Transformer Encoders (BERT), Training Bidirectional Encoders, RoBERTa. In Context Learning, Instruction Fine Tuning, Prompt Engineering. 14 Hours
Text Book(s):	1: "Speech and Natural Language Processing", Daniel Jurafsky and James H. Martin, 3rd edition online, Jan 2025. The more up to date 3rd edition draft is available http://web.stanford.edu/~jurafsky/slp3/ . (text book revision is continuous process according to author)
Reference Book(s):	1: "Introduction to Natural Language Processing", Jacob Eisenstein, MIT Press, Adaptive computation and Machine Learning series, 18th October, 2019. The open source softcopy is available at githubhttps://github.com/jacobeisenstein/gtnlpclass/blob/master/notes/eisenstein-nlp-notes.pdf .
Course Outcome	● Apply text and speech processing techniques to resolve ambiguity. ● Use vector semantics and extract features from speech data. ● Implement sequence models, parsing, and speech recognition. ● Fine-tune transformers, LLMS, prompt engineering and apply coreference resolution techniques for text and speech data.

Course Code	UE23CS351B	Course Title	Cloud Computing				
Program	Core	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	6	Type of Course	Core				
AI Tools /Tools/Languages	Amazon AWS (or equivalent), AWS SkillBuilder, AWS Educate, Qwiklabs, Docker, Kubernetes, Jenkins, Zookeeper, Github, NoSQL database, Flask, Python, Go Lang. AWS Sagemaker	Desirable Knowledge	Computer Networks, Operating Systems.				
Prelude	The cloud computing course introduces not only the various technologies that go into building a cloud native application, but also how cloud systems are designed. The student is introduced to various tools and design techniques/trade-offs. It also gives a flavour for the business relevance/ethics of using cloud computing. This course requires the student to have a desirable knowledge of Computer Networks and Operating System.						
Course Objectives:	● Introduce the rationale behind the cloud computing revolution and cloud native application architecture ● Explore the concepts of Virtualization and Containerization. ● Explore the concepts of Distributed storage and its various techniques ● Design distributed systems for scalability and expose the student to various trade-offs in designing cloud architectures.						

Course Contents	Unit 1 : Cloud Programming Models Parallel computing, Grid computing, Introduction to Cloud Programming Models and service Models, Introduction to technology challenges with Distributed & Cloud computing, Business Drivers - deployment models, Multi-cloud, Cloud architecture and IaaS programming model, Web Services and REST, PaaS Programming Model, Communication using Message queues- Pub Sub model, SaaS Programming model – Microservices and differences with the traditional monolithic model; challenges of migrating monolithic applications. Microservice transactions with Saga pattern 14 Hours Unit 2 : Virtualization Hypervisor - Types, Para virtualization and Transparent virtualization, Software - Trap and Emulate virtualization, Software - Binary translation, Goldberg Popek principles for Virtualization, Hardware - AMDv/Intel, Memory - Shadow page tables, Memory - Nested page tables, IO, VM Migration, Lightweight Virtualization - Containers and Namespaces, Deployment of cloud native applications through Docker – Unionfs, Orchestration and Kubernetes, DevOps, Jenkins pipeline 14 Hours Unit 3 : Distributed Storage Types of Cloud storage - Block, Object stores, Replication, lag, multileader replication, Leaderless replication, Partitioning - key-value data, Consistent hashing, Partitioning - rebalancing partitions, Request routing, Consistency Models, CAP Theorem, Transactions, Two-phase commit. 14 Hours Unit 4 : Cloud Controller, Performance, Scalability and Security Master-slave v/s p2p models, Resource allocation, Scheduling algorithms, Cluster coordination – consensus, Fault Tolerance - faults and partial failures, Failure detection - checkpointing and application recovery, Unreliable communication, Cluster coordination - leader election, distributed locking, Case Study: Zookeeper/Raft - distributed consensus infrastructure. Edge Computing and Fog Computing – principles and paradigms Scaling computation - reverse proxies, Scaling computation - hybrid cloud and cloud bursting, Multitenancy, Multitenant databases, Cloud security requirements - physical/virtual security, security design patterns, Container security features to work with Docker and Kubernetes, Authentication in the cloud: Keystone/IAM, Cloud Threats – DoS, Economic Denial of Sustainability. 14 Hours
Laboratory	1. Migrating a monolithic e-commerce application to a microservices architecture 2. Building a Task Management Application with Raft Consensus Algorithm and MySQL. 3. Microservices communication using RabbitMQ 4. Building a Distributed Key-Value Store with etc 5. Back Up service using docker and Kubernetes 6. Building an E-commerce Microservices Application on Cloud using Docker, Kubernetes, Jenkins, and Git
Text Book(s):	1: “Distributed and Cloud Computing”, Kai Hwang, Jack Dongarra, Geoffrey Fox. ISBN: 978-0-12- 385880-1, Morgan Kaufmann, 2012. 2: “Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems”, Martin Kleppmann. O&#39;Reilly, 2017. 3: “Fog and edge computing : principles and paradigms” / edited by Rajkumar. Buyya, Satish Narayana Srirama. Description: Hoboken, NJ, USA : John Wiley & Sons
Reference Book(s):	1: Docker in Action”, Jeff Nickoloff, Manning Publications, 2016. 2: “Cloud Native DevOps with Kubernetes”, John Arundel and Justin Domingus, OReilly, 2019. 3: “Moving to the clouds: Developing Apps in the new world of cloud computing”, Dinkar Sitaram and Geetha Manjunath. Syngress, 2011.

Course Outcome	• Comprehend the technical and business rationale behind cloud computing. Decide the model of cloud computing to use for solving a particular problem. • Implement Micro service architecture through Containers and Orchestration tools. Analyse virtual machines and containers. • Experiment with cloud storage Models such as object stores, key value stores. • Apply the critical constraints such as Performance, scalability and security to the designed distributed system.
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Course Code	UE23CS352B	Course Title	Object Oriented Analysis and Design				
Program	Core	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	2	5	5
Semester	6	Type of Course	Core				
AI Tools /Tools/Languages	Star UML, Object Oriented Programming Language(Java)	Desirable Knowledge	Java				
Prelude	In this course students will learn to perform Analysis on a given domain and come up with an Object-Oriented Design (OOD). Various techniques will be discussed and practiced which are commonly used in analysis and design phases in the software industry. Unified Modelling Language (UML) will be used as a tool to demonstrate the analysis and design ideas and an object-oriented programming language such as Java would be used to implement the design. The theory is supplemented with implementations which are demonstrated/ practiced in class which provides the hands-on experiences of implementing the patterns.						
Course Objectives:	- Familiarize students with static and dynamic models of object-oriented analysis and modelling using the unified modelling language (UML) - Introduce students to object oriented programming concepts in Java. - Make students appreciate the importance of system architecture design in software development. - Introduce the students to understand the importance of GRASP and SOLID design principles.						
Course Contents	Unit 1: Advanced OO, Object Oriented Analysis and Static Models and Diagrams OOA: Requirements, Modelling and Analysis, Introduction to UML, Use Case Modelling: Use Cases Diagrams.Class Modelling: UML Class Diagrams, OO relationships, CRC Diagrams, Component model, Deployment model,Activity Modelling: UML Activity Diagrams and Modelling, Guidelines. Behaviour Modelling: Sequence Diagram,UML State Machine Diagrams and Models, Advanced State Models 14 Hours Unit 2: Object Orientated Programming and Architecture design Object-oriented Programming: JVM, Abstraction, Encapsulation, Composition, Class Attributes, Behaviour, Objects, and Methods. Interface and Implementation: Role of Constructors and Destructors, Garbage Collector,Parameter Passing, Value Type and Reference Type, Overloading of Methods, Class Attributes and Behaviour:Difference between Class Methods and Instance Methods, Inheritance: Concepts of Single Rooted Hierarchy andInterface, Abstract Class in Programming Languages, Object Class in Java, Collection, Array, List and Stack; OO Development process, System Design and Frameworks, Architectural patterns, MVC architectural pattern 14 Hours Unit 3: Design principles GRASP and its application to Object Design, Creator, Information Expert, Low Coupling, Controller, High Cohesion, Polymorphism, Pure Fabrication, Indirection and Protected Variations .SOLID:						

	Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, Dependency Inversion OO Design Principles and Sample Implementation of Patterns in Java. Introduction to Design Patterns, selection and usage of a design pattern, Creational Design Patterns Theory and Implementation in Java: Singleton, Factory, Builder and Prototype 14 Hours Unit 4: OO Design Patterns & Anti-Patterns Structural Patterns – Adapter, Façade, Proxy and Flyweight Behavioral Patterns – Chain of Responsibilities, Command, Interpreter, Iterator. Anti-patterns – Introduction and classification, Project Management, Architecture and Development anti-patterns (1 anti-patterns of each type) 14 Hours
Laboratory	• Lab assignment on Use case diagram. • Lab assignment on Class diagram. • Lab assignment on Activity and State diagrams. • Lab assignment on Java fundamentals. • Lab assignment on Java Advanced features (Inheritance, Composition, etc.). • Self-Learning Assignment on Java Serialization and Multithreading • Hands-on Assignment on MVC Framework. • Assignment on Design Patterns. • Mini Project using MVC Framework and incorporating all learning of the course.
Text Book(s):	1: “Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development”, by Craig Larman, 3rd Edition, Pearson 2015. 2: “Software Architecture Patterns” by Mark Richards, 2nd Edition, OReilly 2022
Reference Book(s):	1: “Object-Oriented Modelling and Design with UML”, Michael R Blaha and James R Rumbaugh, 2nd Edition, Pearson 2007. 2: “Design Patterns: Elements of Reusable Object-Oriented Software” by Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides, 1st Edition, Pearson 2015. 3: “Java the Complete Reference”, Herbert Schildt, McGraw-Hill, 11th Edition, 2018.
Course Outcome	• Construct static models, use cases and class models, followed by analyzing the dynamics of the system using activity, sequence, state and process models. Depict the architecture of a software system by using component and deployment models • Use the concepts of classes and objects of object-oriented programming in Java to model a complex system • Use GRASP and SOLID principles in the design of software application and apply Creational software design patterns for variety of application scenarios • Apply Structural and Behavioural software design patterns for variety of application scenarios. Understand Anti- patterns

Course Code	UE23CS341B	Course Title	Compiler Design				
Program	Core	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Core				
AI Tools /Tools/Languages	Lex/flex and YACC/Bison. Compiler Explorer + AI Explanation	Desirable Knowledge	Data Structures and its Application, Automata Formal Languages and Logic				
Prelude	Language design and implementation is an active topic in programming, and will likely always be. How we program and the tools we use, changes constantly. We try new ideas and come up with better or alternative approaches frequently. Any language that doesn't continue to adapt will fall into disuse, and any tool chain that remains stagnant will be forgotten. Hence knowledge of compilers in order to tweak these changes in the language design is a must for a Computer Science Engineer. This course requires the student to have a desirable knowledge of Data Structures and its Applications and Automata Formal Languages and Logic.						
Course Objectives:	Introduce the major concept areas of language translation and compiler design. - Develop a greater understanding of the issues involved in programming language design and implementation. - Provide practical programming skills necessary for constructing a compiler. Develop an awareness of the function and complexity of modern compilers. - Provide an understanding of the importance and techniques of optimizing code from the compiler's perspective.						
Course Contents	Unit 1: Compilers: Introduction, Lexical Analysis, Top-down Parsers The Language Processing System, The Phases of a Compiler, The Grouping of Phases into Passes. Lexical Analysis: The Role of the Lexical Analyzer, Input Buffering, Specification of Tokens, Recognition of Tokens, Design of a Lexical Analyzer Generator. The role of the parser, Syntax Error Handling, and Error-Recovery Strategies. Top-down parsing: LL (1) Parser. 14 Hours Unit2: Syntax Analysis: Bottom-up Parsers, Syntax-Directed Translation Bottom-up parsing: Shift-Reduce Parsing, viable prefixes, CLR, LALR, Error recovery for Bottom-up parsers Syntax- directed definitions, Evaluation orders for SDD's: S-attributed SDD, L-attributed SDD. 14 Hours Unit3: Implementation of Syntax-Directed Translation Schemes and Intermediate Code Generation Applications of Syntax-Directed Translation - SDD for Syntax Trees, Expressions, Basic Types and flow control statements, Type Checking, Syntax-directed Translation Schemes – Parser Stack Implementation of Postfix SDT's, SDT's with actions inside Productions, SDT's for L-Attributed Definitions. Implementing L-Attributed SDD's: Bottom-Up Parsing. Variants of Syntax Trees –						

	Directed Acyclic Graphs for Expressions, Three-Address Code – Addresses and Instructions, Quadruples, Triples, Indirect Triples, SSA Form, Control Flow Graph. 14 Hours Unit 4: Machine Independent Code Optimization, Code Generation and Run Time Environment Machine Independent Optimization: Different Optimizations, Optimization of Basic Blocks. Data Flow Analysis: Live-variable analysis, Next-use algorithm. Storage Organization, Different Allocation Strategies, Stack Allocation of space, Access to Non-local Data on the stack. Code Generation: Issues in the design of a code generator, the target language, addresses in the target code, static allocation, stack allocation, run-time addresses for names. A Simple Code generator - The Code generation algorithm, Register allocation problem (Graph coloring) 14 Hours
Text Book(s):	1: “Compilers–Principles, Techniques and Tools”, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffery D. Ullman, 2nd Edition, Pearson Education, 2009.
Reference Book(s):	1: “Modern Compiler Design”, Dick Grune, Kees van Reeuwijk, Henri E. Bal, Criel J.H. Jacobs, Koen Langendoen, 2nd Edition, 2012. 2: AW Appel, J Palsberg, Modern Compiler Implementation in Java, Cambridge University Press, 2002.
Course Outcome	• Use the knowledge of patterns, tokens, and regex for solving the problems in the field of data mining • Analyze and design the semantic behavior of a compiler. Choose the appropriate compiler internal representation for different kinds of compiler tasks. • Translate a source-level language into a low-level compiler internal representation. • Optimize the performance of a program in terms of speed and space using new code optimization techniques.

Course Code	UE23CS342BA1	Course Title	Supply Chain Management for Engineers				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	SCM applications and tools, ML tools, Case studies, Web resources	Desirable Knowledge	Enterprise Business Systems.				
Prelude	This course explores supply chain management (SCM) processes. It aims for students to develop the skills to understand, design and evaluate techniques in the realm of supply chain management. It covers fundamentals relevant to SCM, current SCM challenges and trends in the industry including planning and prediction, techniques and tools using AI technologies.						
Course Objectives:	of this course, the student should be able to, • Understand the fundamentals of supply chain management. • Learn the various SCM stages and strategies of planning, sourcing, inventory, distribution and logistics. • Explore case studies of SCM applications to understand various industrial scenarios. • Explore current trends and AI techniques applied to solve real-world SCM challenges.						
Course Contents	Unit 1: Introduction to SCM Overview of supply chain management and it’s key Objectives. Understand supply chain scope, strategic fit, its drivers and performance metrics. Building a strategic framework to analyze Supply Chains. Managing Material and Information flows. Intro to supply chain networks and their design. Various strategies of planning, sourcing, inventory, distribution and logistics related to SCM – Opportunities, Competitive advantage. Value Stream Mapping. Key Drivers - Inventory, Transportation/Logistics, Facilities (Warehousing, Manufacturing plants, Docks), Sourcing/Procurement, Pricing Strategies. 14 Hours Unit 2: Procurement and Supply Chain Planning (Demand & Supply) Procurement processes, Supplier selection and performance assessments. Intro to Demand Forecasting models, predictive analytics for supply chain planning. Techniques for aggregate planning of demand and supply in supply chains, managing demand variability and accuracy. Case studies. 14 Hours Unit 3: Planning and Managing Inventory and Logistics Management in SCM Introduction to planning and management of Inventory in an enterprise. Techniques for warehouse management and logistics management for distribution. Logistics stakeholders and modern logistics concepts, Outsourcing logistics, Logistics KPIs like On-Time Delivery, Fill Rate, Inventory Turns, etc. Intro to Reverse Logistics, and Lean Logistics. Enhancing transportation and logistics with real-time tracking and monitoring using IoT and AI-driven route planning and last-mile delivery. Case studies in inventory and logistics management. 14 Hours						

	Unit 4: Drivers, Analytics and Current Trends in Supply Chain Management Understanding and managing cross functional drivers in supply chain. Decision making in supply chains - sourcing, pricing, risk management and compliance. Basics of AI, Blockchain, IoT concepts applied to SCM. Current trends in SCM: E-procurement, Sustainability, Green SCM, Lean and Global SCM, Real-time, Ethical and Transparent Supply Chains, SCM Analytics, Digital Twin Tech. Exploration of AI & Gen-AI techniques, tools and technologies for enhancing SCM like AI driven Demand Forecasting – Case studies. Awareness of current challenges and research areas of SCM in the industry like Ethical and transparent operations. 14 Hours
Text Book(s):	1: Supply Chain Management: Strategy, Planning and Operation, Sunil Chopra, Pearson, 7th Revised Edition, 2024. 2: Supply Chain Logistics Management, Donald Bowersox, et al, Mc-Graw Hill Publishing, 5th Edition, 2024.
Reference Book(s):	1: Essentials of Supply Chain Management, Michael H Hugos, John Wiley & Sons, 3rd Edition, 2018. 2: Principles of Supply Chain Management: A Balanced Approach, Joel D Wisner, Cengage, 5th Ed, 2019 3: Strategic Supply Chain Management: The Five Core Disciplines for Top Performances, Cohen Shoshanah & Joseph Roussel, Mc-Graw Hill, 2nd Ed, 2013. 4: Logistics and Supply Chain Management, Martin Christopher, Pearson, 2022 5: Supply Chain Management: A Logistic Approach, John J Coyle, Cengage, 10th Ed, 2019. 6: A Roadmap to Green Supply Chains: Using Supply Chain Archaeology and Big Data Analytics, Kevin L Lyons, Industrial Press, 2015 7: Lean Supply Chain and Logistics Management, Paul Myerson, Mc-Graw Hill, 2012. 8: Supply Chain Analytics, T. A. S. Vijayaraghavan, Wiley, 2021.
Course Outcome	• Identify various stages of SCM including Logistics. • Evaluate KPIs and select suitable strategies for optimized supply chain performance. • Evaluate and select the strategies for planning and managing Inventories and Logistics. • Apply ML techniques to predict trends, optimize inventory, and improve logistics.

Course Code	UE23CS342BA2	Course Title	Algorithms for Information Retrieval				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	Scikit, Tensorflow, Solr, Lucene Search Engines/ Python Programming Languages	Desirable Knowledge					
Prelude	The course explores the data analytics lifecycle: question formulation, data collection and cleaning, exploratory, analysis, visualization, statistical inference, prediction, and decision-making. Focuses on building analytical models using key principles and techniques. This course requires the student to have a desirable knowledge of Statistics for Data Science.						
Course Objectives:	- Understand the architecture, models, and algorithms used in Information Retrieval. - Understand the basic principles and implementation of Indexing and Search. - To explore machine learning techniques for Information Retrieval, Web, and multimodal applications. - To study neural models for Question Answering and recent advancements in the field.						
Course Contents	Unit-1: Basics of Information Retrieval Introduction to Information Retrieval, Background, Architecture and Strategies of Information Retrieval (IR) Systems, IR Models, Boolean and Extended Boolean Models, Dictionary, Vocabulary, Positional Postings, Phrase Queries and Tolerant Retrieval, Indexing and Vector Space Model, Evaluation of IR Algorithms for Indexing and Index Compression, Vector Space Model for Scoring, tf-idf and Variants. 14 Hours Unit-2: Ranking, Web Search Basics and Link Analysis, Efficient Scoring and Ranking, Parametric and Zone Indexes, Tiered Indexes, Query Term Proximity, Query Parser, Aggregating Scores, Performance Measurement, Web Applications and Search Algorithms, Relevance Feedback, Query Expansion, Other IR Models, Web Search Basics. Economic Model of Web Search, Improving Search Results, Link Analysis, The Page Rank Algorithm, Other Search Algorithms, Scalability Issues in Search. Search User Experience, Web Crawling and Indices, Link analysis, Building a Complete Search System, Lucene as a Search Engine, and Other search engines like Solr, Everything, and Google 14 Hours Unit-3: Multimodal Information Retrieval Multimodal Information Retrieval content-based visual information retrieval: metadata searching, Query by Example, Semantic Retrieval, Machine Learning Approaches, content comparison using image distance measures. Multimedia and Multimodal Information Retrieval: Introduction, Requirements, Applications, challenges, Architecture, Metadata, Techniques for						

	Content Processing. Examples of MIR Query languages, Examples of research and commercial MIR solutions 14 Hours Unit-4: Question Answering, neural models for IR Question Answering, neural models for IR Question Answering, Neural models for IR QA as an Information Retrieval task, Factoid QA models, Entity Linking models, Knowledge-based QA, and Pretrained models for QA. Recent and trending research papers in QA. 14 Hours
Text Book(s):	1: "Introduction to Information Retrieval", Christopher D. Manning, Prabhakar Raghavan, Hinrich Schutze, ISBN: 9781107666399, Cambridge University Press, 2009. 2: "Search Computing", Challenges and Directions: chapter 8: Multimedia and Multimodal Information Retrieval. ISBN: 9783642123108, 3642123104. Marco Brambilla, Stefano Ceri. Springer Berlin Heidelberg, 2010 3: "Speech and Language Processing", Third Edition, Daniel Jurafsky, James H. Martin, Chapter 14: Question Answering and Information Retrieval, 2023.
Reference Book(s):	1. "Algorithms of the Intelligent Web", Haralambos Marmanis, Dmitry Babenko, Manning Publishers, 2011
Course Outcome	• Implement an efficient index for document collection. • Perform searches on a document collection, rank, and evaluate results. • Apply machine learning techniques for Information Retrieval, Web, and multimodal applications. • Build and evaluate modern Question Answering systems.

Course Code	UE23CS342BA3	Course Title	AI for Sustainable development				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	OpenAI API/ChatGPT, Streamlit, Google Earth Engine, UN SDG Tracker, Google Colab.	Desirable Knowledge	Basics of AI and Machine Learning				
Prelude	This course explores the transformative potential of Artificial Intelligence (AI) in achieving the United Nations Sustainable Development Goals (SDGs). It integrates core AI methodologies with sustainability applications across sectors like climate, agriculture, health, energy, and smart cities. Students will gain insights into ethical AI practices, responsible innovation, and the design of intelligent systems to address complex global challenges. Through case studies, hands-on, and interdisciplinary discussions, the course empowers learners to apply AI not only as a tool of automation but as a driver for inclusive, equitable, and sustainable development.						
Course Objectives:	- To introduce the intersection of Artificial Intelligence and Sustainable Development Goals (SDGs). - To equip students with the technical knowledge of AI tools applicable to environmental and societal sustainability. - To develop the ability to model, analyse, and solve sustainability challenges using AI algorithms and data-driven insights. - To foster an understanding of ethical implications and policy considerations in deploying AI for social good.						
Course Contents	Unit 1: Foundations of Sustainable Development and AI Introduction to the UN Sustainable Development Goals (SDGs), Role of AI in addressing SDGs, AI lifecycle and ethical considerations, Basics of Machine Learning and AI techniques relevant to sustainability, Types of data in sustainability (satellite, sensor, survey, etc.), Responsible AI and environmental impact of AI models Case Studies: Google’s AI for Social Good, IBM’s Green Horizon Project 14Hours Unit 2:Foundations of Sustainable Development and AI Introduction to the UN Sustainable Development Goals (SDGs), Role of AI in addressing SDGs, AI lifecycle and ethical considerations, Basics of Machine Learning and AI techniques relevant to sustainability, Types of data in sustainability (satellite, sensor, survey, etc.), Responsible AI and environmental impact of AI models Case Studies: Google’s AI for Social Good, IBM’s Green Horizon Project 14 Hours						

	Unit 3: AI in Smart Cities, Energy, and Waste Management AI for traffic optimization and urban mobility, Energy consumption prediction and smart grid optimization, AI in renewable energy (e.g., solar/wind prediction), Waste segregation and recycling using computer vision, Internet of Things (IoT) and AI integration in smart city infrastructure. Case Study: <i>Google DeepMind – Energy Efficiency in Data Centers</i> 14Hours Unit 4: AI for Social Good, Health, and Inclusive Development AI for early disease detection in under-served areas, AI in education: personalized learning and dropout prediction, Financial inclusion through AI-driven credit scoring, Gender and racial bias in AI: detection and mitigation, Policy frameworks and global initiatives using AI. Case Study: AI-Powered Health Diagnostics – Aravind Eye Care System, India 14Hours
Text Book(s):	1. <i>Artificial Intelligence for Sustainable Development</i> by Valli Manickam & R. Venkatesh CRC Press (2023) 2. <i>Artificial Intelligence and Conservation</i> by Fei Fang, Milind Tambe, Bistra Dilkina Cambridge University Press (2021)
Reference Book(s):	1. <i>Sustainability and Artificial Intelligence: A Societal Perspective</i> by Henrik Skaug Sætra, Routledge (2022) 2. <i>Artificial Intelligence for a Better Future: An Ecosystem Perspective on the Ethics of AI and Emerging Digital Technologies</i> by Bernd Carsten Stahl, Springer (2021) 3. <i>Smart Cities and Artificial Intelligence: A Sustainable Development Perspective</i> by Christopher Grant Kirwan, Fu Zhiyong, Elsevier (2020) 4. <i>AI and Climate Change: The Role of Artificial Intelligence in Tackling the Climate Crisis</i> by David A. L. Swanson World Scientific (2023)
Course Outcome	• Analyse the role of AI in various SDGs and identify real-world applications across environmental and social domains. • Apply supervised and unsupervised learning techniques to sustainability-related datasets using contemporary AI tools. • Design and evaluate intelligent systems for climate monitoring, smart cities, energy optimization, and public health. • Demonstrate awareness of ethical AI development, responsible data usage, and sustainability-focused innovation practices.

Course Code	UE23CS342BA4	Course Title	Generative AI and its Applications				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	Python, Hugging Face, LM Studio, Kaggle, Google Colab, DeepEval, TruLens	Desirable Knowledge	Machine Learning				
Prelude	Large Language Models (LLM) in Generative AI represent a significant leap forward in the field of artificial intelligence, leveraging vast amounts of data and advanced algorithms to understand, generate, and interact with human language in a highly sophisticated manner. These models are capable of producing human-like text, understanding context, and even creating content that is coherent and contextually relevant. Their applications are diverse and far-reaching, encompassing areas such as customer service automation, where they can handle inquiries and provide support; content creation, where they generate articles, stories, and marketing copy; and education, where they assist in tutoring and personalized learning experiences and many more applications						
Course Objectives:	● Gain an in-depth understanding of Large Language Models (LLMs), their architecture, evolution, and applications across different domains. ● Develop hands-on skills in implementing and fine-tuning LLMs for various NLP tasks, using state-of- the-art tools and frameworks. ● Learn and apply advanced techniques in prompt engineering, retrieval-augmented generation (RAG), and multimodal LLMs to solve complex problems. ● Understand and address the ethical implications, potential biases, and privacy concerns associated with the use of LLMs.						
Course Contents	UNIT 1: NLP Basics, Pre-Trained Models Introduction to GenAI and its Applications, Large Language Models (LLMs), Evolution of LLMs. Tokenization-Introduction, Techniques-BPE, Word Piece, Sentence Piece, Numerical Problems. Word Embeddings - Introduction, Types of Word Embeddings: TF-IDF, Word2Vec - CBoW & SkipGram, GloVe, Numerical Problems. Transformer Anatomy and Architecture, Numericals on Self and Cross Attention Score computation. Introduction to Pre-trained Models, Exploring BERT and GPT Architectures, Variants of BERT: ELMo, ROBERTa, BART. NLP Basics: Terminologies, N-Gram Models, POS, NER. NLP Tasks: Text Classification, Summarization, Question Answering, Machine Translation. 14 Hours UNIT 2: Prompt Engineering and RAG Introduction to Prompt Engineering – Overview and Applications. Prompt Structure, Prompt Tuning, Hard and Soft Prompts, Prompt Engineering Techniques - Zero Shot, One Shot, Few Shot Prompting, Chain of Thought, Tree of Thought, Graph of Thought Prompting, Vector Stores, RAG- Introduction, Need for RAG, Naïve RAG, Sparse Retrieval(BM25), Challenges Indexing – Flat Index(Brute-force Search), Inverted File Index (IVF), Hierarchical Navigable Small World (HNSW), Product Quantization (PQ) 14 Hours						

	UNIT 3: Advanced RAG and Fine Tuning Advanced RAG- Introduction, Need, Retrieval Enhancement Techniques – Introduction, Sparse Retrieval (BM25), Dense Retrieval (ColBERT, SBERT), Hybrid Retrieval (BM25+SBERT), Re-Ranking, Query Expansion), Numericals on Hybrid Search, Generation Enhancement Techniques- Introduction, Techniques. Fine Tuning - PEFT: Introduction to LLM Fine Tuning, Data Resolution/Precision, Quantization, Generative AI Lifecycle, LLM Pre-training, Fine Tuning Principles, Fine Tuning using LoRA and QLoRA, Prompt Tuning, Instruction Fine Tuning. Mixture of Experts (MoE)- Introduction, Need, Components, Bias and Toxicity Evaluation. 14 Hours UNIT 4: Multimodal LLMs, Agentic Workflow, Evaluation Introduction to Multimodal Models, Exploring CLIP and BLIP, Whisper AI–Architecture, Applications. Introduction to Agentic Workflows, Design Patterns, Frameworks for building Agentic Workflow, LLM Evaluation Techniques using DeepEval and TruLens, Data behind LLM, LLM Ethics and Security. Recent Developments in LLM. 14 Hours
Text Book(s):	1. Quick Start Guide to Large Language Models: Strategies and Best Practices for Using ChatGPT and Other LLMs by Sinan Ozdemir, Addison-Wesley Professional, Edition 1 2023
Reference Book(s):	1: Pretrain Vision and Large Language Models in Python: End-to-end techniques for building and deploying foundation models on AWS, by Emily Webber, packtpublishing, Edition 1, 2023 2. Transformers for Natural Language Processing: Build innovative deep neural network architectures for NLP with Python, PyTorch, TensorFlow, BERT, RoBERTa, and more by Denis Rothman (Author) ,publishing, Edition 1, 2023 3. Understanding Large Language Models: Learning Their Underlying Concepts and Technologies by Thimira Amaratunga, APress, Edition 1 , 2023 4. Retrieval-Augmented Generation (RAG): Empowering Large Language Models (LLMs) by Dr. Ray Islam (Author) ,Edition 1, 2023 5. (Paper) Generative Adversarial Nets (2014) by Ian J. Goodfellow 6. (Survey Paper) An Overview of Diffusion Models: Applications, Guided Generation, Statistical Rates and Optimization. 7. (Paper) A Study of BFLOAT16 for Deep Learning Training. 8. (Paper) Using Soft-Prompt Tuning to Evaluate Bias in Large Language Models 9. (Paper) Improving Language Understanding by Generative Pre-Training
Course Outcome	• Demonstrate a solid grasp of LLM architecture, including transformer models like BERT, GPT, ELMo, RoBERTa, and BART, as well as advanced techniques in prompt engineering and RAG. • Successfully implement and fine-tune LLMs for specific NLP tasks, utilizing frameworks like HuggingFace, and develop multimodal applications that integrate text and image processing. • Create innovative solutions by applying advanced prompt engineering, reasoning techniques, and agentic workflows, enhancing the capabilities of LLMs in real-world applications. • Evaluate and mitigate bias, toxicity, and privacy issues in LLM outputs, ensuring responsible and ethical use of AI technologies in various applications.

Course Code	UE23CS342BA5	Course Title	Blockchain				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	Solidity, Ganache, Meta mask. RemixID, Ganache tool	Desirable Knowledge	Data Structures & Its Applications				
Prelude	This course offers a comprehensive understanding of Blockchain technology, covering its foundational concepts, data integrity, cryptographic techniques, and network structures. Students will explore key blockchain models, consensus algorithms, smart contracts, and second-generation applications like DApps and DAOs. Through hands-on demonstrations, including wallet creation and transactions, learners gain practical skills in working with platforms like Etherscan and Hyperledger Fabric. The course also addresses security challenges and real-world use cases across industries, preparing students for research and professional applications in blockchain.						
Course Objectives:	- To provide foundational knowledge of blockchain technology, including its architecture, data integrity principles, and key components such as nodes, tokens, and ledgers. - To explore various consensus algorithms and their roles in achieving distributed agreement in blockchain networks. - To introduce smart contracts and decentralized applications (DApps) using programming tools like Solidity and platforms such as Ethereum. - To examine blockchain security, vulnerabilities, and real-world use cases in sectors like finance, supply chain, and public administration.						
Course Contents	Unit 1: Introduction and Data flow and Integrity Key Blockchain Concepts, Nodes, Cryptocurrency, tokens, Public Ledger, Peer to peerNetwork,Types of Blockchain, Permissioned Blockchain model, Permission-less Blockchain model, Demonstration of Blockchain Construction steps, Demonstration of ether scan and Goerli/Sepoliaetherscan. Types of cryptography, Digital signatures, Hash functions, SHA-256, Patterns of hashing, Hash Pointer, Merkle tree. Case study: Bitcoin Blockchain Network, Creation of metamask wallets and performing transaction. 14 Hours Unit 2 : The structure of the network: consensus algorithm Introduction to distributed consensus: What, why, Challenges, Proof of Work, Proof of Stake, Delegated Proof of Stake, Proof of Authority, Proof of Elapsed Time, Proof of Scope, Proof of Space, Proof of Burn, RAFT, PAXOS, Byzantine Fault Tolerance System, PBFT. Smart contracts: origins and how they function, Creating and deploying smart contracts. Decentralized applications, Decentralized Autonomous Organizations (DAOs)- Need, Principal agent dilemma, components, The DAO Story, Legality of DAPPs and DAOs. 14 Hours						

	Unit 3: Second generation applications of Blockchain technology Solidity- Variable, Functions, Arrays/Strings, enum/structs, mapping, function visibility - public, private, internal, external, modifiers, view, pure, fallback, overloading, in-built mathematical and cryptography functions, Withdrawal pattern, Restricted Access, ether units. Any real-world problem hands-on. 14 Hours Unit 4: Blockchain Security and use cases Hyperledger Fabric: Blockchain-as-a-service (BaaS), Architecture and core components, Hyperledger fabric model, Creation of a simple DAPP. Blockchain vulnerabilities, Smart contract vulnerabilities, Blockchain on CIA security triad, Blockchain based DNS security platform, deploying blockchain based DDOS protection. Use cases: Public Sector, Finance, Supply Chain. Research Aspects in Blockchain 14 Hours
Text Book(s):	1. Introduction to Blockchain Technology, Tiana Laurence, 1st edition, Van Haren Publishing, 2019. 2. Blockchain Technology from Theory to Practice, Sudeep Tanwar, 1st edition, Springer, 2022.
Reference Book(s):	1. Hands-On Cyber security with Blockchain: Implement DDOS protection, PKI-based identity, 2FA, and DNS security using Blockchain, Rajneesh Gupta, 1st edition, 2018. 2. Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Narayanan, Bonneau, Felten, Miller and Goldfeder, Princeton University Press, 2016. 3. Drescher, Daniel. Blockchain Basics: A Non-Technical Introduction in 25 Steps. Germany: Apress, 2017.
Course Outcome	• Understand and explain the structure, types, and functioning of blockchain systems and cryptographic techniques used for data integrity. • Analyze and compare different consensus mechanisms and their applicability in distributed networks. • Design, code, and deploy smart contracts and DApps using Ethereum-based tools like Metamask and Solidity. • Evaluate blockchain's impact and applicability in various industries, along with identifying and mitigating potential security vulnerabilities.

Course Code	UE23CS342BA6	Course Title	Digital Forensics and Incident Response				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	Open source tools on Forensics. Cellebrite,Deepfake Detection Tools	Desirable Knowledge	Machine Learning				
Prelude	Digital Forensics is the core set of principles and processes necessary to produce usable digital evidence and uncover critical intelligence. This course provides a deep understanding of techniques to gather, protect, analyze, and report digital confirmations through a combination of theoretical foundation and practical exercises. The content is organized into four units, each with a focused objective and corresponding outcomes						
Course Objectives:	• Introduce the history, evolution, basic principles, challenges, and incident response techniques in digital forensics. • Develop expertise in Windows file systems, including file structure analysis, registry examination, and related forensic recovery tools. • Build foundational skills in Linux and macOS systems while integrating network forensics and browser-based analysis. • Equip students with specialized knowledge in email forensics, mobile device investigations, anti-forensics techniques, and professional report writing.						
Course Contents	Unit 1: Introduction to Digital Forensics and Incident Response Foundations of Digital Forensics: Introduction to forensic science and digital forensics, Goals, categories, forensic readiness, life cycle, and challenges, Role of the examiner; types of investigations; importance of international standards Incident Response: Attack lifecycle and introduction to incident response, Building an Incident Response Team and drafting an Incident Response Plan (IRP), Forensic tools and methodologies for initial response, Investigations Triad Evidence Acquisition and Analysis: Types of digital evidence (HDD, SSD, RAM), Data recovery, allocated vs. unallocated space, and persistence (page/swap files), Major Concepts, Essential Technical Concepts Initial responder tasks: search and seizure, documentation, packaging, and transporting evidence Labs & Activities: • Activity 1: Understanding and familiarization with Digital Forensic • Lab 1: Acquiring digital evidence (HDD/SSD and RAM) with open-source tools • Lab 2: Analysing digital evidence (HDD/SSD and RAM) with open-source tools 14 Hours Unit 2: Windows Forensics – Filesystems and Registry Analysis (14 Hours) Windows File Systems: Overview of FAT and NTFS file systems, Examination of file allocation structures and metadata, Techniques for timeline analysis and file recovery Windows Registry Analysis: Key artifacts: Deleted data, hibernation file (Hiberfile.sys), print spooling, recycle bin, metadata, thumbnail cache, Most Recently Used (MRU) lists, restore points, and shadow copies, Methods for recovering deleted registry keys and performing registry forensics Labs & Activities: • Activity 2: Case study using tools like Autopsy for practical Windows forensics • Lab 3: Data recovery and forensic analysis using EXIF Tool • Lab 4: Examination of the Windows Registry with REGEDIT and open-source forensic tools						

	14 Hours Unit 3: Linux, macOS, Network, and Browser Forensics (14 Hours) Linux Forensics: Linux file systems (Ext2/Ext3), structure layers (file, metadata, data unit), and artifacts, Concepts such as deleted data recovery, logical volume management, boot processes, and principles of file carving macOS Forensics: Exploration of OS X file system artifacts, including HFS+ structures and OS X system logs Network Forensics: Fundamentals of networking and types of networks, Techniques for securing network infrastructures, analyzing network evidence, and applying forensic science to networks, Tools and processes for the seizure and analysis of networking devices Browser Forensics: Overview of web browsers (IE, Firefox, Chrome), Forensic analysis of browser artifacts and investigation tools Labs & Activities: • Activity 3: Practical session on understanding Linux artifacts using both CLI and GUI tools • Lab 5: Analysis of PCAP files and network forensic investigations • Lab 6: Browser forensic investigations using open-source tools 14 Hours Unit 4: Email, Mobile Forensics, Anti-Forensics, and Report Writing (14 Hours) Email Forensics: Examination of email communications, protocols, and header analysis, Techniques for recovering and analyzing email evidence Mobile Device Forensics: Architecture, boot processes, and the forensic significance of cellular network data, Procedures related to SIM file systems, device locks, rooting, and jailbreaking Methods for evidence acquisition and challenges associated with mobile devices Anti-Forensics: Overview and classification of anti-forensics techniques, Practices including data wiping, shredding, obfuscation, encryption, and data hiding Report Writing: Preparing and structuring forensic reports best practices for report design, documentation, legal acceptance, and presenting digital evidence Labs & Activities: • Lab 7: Email forensics using open-source tools • Lab 8: Demonstration of mobile forensics techniques Activity 4: Collaborative report writing exercise and peer review 14 Hours
Text Book(s):	1: "The basics of digital forensics: the primer for getting started in digital forensics", Sammons, J. Elsevier, 2012. 2: "Digital Forensics Basics: A Practical Guide Using Windows OS", Hassan, N. A. A press 2019. 3: "Practical Cyber Forensics- An Incident-Based Approach to Forensic Investigations", by Niranjana Reddy, A Press, 2019. 4: "Introductory Computer Forensics-A Hands-on practical Approach", by Xiaodong Lin, Springer, 2018.
Reference Book(s):	1: "Guide to Computer Forensics and Investigations", Bill Nelson, Amelia Phillips and Christopher Steuart, Course Technology, Cengage Learning, 2010. 2: "Digital forensics with open-source tools", Altheide, C., & Carvey, H. Elsevier, 2011. 3: "Digital Forensics Workbook_-Hands-on Activities in Digital Forensics", Michael K Robinson, CreateSpace Independent Publishing Platform, 2015.
Course Outcome	• Students will understand the fundamental principles of digital forensics, be familiar with its historical evolution, and perform initial incident response and evidence acquisition. • Students will be able to analyze Windows file systems and registry artifacts effectively, perform timeline analyses, and recover deleted data. • Students will acquire and analyze forensic data from Linux and macOS environments and demonstrate proficiency in network and browser forensics. • Students will conduct mobile device and email investigations using proper tools, recognize anti-forensics practices, and compile legally sound forensic reports.

Course Code	UE23CS342BA7	Course Title	Digital Twins				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	C/C++/JAVA/Python using OpenGL. Luma Labs Genie, 3D CSM, Skybox Blockade Labs, SIMENS Digital Twin Tool.	Desirable Knowledge	Data Structures, Design and Analysis of Algorithms				
Prelude	Digital Twins are revolutionizing how we design, monitor, and optimize complex systems by creating dynamic virtual replicas of physical entities. This course introduces the foundational concepts, modeling techniques, and simulation strategies needed to build and analyze Digital Twins across domains such as healthcare, manufacturing, and transportation. Students will gain hands-on experience with tools like Unity, Blender, and OpenGL, while also exploring critical topics like 3D view synthesis and cybersecurity. By the end of the course, students will be equipped to conceptualize, build, and secure intelligent Digital Twin systems.						
Course Objectives:	- Develop fundamental understanding of Digital Twins and its architecture. With help of case studies provide a structured way to conceptualize any Digital Twin. - Develop understanding on various techniques for structural twinning and 3D view Synthesis. - To build skill in modelling-based system engineering, focusing on discrete event systems - Collaborate with peers on practical projects to gain hands-on experience in developing intelligent systems. Develop understanding about secure designs and privacy concerns in the context of Digital twin						
Course Contents	Unit 1: Digital Twin Essentials The Big Picture of Digital Twins, History of the Digital Twin, Origin of the Digital Twin concept, Digital Twin and Product Life cycle Relationship, Types of Digital Twin: Discrete Vs Composite, Product versus facility, Types of Digital Twin: Simulation versus operational, Analytics versus physics-based, Characteristics of a Digital Twin, Digital Twin Architecture, Industrial Digital Twin applications, Examples of mock, functional, and executable twins, Data Driven Modelling, Physics Driven Modelling, Hybrid (Data-Physics) Modelling, Examples of Physics Driven Modelling: DT of a Propeller of a Drone, DT of a ceiling fan, Bio-Mechanical DT of human body; Examples of Data Driven Modelling: DT for RUL of a Battery, DT for traffic Mobility in Bangalore, DT of RUL of GUT; Hybrid Modelling: Heart Digital Twin, DT for Solar Array, DT for CAR using OBD., Case Study: Manufacturing, Case Study: Healthcare, Case Study: Buildings, Case Study: Transportation & Logistics 14 Hours Unit 2: Foundations and Techniques in 3D View Synthesis Manual 3D modelling and mesh editing, Point cloud to mesh conversion, Poisson surface reconstruction, Ball-pivoting algorithm, Alpha shapes and Delaunay triangulation, Camera models and calibration, Epipolar geometry and depth estimation, Structure from Motion (SfM), Multi-View Stereo (MVS), Depth Image-Based Rendering (DIBR), Forward and backward image warping, Neural Radiance Fields (NeRF) fundamentals, Volume rendering with MLPs, 3D Gaussian Splatting for real-time rendering 14 Hours Unit 3: System Modelling and Simulation Introduction to Simulation, System and System Environment, Discrete Event System Simulation, General Principles, The Event Scheduling/Time Advance Algorithm, Input Modelling, Estimation of Absolute						

	Performance, Verification, Calibration and Validation 14 Hours Unit 4: Digital Twin & Cyber Security. Digital twins and cybersecurity, Security Framework, Digital twins threat modelling, Common attacks on digital twins, Common attacks on digital twins, Digital twin authentication and identification challenge, IDS, IPS, Authentication Methods, Communication Channel Protection, building cyber resilience in digital twins, Privacy Framework, Lack of Privacy, and trust, Privacy by Design, Enhancing trust with block chain integration. 14 Hours
Text Book(s):	1: Building Industrial Digital Twins by Shyam Varan Nath & Pieter van Schalkwyk, by Packt Publishing Ltd. 2: Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol: Discrete-Event System Simulation, 5th Edition, Pearson Education, 2010. 3: Girod, Bernd, Günther Greiner, and Heinrich Niemann, eds. Principles of 3D image analysis and synthesis. Vol. 556. Springer Science & Business Media, 2013.
Reference Book(s):	1: Diab, W. W., A. Ferraro, B. Klenz, S. W. Lin, E. Liongosari, W. E. Tannous, and B. Zarkout. "Industrial IoT Artificial Intelligence Framework." (2022): 1-59. 2: El Saddik, Abdulmotaleb, ed. Digital Twin for Healthcare: Design, Challenges, and Solutions. Elsevier, 2022.
Course Outcome	• conceptualize a Digital Twin and build its characteristics and input output parameters • use either manual or algorithmic techniques to model the structural twin of the DT. • translate the conceptualized Digital twin application into a Simulation model – a Time advanced discrete event system simulation. • collaborate effectively with peers on practical projects and demonstrate the ability to apply Digital Twin and Extended Reality solutions to real-world problems, such as improving operational efficiency, enhancing security, and reducing costs.

Course Code	UE23CS342BA8	Course Title	Cloud Security				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -3				
AI Tools /Tools/Languages	C/ C++/ JAVA/ Python .AWS cloud services	Desirable Knowledge	Computer Networks , Operating System				
Prelude	This course introduces fundamental cloud security concepts, covering threats, access management, encryption, compliance, and incident response. Through hands-on labs in AWS, students will learn to design and implement secure cloud architectures while integrating DevSecOps practices.						
Course Objectives:	• Introduce cloud computing models, security challenges, and regulatory frameworks relevant to cloud environments. • Develop understanding of secure cloud architecture, identity management, and data protection across hybrid and multi-cloud setups. • Apply secure software development practices, including DevSecOps, container security, and cloud configuration management. • Equip learners with skills for cloud threat detection, incident response, and disaster recovery planning.						
Course Contents	Unit 1: Cloud Security Introduction and Access Management Introduction to Cloud Computing & Security Challenges: Cloud Computing Models (IaaS, PaaS, SaaS), Deployment Models (Public, Private, Hybrid, Multi-Cloud), Security Challenges in Cloud Environments; Cloud Security Frameworks & Compliance: Shared Responsibility Model, Key Regulations: GDPR, SOC 2, PCI DSS, HIPAA, Cloud Security Standards: ISO 27017, NIST 800-144; Cloud Threats & Risk Assessment: OWASP Cloud Threats, Risk Management Frameworks (FAIR, NIST); Identity & Access Management (IAM) in Cloud: Role-Based Access Control (RBAC) & Attribute-Based Access Control (ABAC), Identity Federation & Single Sign-On (SSO), Multi-Factor Authentication (MFA) & Zero Trust Security, IAM in multicloud. 14 Hours Unit 2: Cloud Security Architecture & Access Control Secure Cloud Networking & Data Protection: Virtual Private Cloud (VPC) & Software-Defined Perimeters, Encryption Techniques: Data at Rest, Data in Transit, Data in Use, Cloud Key Management Systems (AWS KMS); Container & Serverless Security: Securing Docker & Kubernetes, Serverless Security Best Practices (AWS Lambda), Container Runtime Security & Image Scanning, Multi Cloud Security Architecture. 14 Hours Unit 3: Secure Software Development in Cloud Secure Software Development in Cloud: DevSecOps Pipelines & CI/CD Security; Cloud Configuration Management: Configuration drift detection across clouds, Cloud Security Posture Management (CSPM), Infrastructure as Code (IaC) & Security Automation, Supply Chain Security & Code Vulnerability Scanning; Cloud Application Security: Secure API Development & Authentication Methods, Web Application Firewalls (WAF) & API Gateways, Security Best Practices for SaaS Applications, Challenges of Multi cloud applications. 14 Hours						

	Unit 4: Cloud Threats, Incident Response & Security Operations Cloud Security Monitoring & Threat Detection: SIEM in Cloud, Intrusion Detection & Prevention in Cloud Environments, Security Logging & Monitoring (CloudTrail, CloudWatch); Incident Response in Cloud: Root Cause Analysis of Cloud Breaches, Real-World Cloud/Multi Cloud Attack Case Studies; Business Continuity & Disaster Recovery in Cloud. 14 Hours
Reference Book(s):	1: Mastering AWS Security: Create and maintain a secure cloud ecosystem by Albert Antony, Oct 2017, 1st Edition, published by Packt. 2: Practical Cloud Security by Chris Dotson Copyright © 2019 Chris Dotson. Published by O'Reilly Media. 3: Empirical Cloud Security: A Guide To Practical Intelligence to Evaluate Risks and Attacks by Aditya K. Sood, Apr 15, 2021, 1st Edition, published by Packt 4: Implementing DevSecOps Practices: Understand application security testing and secure coding by integrating SAST and DAST by Vandana Verma Sehgal, Dec 22, 2023, 1st Edition, published by Packt
Course Outcome	• Understand the core security challenges in cloud environments and apply identity and access management (IAM) principles to secure cloud resources. • Implement network security measures, encryption techniques, and secure virtualization methods. • Design secure cloud architectures, integrate DevSecOps practices to enhance security in cloud-native applications. • Analyze, detect and mitigate cloud security threats using security monitoring tools.

Course Code	UE23CS343BB1	Course Title	Heterogenous Parallelism				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languag es	Pthread, OpenMP, CUDA, Vivado HLS	Desirable Knowledge	Problem Solving with C, Microprocessor and Computer Architecture				
Prelude	This course focuses on parallel heterogeneous architecture as well as programming models and imparts pragmatic skills to program using parallel programming languages and frameworks trending in the industry.						
Course Objectives:	- Familiarize with various parallel heterogeneous architectures ILP, and Optimization techniques and cache-efficient programming. - Explore multi-core architectures, parallel algorithms, and GP-GPU-based computing models. - Learn OpenMP, memory models, and Acquaint with Concurrency Bugs and Resolution debugging techniques for parallel programming. - Study CUDA programming, hardware accelerators (FPGAs & TPUs), and to explore HLS programming						
Course Contents	Unit 1: Fine Grained Parallelism Review on Parallelism and Performance, Instruction Level Parallelism and Enhancement Techniques, Prediction and Speculation, Cache optimized Programming. Laws of Parallelism, Data, Task and Pipeline Parallelism, Pthreads. 14 Hours Unit 2: Coarse Grained Parallelism Parallel Algorithms for regular architecture – Mesh, Pyramids, Task Decompositions and Mapping, Multithreaded and Multi-Core architectures, Many-Core Heterogeneous Architectures, GPU Architectures. 14 Hours Unit 3: Parallelism Frameworks OpenMP, Race Conditions, Deadlocks & Debugging, Memory Models for Parallel Programming, Memory Consistency Models. 14 Hours Unit 4: GPU Programming & Hardware Accelerators CUDA Programming Frameworks, Hardware acceleration platforms (FPGAs and TPUs), Architecture and organization of modern FPGAs, HLS Programming, Case Studies. 14 Hours						

Text Book(s):	1. “Multiprogramming and GPU Programming – An Integrated approach” 2nd Edition by Gerassimos Barlas- Morgan Kaufmann Publications. 2. “Parallel Programming: for Multicore and Cluster Systems”, Thomas Rauber, Gudula Runger, 3rd Ed., Springer, 2023.
Reference Book(s):	1. “Computer Architecture: A Quantitative Approach: John Hennessy, David Patterson”, 6th Edition, Morgan Kaufmann, 2017. 2. “Programming Massively Parallel Processors”, David Kirk and Wen-mei Hwu, 3rd Ed, Morgan Kaufmann, 2016. 3. “Computer Systems: A Programmer's Perspective”, Randal E. Bryant, David R. O'Hallaron, Pearson, 2nd Ed, 2016. 4. “Parallel Programming for FPGAs”, Ryan Kastner, Janarбек Matai, and Stephen Neuendorffer, Creative Commons, 2018. 5. Introduction to Parallel Computing, Second Edition-Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar-Pearson Publication
Course Outcome	• Understand the underpinnings of Parallel Heterogeneous Architectures and Techniques, Memory Consistency Models and Coherence Techniques. • Develop and analyze parallel algorithms using many-core architectures. • Program using popular Parallel Programming Frameworks and Languages trending in the industry. • Implement parallel programs on GPUs and analyze real-world acceleration techniques.

Course Code	UE23CS343BB2	Course Title	Topics in Deep Learning				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages	Pytorch, Tensorflow, Kaggle, Google Colab, Hugging Face models	Desirable Knowledge	Machine Learning				
Prelude	Deep Learning has received a lot of attention over the past few years and has been employed successfully by companies like Google, Microsoft, IBM, Facebook, Twitter etc. to solve a wide range of problems in Computer Vision and Natural Language Processing. In this course we will learn about the building blocks used in these Deep Learning based solutions. of this course students would have knowledge of deep architectures used for solving various Vision and NLP tasks.						
Course Objectives:	- Understand key CNN architectures and the fundamentals of transfer learning. - Learn sequence modeling using RNN, LSTM, GRU, and Transformer-based architectures. - Gain understanding of generative models, graph neural networks, and meta learning techniques. - Attain knowledge of reinforcement learning, diffusion models, federated learning, and latest deep learning advancements.						
Course Contents	Unit 1: Introduction to Deep Learning Introduction, Activation functions, Loss functions, Batch Normalization, Regularization and Optimization. Convolutional Neural Network(CNN): Introduction, Filters, Feature Maps, Max- Pool Layers, Other Pooling Types, Back Propagation. Convolution Architectures - Alexnet, ZFNet, VGGNet, GoogleNet, ResNet. RCNN, FRCNN, Faster RCNN, YOLO. Transfer Learning: Introduction, Motivation, Variations, TL Architecture of CNNs. Hands-on: Assignment on CNN & TL. 14 Hours Unit 2: Recurrent Neural Networks (RNN) and Transformers: Introduction-Recurrent Neurons, Memory Cells, Variable-Length Input-Output Sequences, RNN Architecture, Sequence learning problem, BPTT-Back Propagation Through Time, truncated BPTT, Vanishing and Exploding Gradient, Bidirectional RNN, LSTM Cell and GRU Cell, Text Classification with RNN, Encoder/Decoder architecture, Seq2Seq model with Attention , Transformer model and BERT architecture, Transformer Attention and its implementation, GPT Architecture. Hands-on: Problem using RNNs, Transformers and TL / predefined model 14 Hours Unit 3: Generative Models & Meta Learning: Introduction to Autoencoders , Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contrastive autoencoders , Variational Auto Encoders(VAEs). Generative Adversarial Networks(GANs)- Architecture and Training Methods, Image Generation, DCGAN, Style GAN, WGAN, Applications. Graphical Neural Networks(GNN): Introduction to GNNs, Graph Convolution Networks, Applications. Meta Learning: Introduction to Meta Learning, MAML, FOMAML, Adaptive Neural inductive Learning, Siamese Networks. Hands-on: Problem on Fashion MNIST dataset for generating apparel. 14 hours Unit 4: Reinforcement Learning, Diffusion Models, Federated Learning and Overview of Latest Deep Learning Models: Introduction, Basic Framework of RL, Learning to Optimize Rewards, Credit Assignment Problem, Temporal Difference, Learning and Q Learning. Deep RL: Deep Q Learning, Training and Testing. Diffusion Model, Stable diffusion architectures, Introduction to Vision Transformers. Federated Learning: Horizontal, Vertical and FTL (Federated Transfer Learning). Explainable AI. Hands-on : Flower Architecture for MIMC3 dataset OR Learning to play a simple game using Double Deep (DD) Q-learning implementation on OpenAI gym. 14 Hours						

Text Book(s):	1: “Advanced Deep Learning with Python” - Ivan Vasilev, Packt Publishing, 2019. 2: “Neural Network and Deep learning” by Charu C Agarwal. Springer International Publishing, 2 nd edition, 2023.
Reference Book(s):	1: “Hands-on Machine Learning with Scikit-Learn and TensorFlow”, Aurelian Geron, O'REILLY, 3rd Edition, 2022. 2: “Deep Learning with Keras”, Antonio Gulli and Sujit Pal, Packt Publishing, 1st Edition, 2017. 3: “Pattern Recognition and Machine Learning”, Christopher Bishop, Springer, 1st Edition, 2011 (Reprint). 4: Handouts: Transfer Learning / Latest Deep Learning Techniques / Vision Transformers/ GPT / FL
Course Outcome	• Implement a Neural Network using Tensor Flow and Keras • Classify images using CNN. • Solve time-series related problems with RNN. • Generate data in the form of images using GAN.

Course Code	UE23CS343BB3	Course Title	Database Technologies				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages	MySQL, postgres, Oracle, Apache Spark, Apache Kafka, Amazon Kinesis.	Desirable Knowledge	Database Management Systems.				
Prelude	The last decade saw enormous advancements in the design of large-scale data processing systems due to the rise of Internet services. This course covers the architecture of modern data management systems. Topics include storage management, query optimization, distributed, parallel and stream processing, with a focus on the key design ideas shared across many types of data intensive systems.						
Course Objectives:	• Data Storage Techniques, Indexing Mechanisms. • Query Processing and Query Optimization Techniques. • Parallel and Distributed Databases. • Data Streaming Systems, current Trends in Database Systems.						
Course Contents	Unit 1- Relational Data Model and Storage Formats and Indexing Review of Relational Design Theory – Relational Algebra. Secondary Storage Management: The Memory Hierarchy, Flash Storage and Database Buffer, Architecture, Accelerating Access to Secondary Storage, Disks and Disk Failures, Arranging Data on Disk, Block and Record Addresses, Variable-Length Data and Records – Column Stores, Record Modifications. Index Structures: Concepts, Types of Indexes, B+ Tree Indexes, Hash Tables, Multidimensional Data and Indexes and It’s Applications, Multiple-key Indexes and R-trees, Bitmap Indexes, Indexes in SQL. 14 Hours Unit 2 - Query Processing and Optimization Query Execution: Physical-Query-Plan Operators, One-Pass Algorithms and Two-Pass Algorithms (based on Sorting and Hashing), Clustered and Non-clustered indexes, Buffer Management. The Query Compiler: Parsing and Preprocessing, Algebraic Laws for Improving Query Plans, From Parse Trees to Logical Query Plans, Cost based Plan Selection, Choosing an Order for Joins. 14 Hours Unit 3- Parallel and Distributed Databases Avenues for Parallelism: Parallel Database Architectures, Models of Parallelism, Intra-query and Inter-query Parallelism, Parallel Algorithms on Relations. Distributed Databases – Distributed Data Storage, Types, Advantages, Data Fragmentation and Replication, Distributed Transactions, Distributed Query Processing, Distributed Commit – Two-phase Commit, Distributed Locking, Distributed Catalog Management, Concurrency Control, Peer-to-Peer Distributed Search – Chord Circles. 14 Hours Unit 4 - Data Stream Management & Trends in Database Systems Introducing Stream Processing, Stream Processing Model, Streaming Architectures, Apache Spark as a Stream Processing Engine, Spark’s Distributed Processing Model, Spark’s Resilience Model, Apache Kafka, and Amazon Kinesis. Introduction to Vector databases, Vector Indexing and Applications in the Business World. Current Trends in Database Systems and Decision Support Systems; Data Warehousing and Data Mining Applications; Introducing Data Lake and Data Fabric. 14 Hours						

Text Book(s):	1: "Database Systems: The Complete Book", H Garcia-Molina, JD Ullman and J Widom, 2nd Ed., Pearson, 2018. 2: "Stream Processing with Apache Spark", Gerard Maas & Francois Garillot, O'Reilly, June 2019
Reference Book(s):	1: "Fundamentals of Database Systems", Elmasri and Navathe, Pearson Education, 7th Ed., 2016. 2: "Streaming Systems" by Tyler Akidau, Slava Chernyak, Reuven Lax, O'Reilly, July 2018.
Course Outcome	• Design and Deploy Storage Solutions and Indexing mechanisms for optimized performance of databases. • Perform Query Optimization. • Apply Parallel and Distributed Database approaches to solve problems of large databases. • Manage "Data Streams" using latest tools

Course Code	UE23CS343BB4	Course Title	Machine Learning on Graphs				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages	NetworkX for statistical features of graphs, Tensorflow, Keras, and Scikit Learn for traditional graph ML, and Pytorch Geometric for Graph Neural Networks.	Desirable Knowledge	UE22CS352A – Machine Learning, UE23CS343AB3 - Graph Theory and its applications (Not mandatory)				
Prelude	This course introduces the students to the three meta landscapes of representation, optimization, and objective for the learning tasks on graph data. Graphs are ubiquitous in almost everything today –in social graphs, large-scale biological systems, road networks, polypharmacy, and user-item relations. Graphs, being non-Euclidean, require different machine-learning approaches. This course starts by exploring graph features for traditional machine learning on graphs and discussing a few noteworthy classical graph ML approaches. The second unit focuses on shallow embedding methods for graphs in general, and then the same for knowledge graphs. The third unit focuses on representation learning using graph neural networks and discusses key algorithms along with extensions like temporal graphs and multi-relational graphs. The fourth unit discusses the deep generative models on the graph and model-building techniques using graph neural networks. The theory input concludes with a discussion on the interdisciplinary applications of Graph Neural Networks, interpretability, and meta-learning perspectives of graph AI.						
Course Objectives:	- To understand the theoretical and practical aspects of graph feature-based traditional machine learning and implement the same - To understand the theoretical and practical aspects of and to implement shallow embedding approaches for graphs and multi-relational knowledge graphs. - To understand the theoretical and practical aspects of and to implement neural methods of representation learning on graphs. - To understand the theoretical and practical aspects GNN modeling techniques and deep graph generative models with interdisciplinary applications.						
Course Contents	Unit 1: Traditional machine learning on graphs Types of complex Graphs and computational tasks on Graphs. Graph Features for traditional machine learning– Node-level features. Graph kernel features- Graphlet & Motif, Weisfeiler-Lehman kernel. Measures for neighborhood overlap. Graph clustering and spectral methods –Graph Laplacians, Graph Cuts and clustering, Spectral clustering. Encoder-Decoder perspective – encoder, decoder, and optimization of the encoder-decoder model. Factorizationbased approaches. 14 Hours Unit 2: Graph Representation Learning Based on Shallow Embedding Methods Random Walk-based embedding –DeepWalk and Node2Vec. Limitations of shallow embedding. Shallow embedding in multi-relational Knowledge Graph- embedding as a reconstruction task, loss function, and decoders. Knowledge Graph Completion Task-TransE, TransR, DistMult, ComplEx. Semi-supervised Learning on Graph - Label Propagation Algorithm and Label Spreading Algorithm 14 Hours Unit 3: Graph Learning Using Graph Neural Network Vanilla GNN - Neural message passing framework. Generalized neighborhood aggregation. Generalized						

	update methods. Graph Convolution Networks (GCN): Spectral methods(GCN, AGCN) and Spatial Methods(DCNN, DGCN, LGCN, and GraphSAGE), Multi-relational GCN and Graph Attention Networks (GAT), Graph Transformer. Graph pooling. Expressiveness, designing Maximally expressive GNNs, and Expressive Power by aggregate functions. Applications and loss functions. 14 Hours Unit 4: Deep Generative Models, GNN on complex graphs, and interdisciplinary applications Deep Generative models on graph – Variational autoencoder on the graph, Generative adversarial network on the graph, Auto-regressive methods. GNN on complex graphs – Heterogeneous GNN, Dynamic GNN, Hypergraph GNN. Application of GNN to Biology and Chemistry, Interdisciplinary Applications of GNN: Unstructured data – Text and Image. Structured Data - Social Network, Recommender System, Cyber Security. An introduction to trustworthy graph AI, Meta-Learning perspectives of graph ML 14 Hours
Text Book(s):	1: “Graph Representation Learning”, William L Hamilton, Morgan and Claypool Publishers, 2020. 2: “Deep Learning on graphs”, Yao Ma and Jiliang Tang, Cambridge University Press, 2021 3. “Introduction to Graph Neural networks”, Zhiyuan Liu and Jie Zhou, Synthesis Lectures on Artificial Intelligence and Machine learning, Morgan and Claypool Publishers, 2020.
Reference Book(s):	1.“Social Media Mining”, Reza Zafarani, Cambridge University Press, 2015 (Asian Economic edition available) 2: “ Network Science” by Albert Barabasi , 2016
Course Outcome	• feature engineering-based graph ML tasks. • downstream machine learning tasks on graphs using shallow embedding. • solve a downstream machine-learning task using a graph neural network. • graph modeling techniques and appreciate deep generative models on graphs.

Course Code	UE23CS343BB5	Course Title	Embedded system programming with micro python				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages	Microcontroller: Pyboard, ESP32, BBC micro:bit Components: LEDs, resistors, buttons, temperature sensors (e.g., TMP36, DHT11), OLED display, potentiometer, buzzer, jumper wires, breadboard, USB cable	Desirable Knowledge	Python for computational problem solving, Microprocessor and Computer Architecture				
Prelude	This course introduces learners to the world of embedded systems using MicroPython, a lightweight implementation of Python 3 designed to run on microcontrollers. Students will develop a strong foundation in embedded programming principles using accessible and readable Python code. The course guides students from basic setup and scripting to more advanced interfacing with sensors, hardware protocols, and simple Internet of Things (IoT) applications. Through practical labs, real-world examples, and mini-projects, learners will gain hands-on experience in building responsive and connected embedded systems.						
Course Objectives:	• Data Storage Techniques, Indexing Mechanisms. • Query Processing and Query Optimization Techniques. • Parallel and Distributed Databases. • Data Streaming Systems, current Trends in Database Systems.						
Course Contents	Unit 1- Relational Data Model and Storage Formats and Indexing Review of Relational Design Theory – Relational Algebra. Secondary Storage Management: The Memory Hierarchy, Flash Storage and Database Buffer, Architecture, Accelerating Access to Secondary Storage, Disks and Disk Failures, Arranging Data on Disk, Block and Record Addresses, Variable-Length Data and Records – Column Stores, Record Modifications. Index Structures: Concepts, Types of Indexes, B+ Tree Indexes, Hash Tables, Multidimensional Data and Indexes and It’s Applications, Multiple-key Indexes and R-trees, Bitmap Indexes, Indexes in SQL. 14 Hours Unit 2 - Query Processing and Optimization Query Execution: Physical-Query-Plan Operators, One-Pass Algorithms and Two-Pass Algorithms (based on Sorting and Hashing), Clustered and Non-clustered indexes, Buffer Management. The Query Compiler: Parsing and Preprocessing, Algebraic Laws for Improving Query Plans, From Parse Trees to Logical Query Plans, Cost based Plan Selection, Choosing an Order for Joins. 14 Hours Unit 3- Parallel and Distributed Databases Avenues for Parallelism: Parallel Database Architectures, Models of Parallelism, Intra-query and Inter-query Parallelism, Parallel Algorithms on Relations. Distributed Databases – Distributed Data Storage, Types, Advantages, Data Fragmentation and Replication, Distributed Transactions, Distributed Query Processing, Distributed Commit – Two-phase Commit,						

	Distributed Locking, Distributed Catalog Management, Concurrency Control, Peer-to-Peer Distributed Search – Chord Circles. Unit 4 - Data Stream Management & Trends in Database Systems Introducing Stream Processing, Stream Processing Model, Streaming Architectures, Apache Spark as a Stream Processing Engine, Spark's Distributed Processing Model, Spark's Resilience Model, Apache Kafka, and Amazon Kinesis. Introduction to Vector databases, Vector Indexing and Applications in the Business World. Current Trends in Database Systems and Decision Support Systems; Data Warehousing and Data Mining Applications; Introducing Data Lake and Data Fabric. 14 Hours
Text Book(s):	1: "Database Systems: The Complete Book", H Garcia-Molina, JD Ullman and J Widom, 2nd Ed., Pearson, 2018. 2: "Stream Processing with Apache Spark", Gerard Maas & Francois Garillot, O'Reilly, June 2019
Reference Book(s):	1: "Fundamentals of Database Systems", Elmasri and Navathe, Pearson Education, 7th Ed., 2016. 2: "Streaming Systems" by Tyler Akidau, Slava Chernyak, Reuven Lax, O'Reilly, July 2018.
Course Outcome	• Design and Deploy Storage Solutions and Indexing mechanisms for optimized performance of databases. • Perform Query Optimization. • Apply Parallel and Distributed Database approaches to solve problems of large databases. • Manage "Data Streams" using latest tools

Course Code	UE23CS343BB6	Course Title	Information Security				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages	SEED Labs VM, Scapy, Burp Suite, Metasploit, Nmap, etc.	Desirable Knowledge	Computer Network Security				
Prelude	This course will present security aspects from a secure software life cycle process – requirement, architecture, design, coding, and testing. Students will have opportunity to dwell well in to technical "how to" with hands-on sessions, assignments, and some case study discussions.						
Course Objectives:	• To understand various cyber threats and attacks and secure software development process. • To learn attack and defence mechanisms for buffer overflow, shellshock attack, etc. • To understand the concept of threat modelling and its application. • To learn about the most common web application security vulnerabilities. • To understand and apply various penetration testing techniques and tools.						
Course Contents	Unit 1: Introduction and Privilege Escalation Attacks Software Threats, Attacks and Vulnerabilities, CIA Triad, OWASP Top 10, CVE, Security and reliability, Security vs. privacy, Cyberattack Types, Anatomy of an Attack, Security Concepts and Relationships. Use cases and Misuse cases, Misuse case legend, Security use case vs Misuse case, Secure Software Development Life Cycle (SDL). Case Study: Target case study. Set-UID program: Need for privileged programs, Set-UID mechanism, Superman story, Attack surfaces, Invoking other surfaces, Principle of least privilege. Environment variables and attacks: Environment variables, Attack surface, Attacks via Dynamic linker, External program, and Library. Lab: Set-UID program & Environment variables and attacks. Shellcode attack on Set-UID and CGI programs. 14 Hours Unit 2: Software Vulnerabilities and Malicious Software Buffer overflow attack: Program memory layout, Stack and function invocation, Stack buffer-overflow attack, Attacks with Unknown address and Buffer size, Shellcode, Countermeasures & Defeating it. Return-to-libc attack: Introduction, Launch the attack part I & part II. Format string vulnerability: Introduction to functions and format string, Vulnerable program, Exploiting the vulnerability, Code injection attack, Countermeasures. Case study: Target case study. Malware and its Types, Malware analysis: Conifer, Morris, Stuxnet worm, Ransomware. 14 Hours Unit 3: Threat modelling and Basic Web Security Threat Modelling, Trust Boundaries, Attack Surfaces, Brainstorming, Modelling Methods, STRIDE model and variants, Defensive tactics, and Technologies, Privacy Threats, Taxonomy and Types. Web security basics, Attacks on HTTP GET and POST services, Cross Site Request Forgery (XSRF/CSRF): Cross-site requests and its problems, CSRF attacks, Countermeasures. Case study: Apple - Privacy vs Safety. 14 Hours Unit 4: Web application security and Penetration Testing (XSS/CSS) Attack: CSS attack, CSS attacks in action, Self-propagation, Preventing CSS attacks. SQL injection attack: Introduction to SQL, interacting with database in web, Launching SQL injection attacks, Countermeasures. Static analysis, Penetration testing: Introduction, Benefits, Drawbacks, Penetration testing tools and Fuzzing, Patching. 14Hours						

Text Book(s):	1. "Computer & Internet Security: A Hands-on Approach", Wenliang Du, 2nd Edition/3 rd Edition.
Reference Book(s):	1: "Computer Security: Principles and Practice", William Stallings and Lawrie Brown, Pearson Education, 3rd Edition, 2014. 2: "Secure Programming with Static Analysis", Brian Chess and Jacob West, Pearson Education, 2007
Course Outcome	• Identify possible misuse cases in the context of software development. • Defend against various attacks and how to write secure code. • Apply threat modelling techniques to expose inherent vulnerabilities in applications. • Design and develop secure web applications. • Exploit software vulnerabilities and launch attacks.

Course Code	UE23CS343BB7	Course Title	Mobile and Autonomous Robot				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languag es	C,C++,Python, ROS, Webots, AirSim, Gazebo	Desirable Knowledge	UE23CS343AB7-Introduction to Autonomous Systems with ROS , UE23MA241B-Linear Algebra and its Applications				
Prelude	This course provides an introduction to autonomous robots, focusing on the principles and techniques behind their design, programming, and control. Students will explore key topics such as perception, planning, and navigation, while gaining hands-on experience through projects and simulation using ROS/ROS2. The course combines theoretical foundations with practical applications across both terrestrial and aerial robotic platforms						
Course Objectives:	- Understand the evolution, current trends, and research directions in autonomous mobile robotics. - Learn the theoretical and practical fundamentals of designing and operating autonomous robots. - Explore core topics including perception, localization, planning, and control. - Gain hands-on experience with robot simulation and programming using ROS/ROS2. - Apply knowledge to develop and test autonomous systems across terrestrial and aerial platforms.						
Course Contents	Unit 1: Introduction to Robotics and Locomotion: Robotics overview: Past, Present and Future; Robot Hardware and Software. AI in Robotics: Machine Learning Basics for Robotics. Locomotion: The basics of SOTA locomotion systems, Legged Robots, Wheeled Robots, Aerial Robots. Robot Kinematics, ROS overview: ROS/ROS2 for robotics, ROS architecture and communication protocols, ROS packages for robot hardware control. 14 Hours Unit 2: Perception Introduction to Perception, Sensors for Robots, Visual and inertial measurements: Gyroscope, accelerometers, IMU, GPS, Range sensors, camera vision and LiDAR. Fundamentals of Computer Vision and Image Processing and filtering, Feature Extraction: Object detection and Place Recognition, Stereo vision and 3D perception, Feature Extraction Based on Range Data (Laser, Ultrasonic). 14 Hours Unit 3: Localization Introduction and Challenges of Localization, To Localize or Not to Localize, Belief Representation, Map representation, Probabilistic Map-Based Localization: Markov localization, Kalman Filter localization, Autonomous Map Building: SLAM, EKF SLAM, Particle filter SLAM, GraphSLAM, Open challenges in SLAM. 14 Hours Unit 4: Planning and Navigation Introduction Path Planning and Navigation, Path planning: Graph search and Potential field path planning. Obstacle avoidance: Bug algorithm, Vector field histogram, bubble band technique, Curvature velocity techniques, Dynamic window approaches. Navigation Architectures. Revisiting Navigation using Reinforcement Learning and Imitation learning: A Robotics Perspective, Applications and Social Implications. 14 Hours						

Text Book(s):	1: Introduction to Autonomous Mobile Robots (Intelligent Robotics and Autonomous Agents series) second edition by Roland Siegwart, Illah Reza Nourbakhsh, Davide Scaramuzza 2: Artificial Intelligence for Robotics Build Intelligent Robots that Perform Human Tasks Using AI Techniques By Francis X. Govers · 2018
Reference Book(s):	1: Introduction to Autonomous Robots: Nikolaus Correll, Magellan Scientific, 2016. 2: ROS Robot Programming, ROBOTIS Co., Ltd. From the basic concept to practical programming and robot application. YoonSeok Pyo, HanCheol Cho, RyuWoon Jung, and TaeHoon Lim. 3: Introduction to Robotics: Mechanics and Control 4th Edition, John Craig, ISBN-13: 978-0133489798, Pearson; 4th edition 4: Probabilistic Robotics By Sebastian Thrun, Wolfram Burgard and Dieter Fox. ISBN-13: 978-0262201629, ISBN-10: 0262201623. Intelligent Robotics and Autonomous Agents series; 1st Edition
Course Outcome	• Explain the principles and applications of autonomous mobile robots, including their key components and system architecture. • Identify and integrate appropriate sensors and actuators for robot perception and control. • Develop and simulate robotic systems using ROS/ROS2 for navigation, planning, and hardware interfacing. • Apply algorithms for localization, path planning, and obstacle avoidance in real or simulated environments. • Design and implement autonomous robotic solutions for terrestrial and aerial platforms.

Course Code	UE23CS343BB8	Course Title	Security for Internet of Things				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages	Wireshark, Yersinia, VoIP Hopper, Bettercap,aircrack-ng	Desirable Knowledge	Internet of Things				
Prelude	This course enables the learner to understand the security threats associated with Internet of Things and perform security testing on connected devices within our homes and enterprise to build a better model for protecting ourselves.						
Course Objectives:	- Understand the threats of IoT devices - Understand vulnerabilities, possible attacks and learn the security testing methodology for networks, IoT hardware and radio protocols. - Enhance knowledge with hands-on experience on IoT security tools and attack analysis. - Understand the various Security measures to be adopted to secure IoT devices						
Course Contents	Unit 1: Introduction to IoT Security Introduction to IoT Security, Traditional security vs IoT security,Basic concepts of IoT Architecture from security perspective, Challenges of IoT Security, OWASP Top 10 security risks and consumer IoT security guidance, Threat Modelling for IoT attack, Common IoT threats, Network Hacking: VLAN hopping in IoT networks. 14 hours Unit 2: Network Security and Hardware Security for IoT devices MQTT authentication, Analysing Network protocols: Wireshark dissector and Nmap Scripting Engine module for the DICOM protocol, exploiting zero-configuration Networking: UPnP, mDNS, DNS-SD, and WS-Discovery, Hardware Hacking: UART, JTAG, and SWD Exploitation- Hacking an STM32F103 microcontroller using UART and SWD.SPI and I2C, Firmware Hacking. 14 Hours Unit 3: Radio Hackingand Smart IoT devices hacking Radio Hacking:Short Range Radio: Abusing RFID, Bluetooth low energy, Medium Range Radio: Hacking Wi-Fi, Long Range Radio: LPWAN, Smart home, Hacking the smart home: Gaining physical entry to a building, cloning a keylock system’s RFID Tag, Jamming the Wireless alarm. 14 Hours Unit 4: Targeting the IoT Ecosystem and Secure Design of IoT Devices Playing back an IP Camera stream, Analysing IP Camera Network traffic, Extracting the video stream, attacking a Smart treadmill, Secure design goals: Mitigate automated attack risks, Secure points of integration, Hardware protection measures, IoT IAM infrastructure: PKI for IoT, Revocation support: OCSP, SSL pinning, Authorization and access control with OAuth 2.0 Cryptographic Controls for IoT Protocols. 14 Hours						

Text Book(s):	1: “Practical IoT Hacking”, Fotios Chantzis, Ioannis Stais, Paulino Calderon, Evangelos Deirmentzoglou, Beau Woods, March 2021, No Starch Press Publishers, ISBN: 9781718500907.
Reference Book(s):	1: “Practical Internet of Things Security” , Brian Russell, Drew Van Duren, Packt Publishers, 2nd Edition
Course Outcome	of this course, the student will be able to: • Identify and describe the variety of IoT systems architectures, essential components and challenges specific to IoT systems. • Analyze various network and hardware security mechanisms for IoT devices • Gain hands-on experience on different tools to target IoT ecosystem. • Analyze and apply appropriate security and privacy solutions for real-world applications.

Course Code	UE23CS343BB9	Course Title	Applied ML in IoT with TinyML				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages	Arduino Nano 33 BLE sense , Arduino IDE , Google Colab	Desirable Knowledge	Machine Learning				
Course Objectives:	- Design and deploy simple Machine Learning models on microcontrollers for various IoT applications. - Preprocess sensor data, optimize code, and evaluate the trade-off between model accuracy and resource consumption. - Utilize tools like TensorFlow Lite Micro and TinyML Kit for TinyML development. - Integrate sensor data acquisition, communication protocols, and data pipelines for real-time TinyML applications.						
Course Contents	Unit-1: Foundations of Machine Learning: Introduction to IoT & TinyML, The ML paradigm, Building blocks of DL, NN, Regression with Dense NN, Classification with Dense NN, Image Classification using CNN, Datasets and Model Performance Metrics, Preventing Overfitting. 14 Hours Unit-2: TensorFlow Lite for Microcontrollers & TinyMLKit: What Is TensorFlow Lite for Microcontrollers?, Build Systems, Supporting a New Hardware Platform, Supporting a New IDE or Build System, Integrating Code Changes Between Projects and Repositories, Contributing Back to Open Source, Supporting New Hardware Accelerators, Understanding the File Format, Porting TensorFlow Lite Mobile Ops to Micro, TinyML Kit Overview, TinyML Kit Setup, TinyML Kit Sensor Testing, Sensor Testing, Sensor Fusion. 14 Hours Unit-3: Applications and Deployment to Microcontrollers (MCUs)-1: TF-Lite, TFL-Micro, TFL-Micro Hello- World example, KeyWord Spotting (KWD) using Edge Impulse, KDW dataset creation, Micro Speech Example, Workflow, Model development & testing, Person-detection example, Image classification 14 Hours Unit-4: Applications and Deployment to Microcontrollers (MCUs)-2: Magic Wand: Building and Application, Training a Model, Building a Weather Station with TensorFlow Lite for Microcontrollers, Voice Controlling LEDs with Edge Impulse. 14 Hours						
Text Book(s):	- Pete Warden and Daniel Situnayake, TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low- Power Microcontrollers, O'Reilly Media, 2020. - Vijay Janapa Reddi at Harvard and open-source collaborators, Machine Learning Systems with						

	TinyML, Open- source collaborative-effort book, 2023-present.
Reference Book(s)	1: Gian Marco Iodice and Ronan Naughton, TinyML Cookbook: Combine artificial intelligence and ultra-low-power embedded devices to make the world smarter, 2022. 2: Francois Chollet, Deep Learning with Python, Manning, Second Edition, 2021
Course Outcome	• Understand the fundamental concepts of ML, Deep Learning (DL), and their application in the IoT domain. • Utilize TensorFlow Lite Micro and TinyML Kit for developing and deploying ML models on microcontrollers. • Design and implement TinyML applications for various functionalities like keyword spotting, image classification, and sensor fusion. • Evaluate and interpret the performance of TinyML models on real hardware.

Course Code	UE23CS343BB10	Course Title	Video Processing and understanding				
Program	B.Tech CSE	Hours per week/ Credit Assigned	L	T	P	S	C
			4	0	0	4	4
Semester	6	Type of Course	Elective -4				
AI Tools /Tools/Languages		Desirable Knowledge	Image Processing and Machine Learning.				
Prelude	The rise of video as a dominant medium in digital communication and content creation has made the ability to analyze and understand video data a vital skill in the age of AI. This course provides a comprehensive introduction to the theory and application of video processing and video understanding, spanning low-level video enhancement and segmentation techniques to high-level video analytics such as semantic scene understanding, and vision-language integration. Students will explore cutting-edge techniques in action recognition and vision-language models, with applications in video captioning, anomaly detection, and visual question answering, preparing them for real-world challenges in video analytics and artificial intelligence.						
Course Objectives:	- To introduce the foundational concepts and techniques used in digital video processing including representation formats, compression techniques, resolution and frame rate. - To explore motion analysis methods such as optical flow, motion tracking, and temporal feature extraction from video streams. - To equip students with techniques for video understanding including object recognition, scene parsing, and semantic analysis. - To provide exposure to deep learning-based approaches and modern AI and for video analytics, including action recognition, anomaly detection and vision-language models.						
Course Contents	Unit 1: Fundamentals of Video Processing Introduction to digital video: Spatial resolution and frame rate, color spaces (RGB, YUV), video enhancement techniques (denoising, sharpening), video sampling and interpolation, video formats and compression techniques (H.264 (MPEG-4 AVC)), frame extraction and representation (I-frames, P-frames, B-frames), separating audio and video from an input video. 14 Hours Unit 2: Video segmentation, motion estimation, and tracking Video segmentation: Scene cut detection, spatio-temporal segmentation, Motion Detection and estimation (frame differencing, background subtraction, optical flow methods: Lucas-Kanade, Horn-Schunck), Spatial Feature extraction: 2D CNN, Skeletal keypoint detection, Temporal feature extraction: 3D CNN, temporal pooling, Object tracking: Kalman filter, DeepSORT. 14 Hours Unit 3: Video Understanding-1 Key frame extraction and scene segmentation, video-based object recognition: YOLOV++, semantic segmentation in video and its challenges: SAM2, pose estimation (for blurring of faces for privacy, analyzing human movement, orienting a robot for navigation, etc.), extraction and processing text content in video (for indexing and retrieval), audio and video semantic analysis, scene understanding.						

	14 Hours Unit 4: Video Understanding-2 Action Recognition: C3D, R2+1 D, MViTv2, Anomaly recognition: Real-time anomaly detection in dynamic environments, GNN for activity recognition, vision language models: video captioning, visual question answering and reasoning, visual dialog: video CLIP, LLAVA. 14 Hours
Text Book(s):	1. M. Tekalp, Digital Video Processing. Signal Processing Series, Prentice Hall, 2015 (2nd edition). 2. Szeliski, Richard. Computer vision: algorithms and applications. Springer Nature, 2022.
Reference Book(s):	1. Awais, Muhammad, et al. "Foundation Models Defining a New Era in Vision: a Survey and Outlook." IEEE Transactions on Pattern Analysis and Machine Intelligence (2025). 2. Jerry D. Gibson and Al Bovik. "Handbook of Image and Video Processing". Academic Press, Inc., USA (2000). 3. "Computer Vision A Modern Approach", D. A Forsyth and J. Ponce, Pearson Education, 2003 4. A. Bovik (Ed.), The Essential Guide to Video Processing. Academic Press, 2009. 5. Reading of relevant, recent research papers prescribed during the course
Course Outcome	• Apply video preprocessing and enhancement techniques to improve video quality and extract frames based on compression methods, • Assess the suitable solution and implement motion estimation and tracking algorithms, including Kalman filter and DeepSORT, for object tracking, • Implement methods to detect and segment objects in video, followed by scene understanding for effective video analysis and • Recognize actions and detect anomalies in video using advanced deep learning models, and apply vision-language models for tasks like video captioning and visual question answering.